

# Quicksort

Übungsblatt mit steigender Schwierigkeit  
(Automatisch generiert)

Algorithmen und Datenstrukturen I

## Hinweise

### Quicksort-Algorithmus (mit erstem Element als Pivot):

1. Wähle das **erste Element** als Pivot
2. Partitioniere: Elemente  $\leq$  Pivot nach links, Elemente  $>$  Pivot nach rechts
3. Platziere Pivot an seiner endgültigen Position
4. Rekursiv auf linke und rechte Teilarrays anwenden

### Partition-Schema (Hoare-ähnlich):

- Zeiger  $i$  startet links (nach Pivot), Zeiger  $j$  startet rechts
- $i$  wandert nach rechts, bis Element  $>$  Pivot gefunden
- $j$  wandert nach links, bis Element  $\leq$  Pivot gefunden
- Wenn  $i < j$ : Tausche Elemente und wiederhole
- Wenn  $i \geq j$ : Tausche Pivot mit Element an Position  $j$

### Markierungen:

- Wert = Element hat seine **endgültige Position** erreicht (grün hinterlegt)
- Unterstrichene Teillisten = Argumente des nächsten rekursiven Aufrufs
- Gelb = aktuelles Pivot-Element

## Aufgabe 1 – Schwierigkeit: ★★

### Standard-Quicksort

Gegeben sei folgende Liste, die in einem Array abgelegt ist:

**12, 3, 56, 18, 32, 19, 1, 21**

Die Liste soll mit dem Standard-Quicksort-Verfahren (keine Variante) in-situ sortiert werden. Als Pivot-Element soll immer das **erste Element** der zu sortierenden (Teil-)Liste genommen werden.

#### Fragen:

1. Geben Sie den Zustand des Arrays nach der **ersten** Partition an.
2. Geben Sie den Zustand des Arrays nach der **letzten** Partition an.
3. Wie viele Vergleiche werden insgesamt durchgeführt?

### Lösung:

#### Schritt-für-Schritt-Lösung:

**Schritt 0:** Ausgangszustand

|    |   |    |    |    |    |   |    |
|----|---|----|----|----|----|---|----|
| 12 | 3 | 56 | 18 | 32 | 19 | 1 | 21 |
| 0  | 1 | 2  | 3  | 4  | 5  | 6 | 7  |

=== Rekursionsebene 0: Partitioniere [0..7] ===

**Schritt 1:** Partition [0..7]: Pivot = 12 (erstes Element)

|    |   |    |    |    |    |   |    |
|----|---|----|----|----|----|---|----|
| 12 | 3 | 56 | 18 | 32 | 19 | 1 | 21 |
| 0  | 1 | 2  | 3  | 4  | 5  | 6 | 7  |

Pivot

**Schritt 2:** Initialisiere Zeiger: i=1, j=7

|    |   |    |    |    |    |   |    |
|----|---|----|----|----|----|---|----|
| 12 | 3 | 56 | 18 | 32 | 19 | 1 | 21 |
| 0  | 1 | 2  | 3  | 4  | 5  | 6 | 7  |

Pivot

i ↓ j ↓

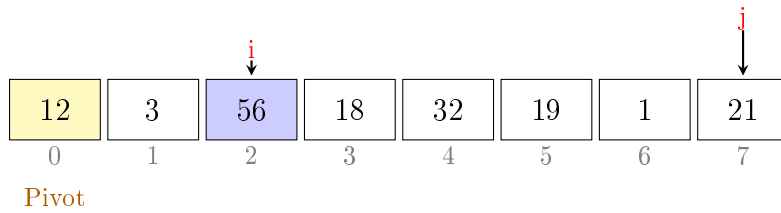
**Schritt 3:** array[1]=3 Pivot=12 → i nach rechts

|    |   |    |    |    |    |   |    |
|----|---|----|----|----|----|---|----|
| 12 | 3 | 56 | 18 | 32 | 19 | 1 | 21 |
| 0  | 1 | 2  | 3  | 4  | 5  | 6 | 7  |

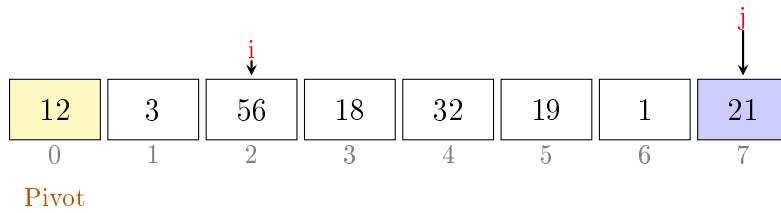
Pivot

i ↓ j ↓

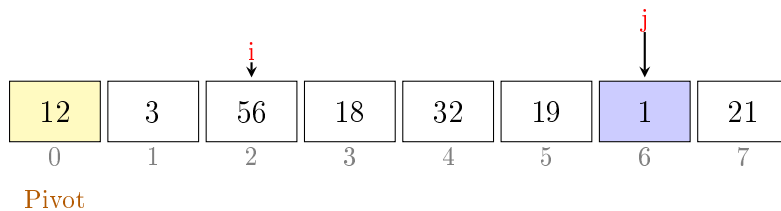
**Schritt 4:**  $\text{array}[2]=56 > \text{Pivot}=12 \rightarrow i$  stoppt



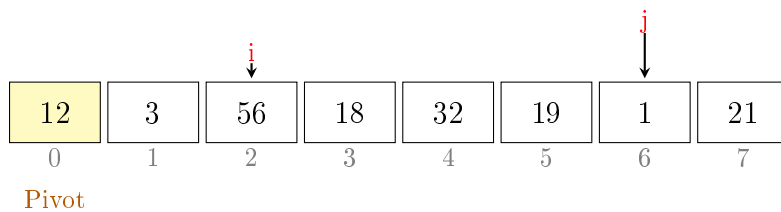
**Schritt 5:**  $\text{array}[7]=21 > \text{Pivot}=12 \rightarrow j$  nach links



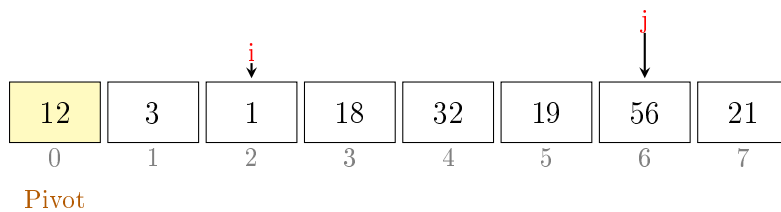
**Schritt 6:**  $\text{array}[6]=1$  Pivot=12  $\rightarrow j$  stoppt



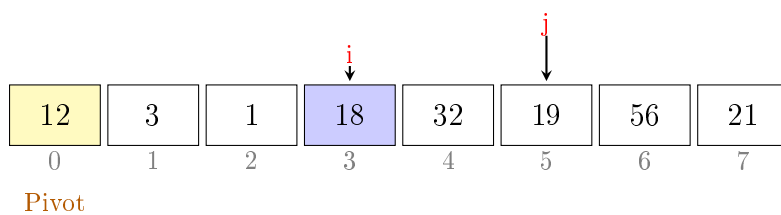
**Schritt 7:** Tausche  $\text{array}[2]=56 \leftrightarrow \text{array}[6]=1$



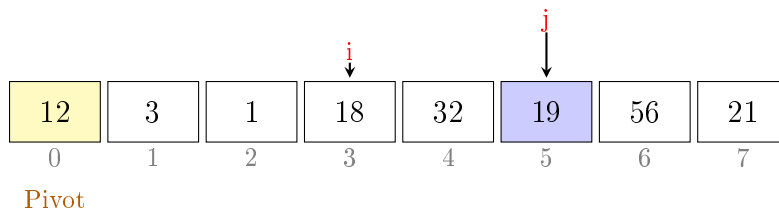
**Schritt 8:** Nach Tausch:  $\text{array}[2]=1$ ,  $\text{array}[6]=56$



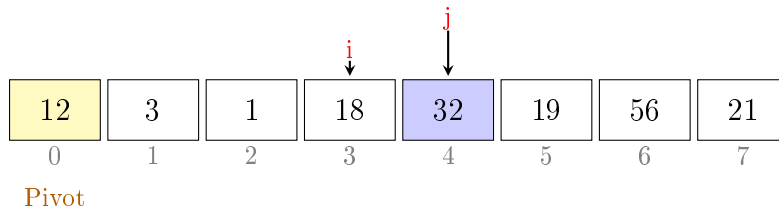
**Schritt 9:**  $\text{array}[3]=18 > \text{Pivot}=12 \rightarrow i$  stoppt



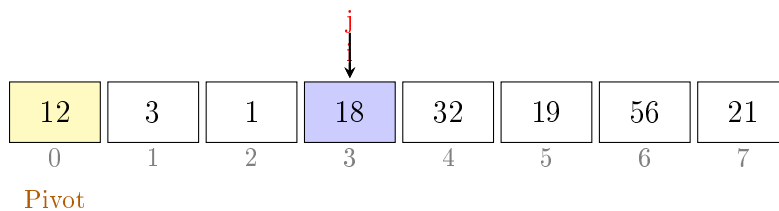
**Schritt 10:**  $\text{array}[5]=19 > \text{Pivot}=12 \rightarrow j$  nach links



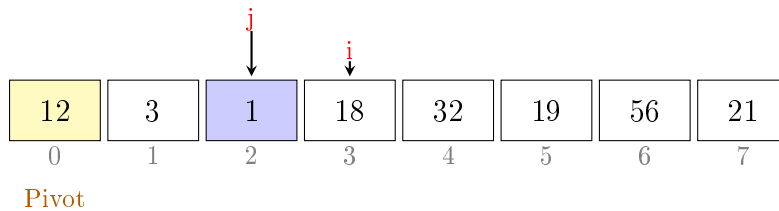
**Schritt 11:**  $\text{array}[4]=32 > \text{Pivot}=12 \rightarrow j$  nach links



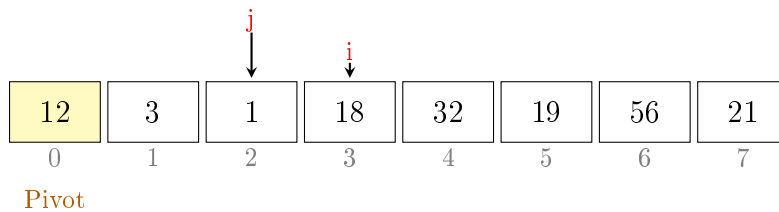
**Schritt 12:**  $\text{array}[3]=18 > \text{Pivot}=12 \rightarrow j$  nach links



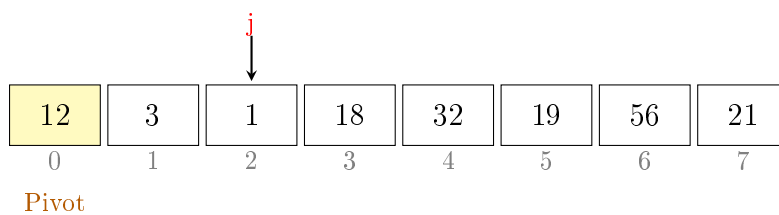
**Schritt 13:**  $\text{array}[2]=1$  Pivot=12  $\rightarrow j$  stoppt



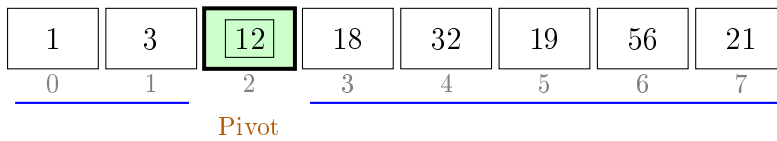
**Schritt 14:**  $i=3$   $j=2 \rightarrow$  Zeiger gekreuzt, Partition beenden



**Schritt 15:** Platziere Pivot: Tausche  $\text{array}[0]=12 \leftrightarrow \text{array}[2]=1$

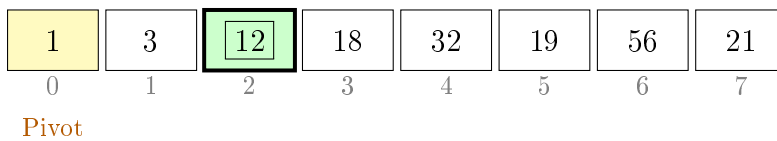


**Schritt 16:** Partition [0..7] fertig: Pivot 12 an Position 2

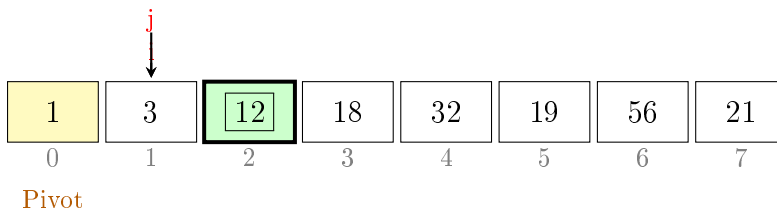


=== Rekursionsebene 1: Partitioniere [0..1], [3..7] ===

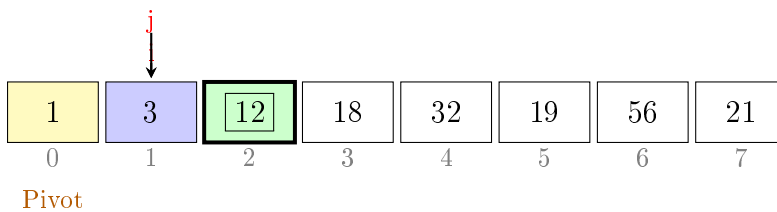
**Schritt 17:** Partition [0..1]: Pivot = 1 (erstes Element)



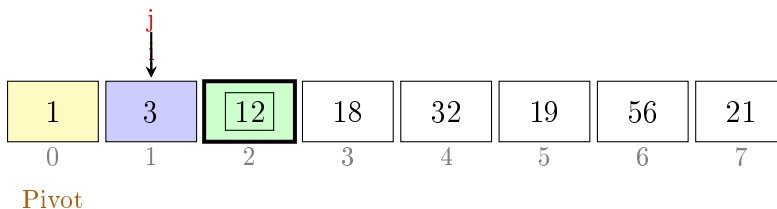
**Schritt 18:** Initialisiere Zeiger: i=1, j=1



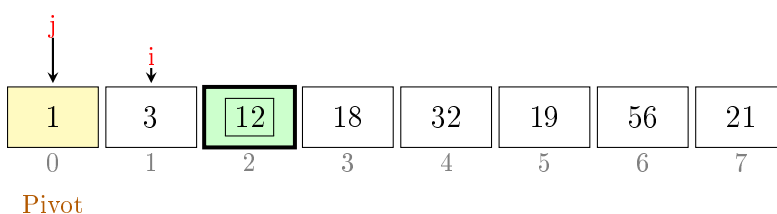
**Schritt 19:** array[1]=3 > Pivot=1 → i stoppt



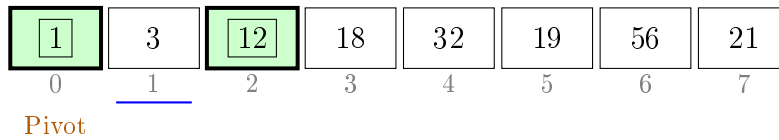
**Schritt 20:** array[1]=3 > Pivot=1 → j nach links



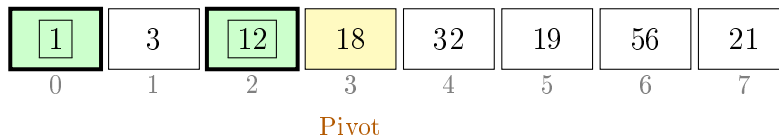
**Schritt 21:** i=1 j=0 → Zeiger gekreuzt, Partition beenden



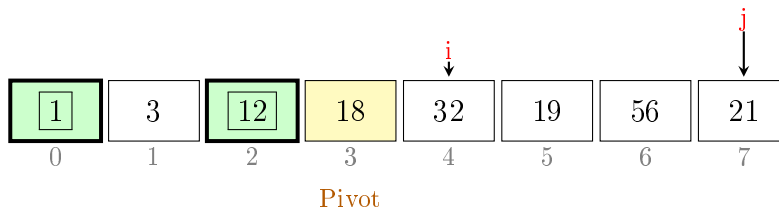
**Schritt 22:** Partition [0..1] fertig: Pivot 1 an Position 0



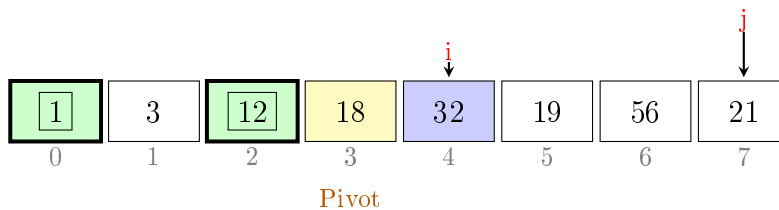
**Schritt 23:** Partition [3..7]: Pivot = 18 (erstes Element)



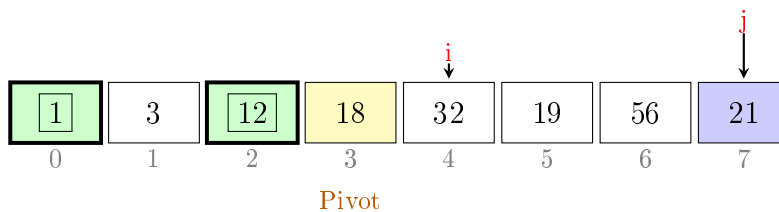
**Schritt 24:** Initialisiere Zeiger: i=4, j=7



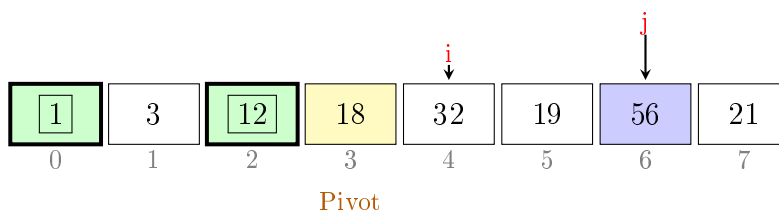
**Schritt 25:** array[4]=32 > Pivot=18 → i stoppt



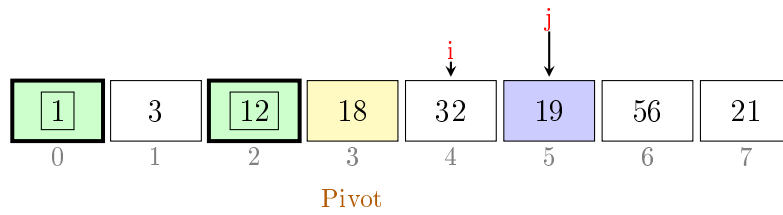
**Schritt 26:** array[7]=21 > Pivot=18 → j nach links



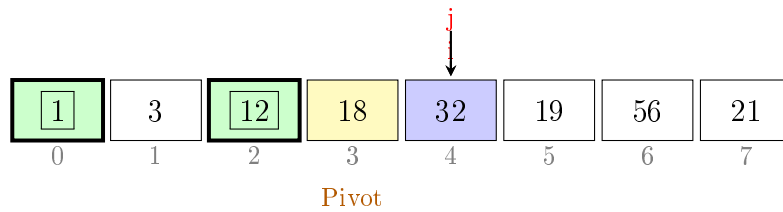
**Schritt 27:** array[6]=56 > Pivot=18 → j nach links



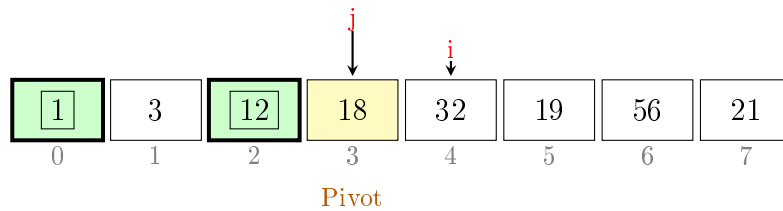
**Schritt 28:** array[5]=19 > Pivot=18 → j nach links



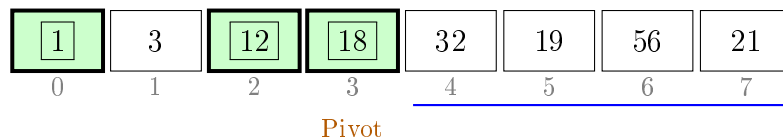
**Schritt 29:**  $\text{array}[4]=32 > \text{Pivot}=18 \rightarrow j$  nach links



**Schritt 30:**  $i=4, j=3 \rightarrow$  Zeiger gekreuzt, Partition beenden

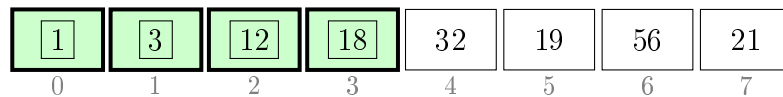


**Schritt 31:** Partition [3..7] fertig: Pivot 18 an Position 3

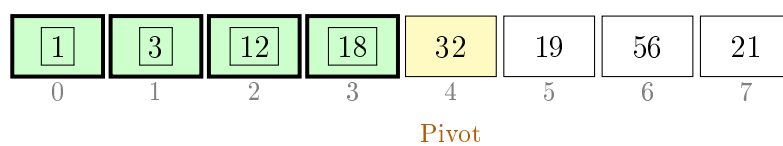


=== Rekursionsebene 2: Partitioniere [4..7]; Basisfall: [1] (Basisfall) ===

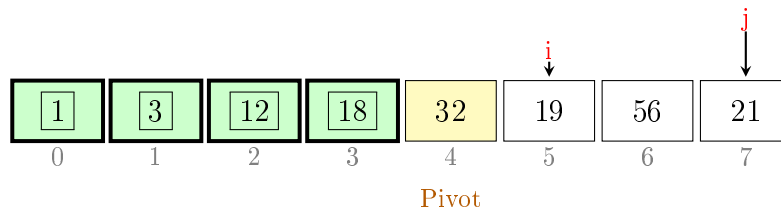
**Schritt 32:** quicksort(1, 1):  $l == r$ , einzelnes Element  $a[1]=3$  ist sortiert



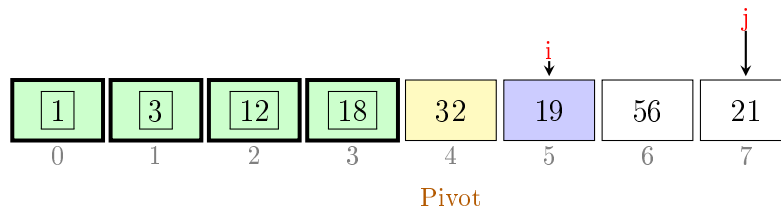
**Schritt 33:** Partition [4..7]: Pivot = 32 (erstes Element)



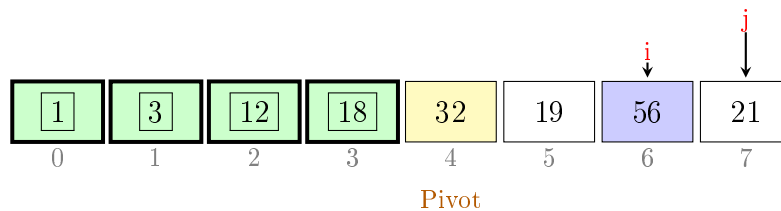
**Schritt 34:** Initialisiere Zeiger:  $i=5, j=7$



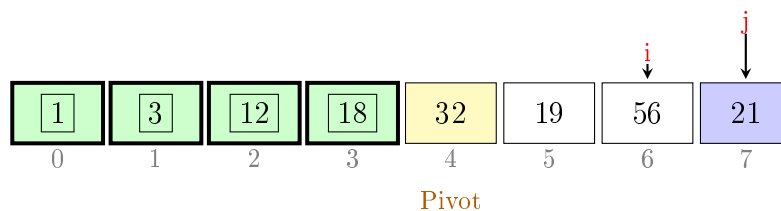
**Schritt 35:**  $\text{array}[5]=19$  Pivot=32  $\rightarrow$  i nach rechts



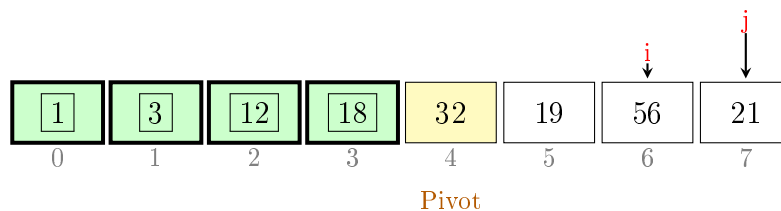
**Schritt 36:**  $\text{array}[6]=56 > \text{Pivot}=32 \rightarrow$  i stoppt



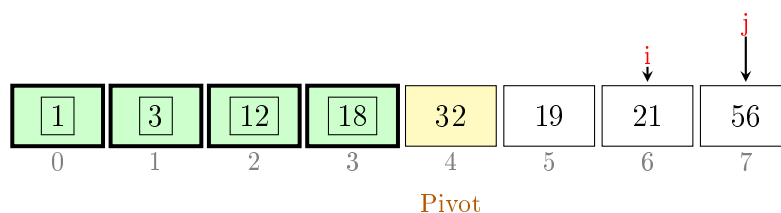
**Schritt 37:**  $\text{array}[7]=21$  Pivot=32  $\rightarrow$  j stoppt



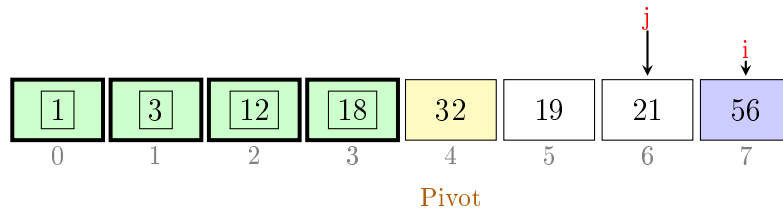
**Schritt 38:** Tausche  $\text{array}[6]=56 \leftrightarrow \text{array}[7]=21$



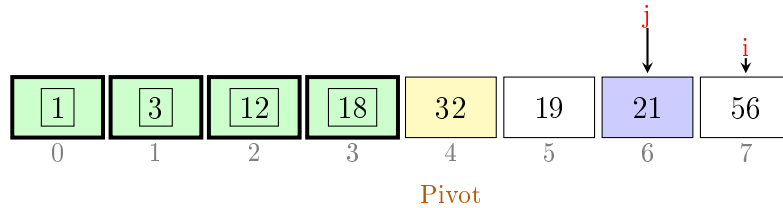
**Schritt 39:** Nach Tausch:  $\text{array}[6]=21$ ,  $\text{array}[7]=56$



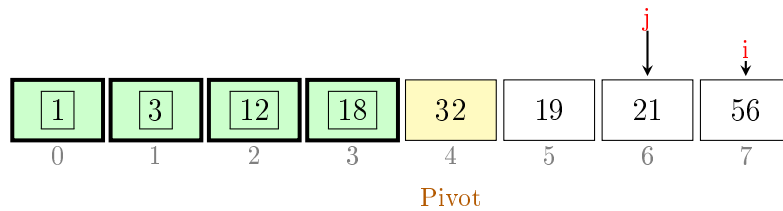
**Schritt 40:**  $\text{array}[7]=56 > \text{Pivot}=32 \rightarrow i$  stoppt



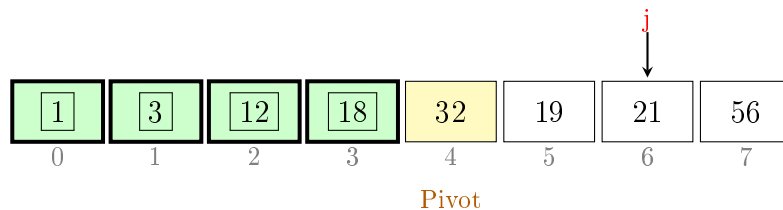
**Schritt 41:**  $\text{array}[6]=21$   $\text{Pivot}=32 \rightarrow j$  stoppt



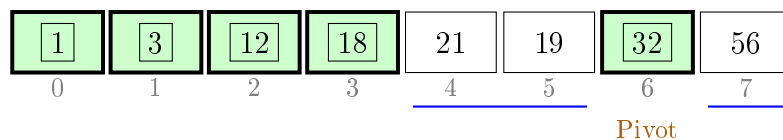
**Schritt 42:**  $i=7$   $j=6 \rightarrow$  Zeiger gekreuzt, Partition beenden



**Schritt 43:** Platziere Pivot: Tausche  $\text{array}[4]=32 \leftrightarrow \text{array}[6]=21$

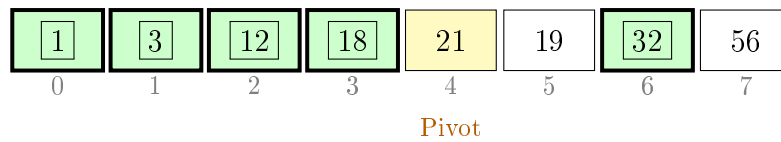


**Schritt 44:** Partition [4..7] fertig: Pivot 32 an Position 6

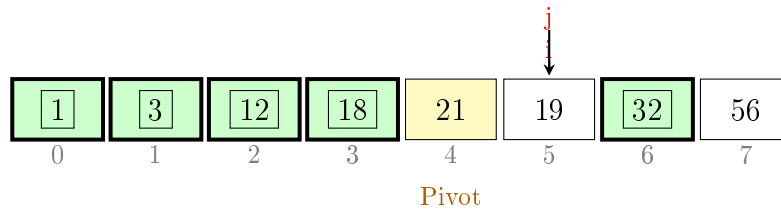


=== Rekursionsebene 3: Partitioniere [4..5]; Basisfall: [7] (Basisfall) ===

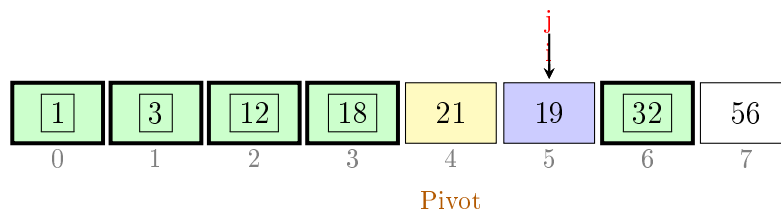
**Schritt 45:** Partition [4..5]: Pivot = 21 (erstes Element)



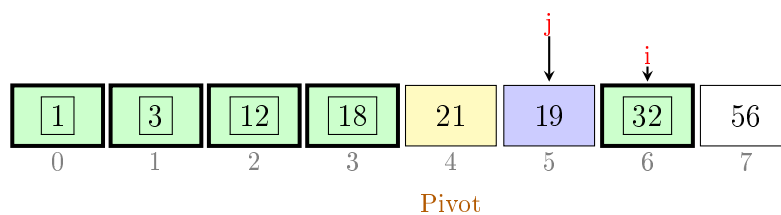
**Schritt 46:** Initialisiere Zeiger:  $i=5, j=5$



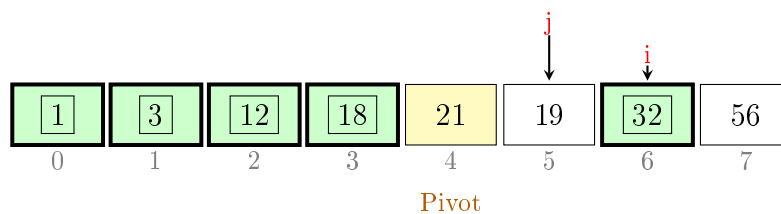
**Schritt 47:**  $\text{array}[5]=19$  Pivot=21  $\rightarrow i$  nach rechts



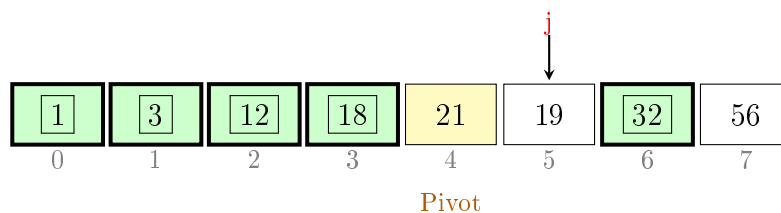
**Schritt 48:**  $\text{array}[5]=19$  Pivot=21  $\rightarrow j$  stoppt



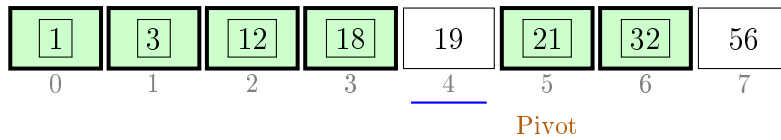
**Schritt 49:**  $i=6, j=5 \rightarrow$  Zeiger gekreuzt, Partition beenden



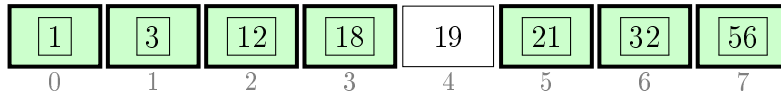
**Schritt 50:** Platziere Pivot: Tausche  $\text{array}[4]=21 \leftrightarrow \text{array}[5]=19$



**Schritt 51:** Partition [4..5] fertig: Pivot 21 an Position 5

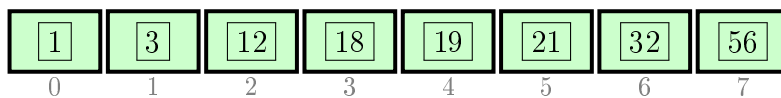


**Schritt 52:** quicksort(7, 7):  $l == r$ , einzelnes Element  $a[7]=56$  ist sortiert

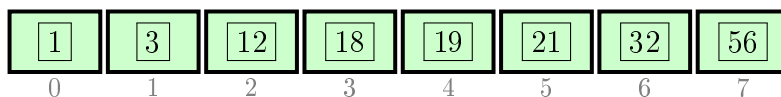


=== **Rekursionsebene 4: Basisfall: [4] (Basisfall)** ===

**Schritt 53:** quicksort(4, 4):  $l == r$ , einzelnes Element  $a[4]=19$  ist sortiert



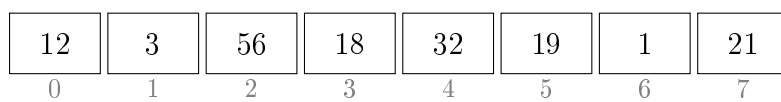
**Schritt 54:** Sortierung abgeschlossen



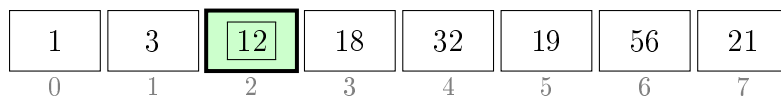
### Übersicht der Rekursionsschritte:

Jede Zeile zeigt den Zustand nach einem Rekursionsschritt. Unterstrichene Teillisten werden im nächsten Schritt parallel verarbeitet. Umrahmte Werte haben ihre endgültige Position erreicht.

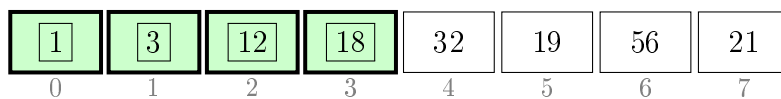
**Zustand 0:** Ausgangszustand



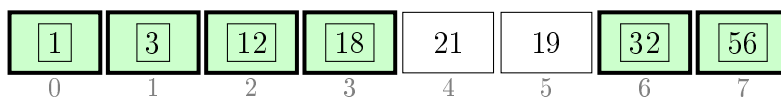
**Zustand 1:** Nach Rekursionsschritt 1



**Zustand 2:** Nach Rekursionsschritt 2



**Zustand 3:** Nach Rekursionsschritt 3



**Zustand 4:** Nach Rekursionsschritt 4

|   |   |    |    |    |    |    |    |
|---|---|----|----|----|----|----|----|
| 1 | 3 | 12 | 18 | 19 | 21 | 32 | 56 |
| 0 | 1 | 2  | 3  | 4  | 5  | 6  | 7  |

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**Antworten:**

1. Nach erster Partition: [1, 3, 12, 18, 32, 19, 56, 21]
2. Nach letzter Partition: [1, 3, 12, 18, 19, 21, 32, 56]
3. Anzahl Vergleiche: **23**

**Endergebnis:** [1, 3, 12, 18, 19, 21, 32, 56]

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## Aufgabe 2 – Schwierigkeit: ★★

### Standard-Quicksort

Gegeben sei folgende Liste, die in einem Array abgelegt ist:

**45, 23, 11, 89, 77, 38, 62, 5**

Die Liste soll mit dem Standard-Quicksort-Verfahren (keine Variante) in-situ sortiert werden. Als Pivot-Element soll immer das **erste Element** der zu sortierenden (Teil-)Liste genommen werden.

#### Fragen:

1. Geben Sie den Zustand des Arrays nach der **ersten** Partition an.
2. Geben Sie den Zustand des Arrays nach der **letzten** Partition an.
3. Wie viele Vergleiche werden insgesamt durchgeführt?

### Lösung:

#### Schritt-für-Schritt-Lösung:

**Schritt 0:** Ausgangszustand

|    |    |    |    |    |    |    |   |
|----|----|----|----|----|----|----|---|
| 45 | 23 | 11 | 89 | 77 | 38 | 62 | 5 |
| 0  | 1  | 2  | 3  | 4  | 5  | 6  | 7 |

=== Rekursionsebene 0: Partitioniere [0..7] ===

**Schritt 1:** Partition [0..7]: Pivot = 45 (erstes Element)

|    |    |    |    |    |    |    |   |
|----|----|----|----|----|----|----|---|
| 45 | 23 | 11 | 89 | 77 | 38 | 62 | 5 |
| 0  | 1  | 2  | 3  | 4  | 5  | 6  | 7 |

Pivot

**Schritt 2:** Initialisiere Zeiger: i=1, j=7

|    |    |    |    |    |    |    |   |
|----|----|----|----|----|----|----|---|
| 45 | 23 | 11 | 89 | 77 | 38 | 62 | 5 |
| 0  | 1  | 2  | 3  | 4  | 5  | 6  | 7 |

Pivot

i ↓ j ↓

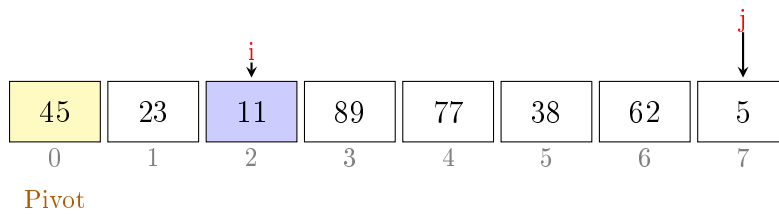
**Schritt 3:** array[1]=23 Pivot=45 → i nach rechts

|    |    |    |    |    |    |    |   |
|----|----|----|----|----|----|----|---|
| 45 | 23 | 11 | 89 | 77 | 38 | 62 | 5 |
| 0  | 1  | 2  | 3  | 4  | 5  | 6  | 7 |

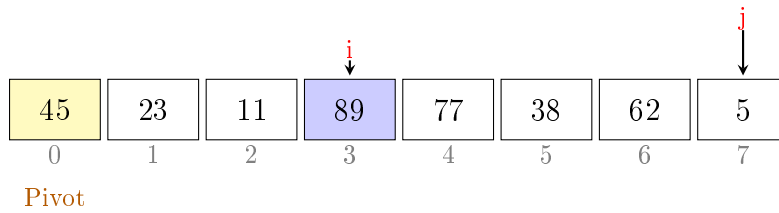
Pivot

i ↓ j ↓

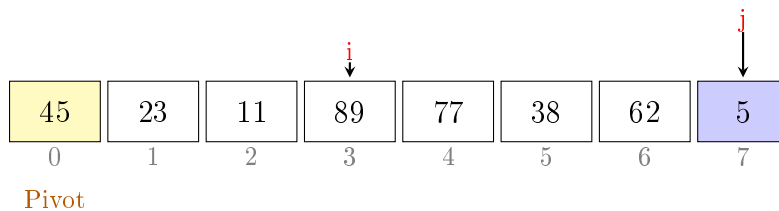
**Schritt 4:**  $\text{array}[2]=11$  Pivot=45  $\rightarrow$  i nach rechts



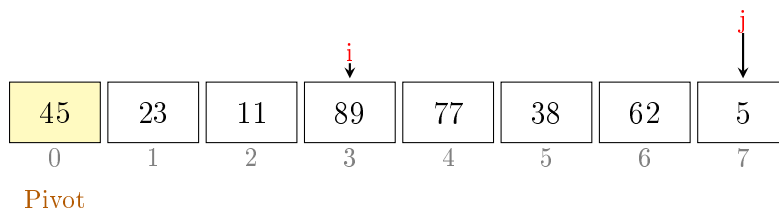
**Schritt 5:**  $\text{array}[3]=89 > \text{Pivot}=45 \rightarrow$  i stoppt



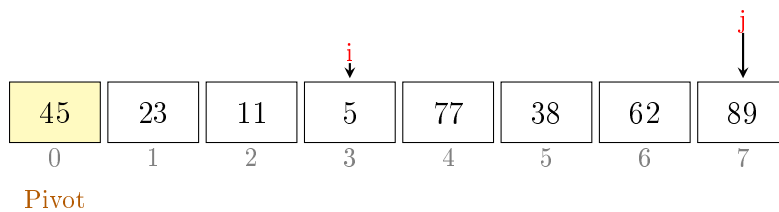
**Schritt 6:**  $\text{array}[7]=5$  Pivot=45  $\rightarrow$  j stoppt



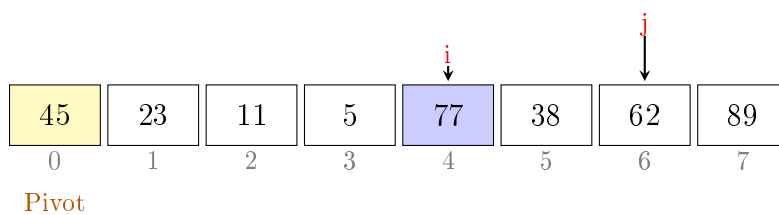
**Schritt 7:** Tausche  $\text{array}[3]=89 \leftrightarrow \text{array}[7]=5$



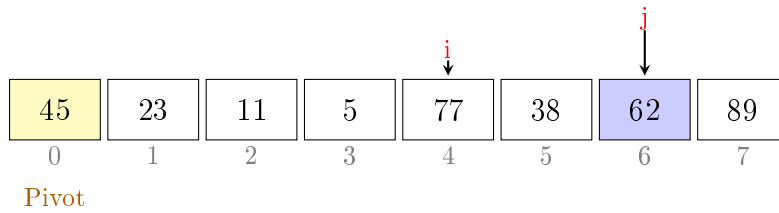
**Schritt 8:** Nach Tausch:  $\text{array}[3]=5$ ,  $\text{array}[7]=89$



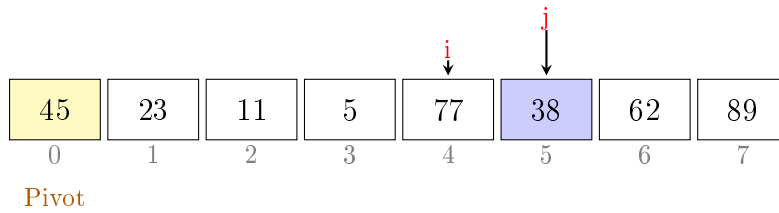
**Schritt 9:**  $\text{array}[4]=77 > \text{Pivot}=45 \rightarrow$  i stoppt



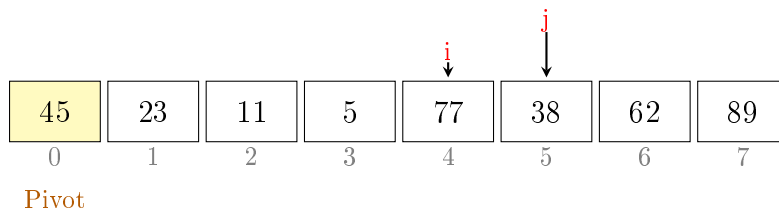
**Schritt 10:**  $\text{array}[6]=62 > \text{Pivot}=45 \rightarrow j$  nach links



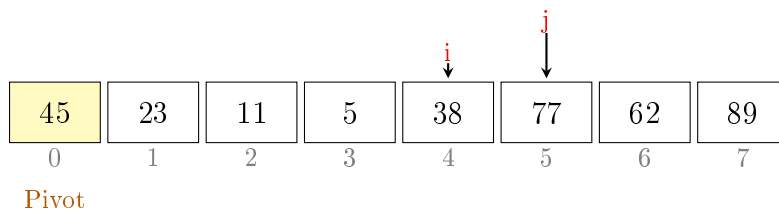
**Schritt 11:**  $\text{array}[5]=38$  Pivot=45  $\rightarrow j$  stoppt



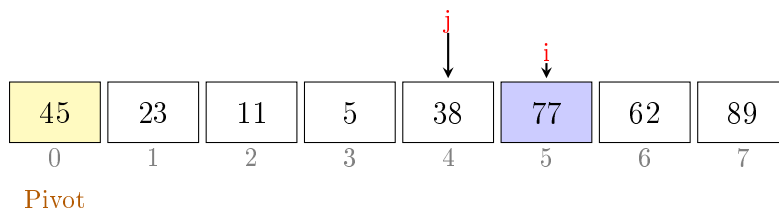
**Schritt 12:** Tausche  $\text{array}[4]=77 \leftrightarrow \text{array}[5]=38$



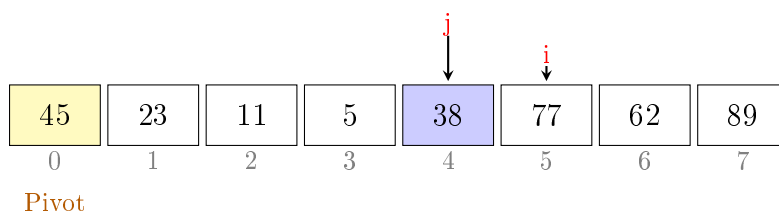
**Schritt 13:** Nach Tausch:  $\text{array}[4]=38$ ,  $\text{array}[5]=77$



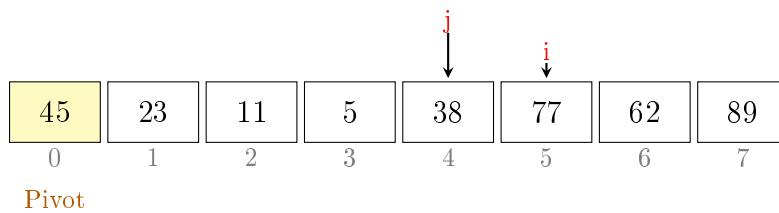
**Schritt 14:**  $\text{array}[5]=77 > \text{Pivot}=45 \rightarrow i$  stoppt



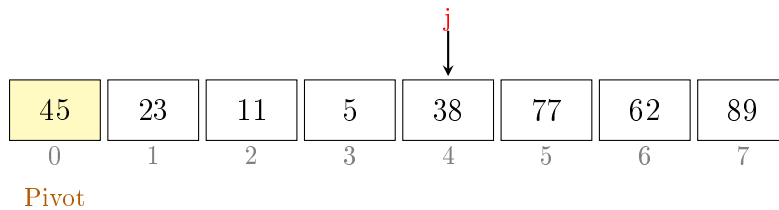
**Schritt 15:**  $\text{array}[4]=38$  Pivot=45  $\rightarrow j$  stoppt



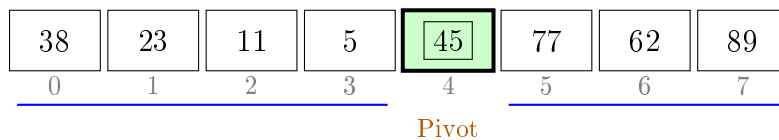
**Schritt 16:**  $i=5$   $j=4 \rightarrow$  Zeiger gekreuzt, Partition beenden



**Schritt 17:** Platziere Pivot: Tausche  $\text{array}[0]=45 \leftrightarrow \text{array}[4]=38$

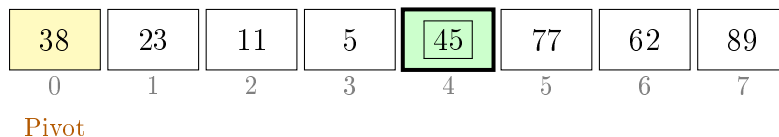


**Schritt 18:** Partition  $[0..7]$  fertig: Pivot 45 an Position 4

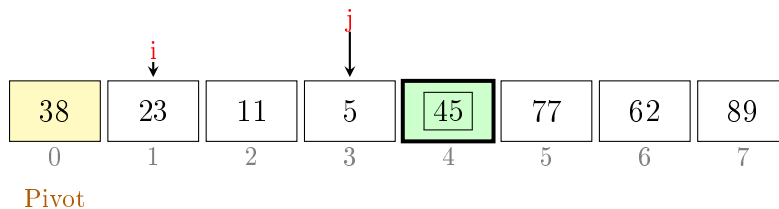


=== Rekursionsebene 1: Partitioniere  $[0..3]$ ,  $[5..7]$  ===

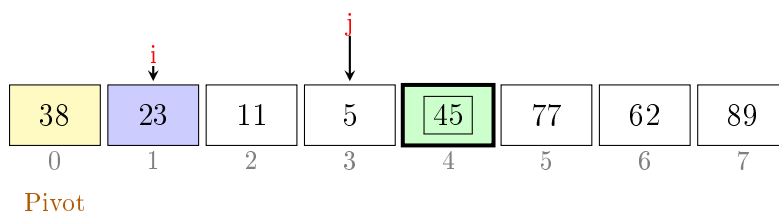
**Schritt 19:** Partition  $[0..3]$ : Pivot = 38 (erstes Element)



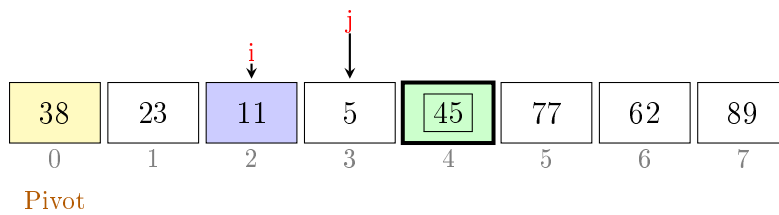
**Schritt 20:** Initialisiere Zeiger:  $i=1$ ,  $j=3$



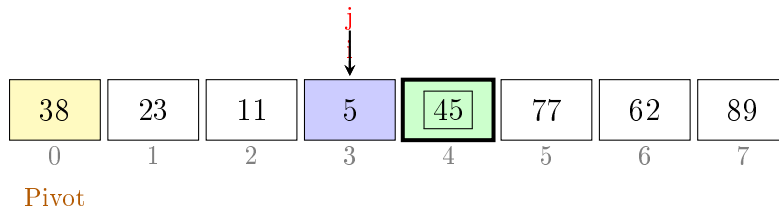
**Schritt 21:**  $\text{array}[1]=23$  Pivot=38  $\rightarrow$  i nach rechts



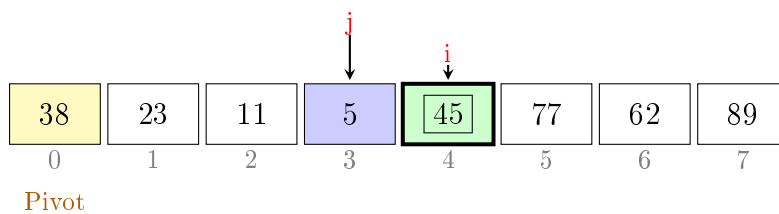
**Schritt 22:**  $\text{array}[2]=11$  Pivot=38  $\rightarrow$  i nach rechts



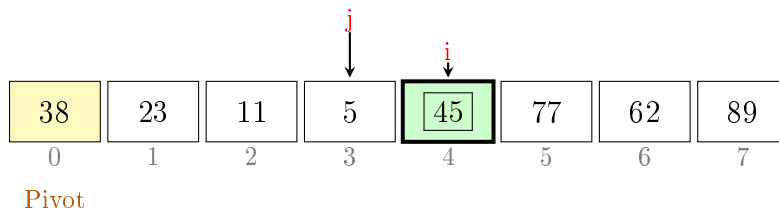
**Schritt 23:**  $\text{array}[3]=5$  Pivot=38  $\rightarrow$  i nach rechts



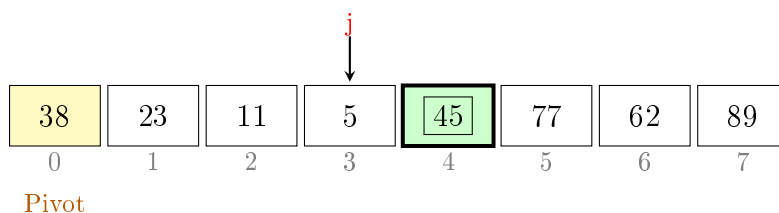
**Schritt 24:**  $\text{array}[3]=5$  Pivot=38  $\rightarrow$  j stoppt



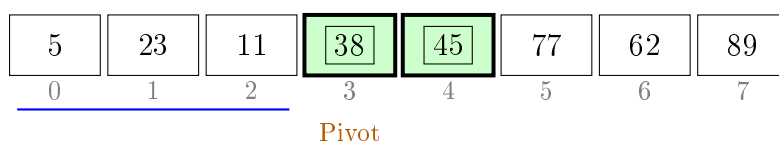
**Schritt 25:**  $i=4$   $j=3$   $\rightarrow$  Zeiger gekreuzt, Partition beenden



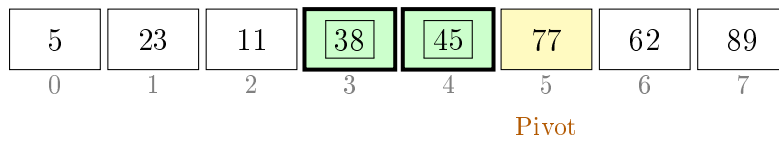
**Schritt 26:** Platziere Pivot: Tausche  $\text{array}[0]=38 \leftrightarrow \text{array}[3]=5$



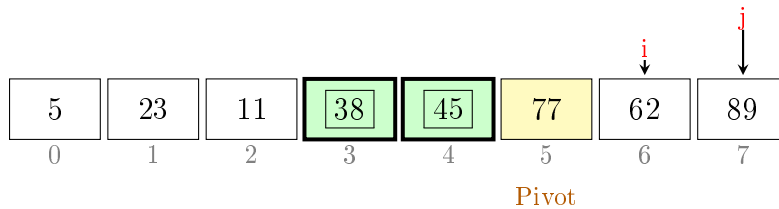
**Schritt 27:** Partition [0..3] fertig: Pivot 38 an Position 3



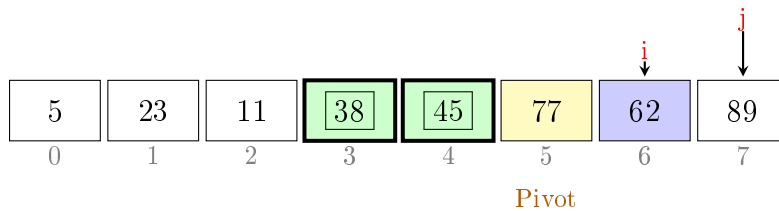
**Schritt 28:** Partition [5..7]: Pivot = 77 (erstes Element)



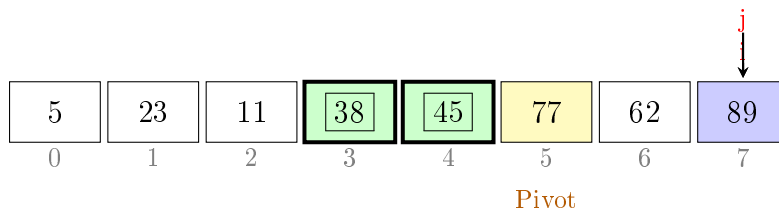
**Schritt 29:** Initialisiere Zeiger: i=6, j=7



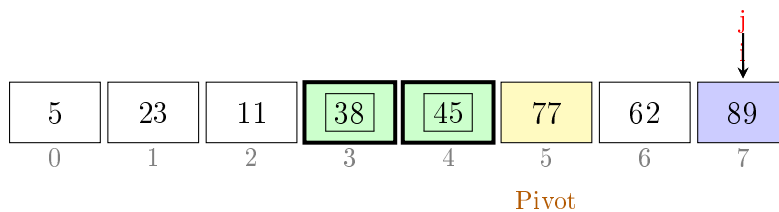
**Schritt 30:** array[6]=62 Pivot=77 → i nach rechts



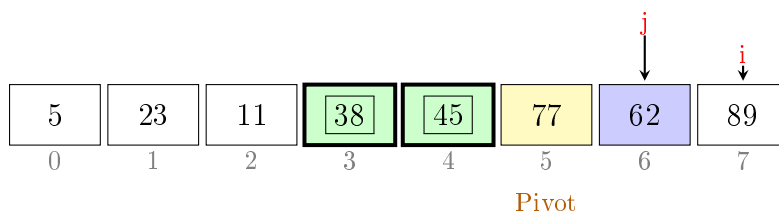
**Schritt 31:** array[7]=89 > Pivot=77 → i stoppt



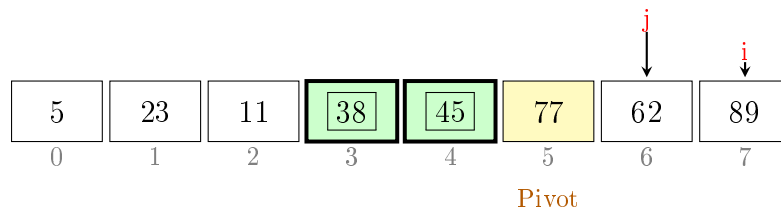
**Schritt 32:** array[7]=89 > Pivot=77 → j nach links



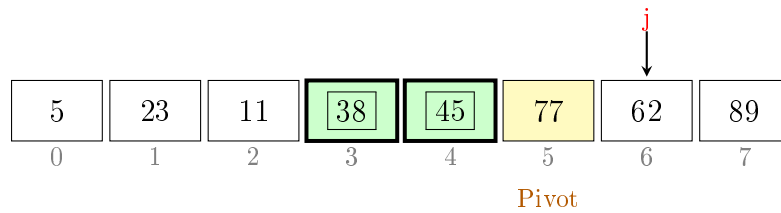
**Schritt 33:** array[6]=62 Pivot=77 → j stoppt



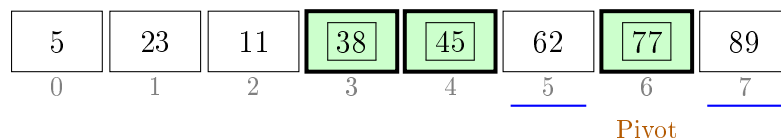
**Schritt 34:**  $i=7$   $j=6 \rightarrow$  Zeiger gekreuzt, Partition beenden



**Schritt 35:** Platziere Pivot: Tausche  $\text{array}[5]=77 \leftrightarrow \text{array}[6]=62$

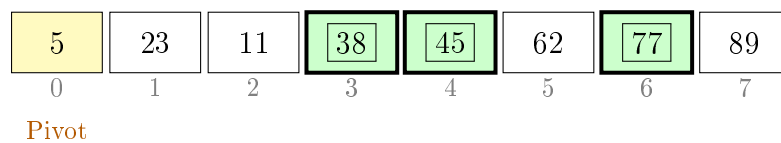


**Schritt 36:** Partition [5..7] fertig: Pivot 77 an Position 6

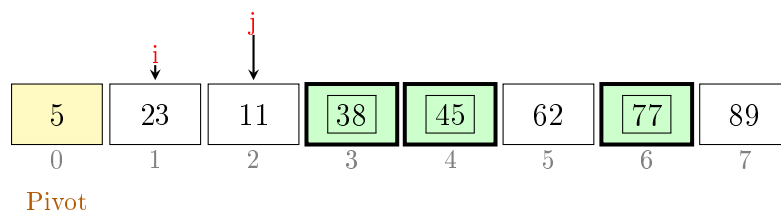


=== Rekursionsebene 2: Partitioniere [0..2]; Basisfall: [5] (Basisfall), [7] (Basisfall) ===

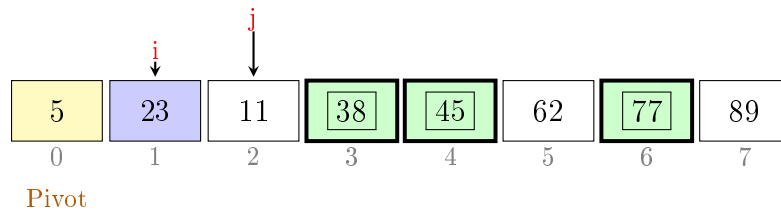
**Schritt 37:** Partition [0..2]: Pivot = 5 (erstes Element)



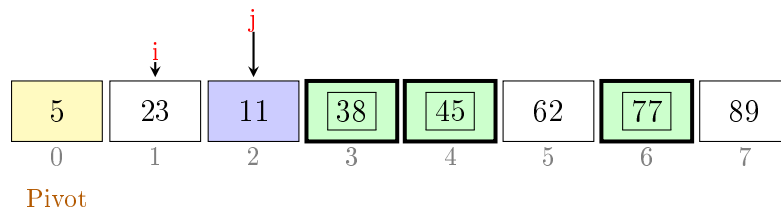
**Schritt 38:** Initialisiere Zeiger:  $i=1$ ,  $j=2$



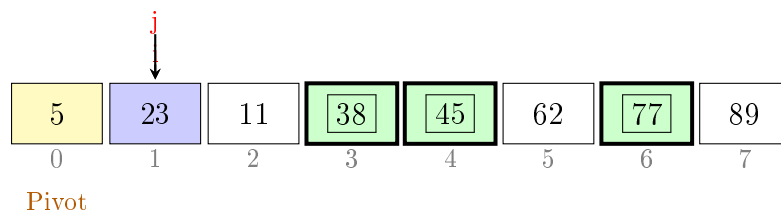
**Schritt 39:**  $\text{array}[1]=23 > \text{Pivot}=5 \rightarrow i$  stoppt



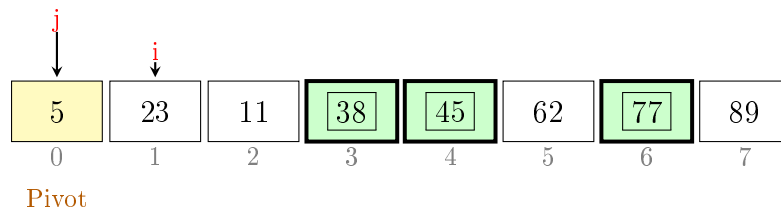
**Schritt 40:**  $\text{array}[2]=11 > \text{Pivot}=5 \rightarrow j$  nach links



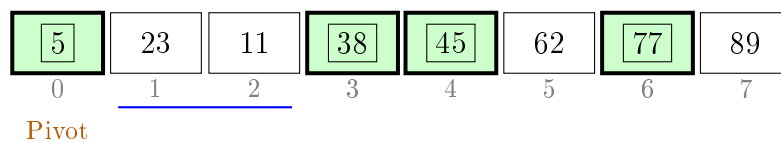
**Schritt 41:**  $\text{array}[1]=23 > \text{Pivot}=5 \rightarrow j$  nach links



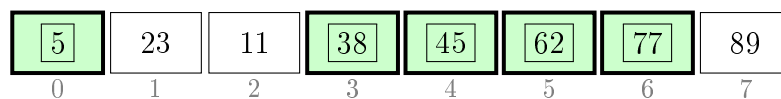
**Schritt 42:**  $i=1 \ j=0 \rightarrow$  Zeiger gekreuzt, Partition beenden



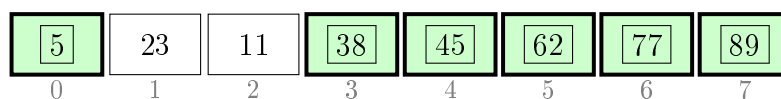
**Schritt 43:** Partition  $[0..2]$  fertig: Pivot 5 an Position 0



**Schritt 44:**  $\text{quicksort}(5, 5)$ :  $l == r$ , einzelnes Element  $a[5]=62$  ist sortiert

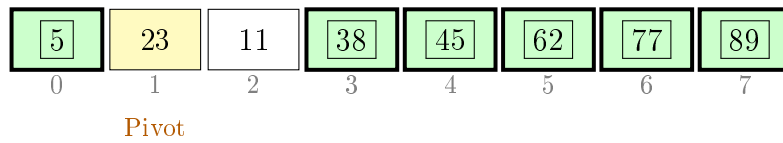


**Schritt 45:**  $\text{quicksort}(7, 7)$ :  $l == r$ , einzelnes Element  $a[7]=89$  ist sortiert

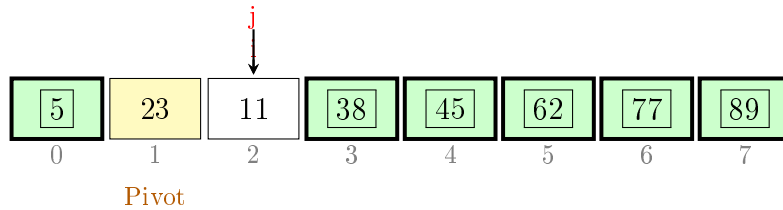


=== Rekursionsebene 3: Partitioniere [1..2] ===

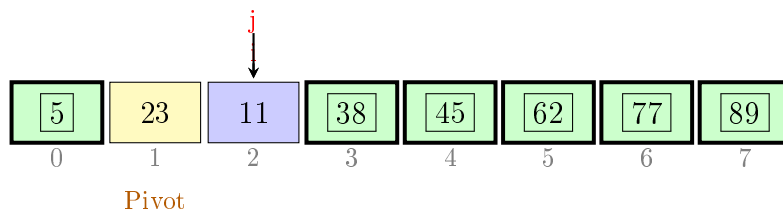
**Schritt 46:** Partition [1..2]: Pivot = 23 (erstes Element)



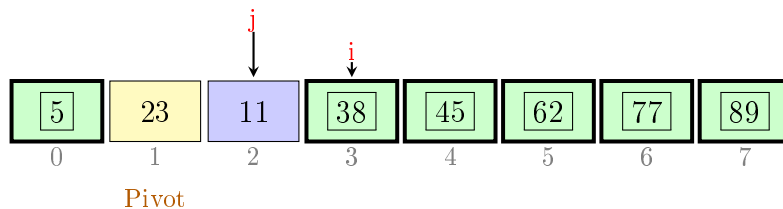
**Schritt 47:** Initialisiere Zeiger:  $i=2$ ,  $j=2$



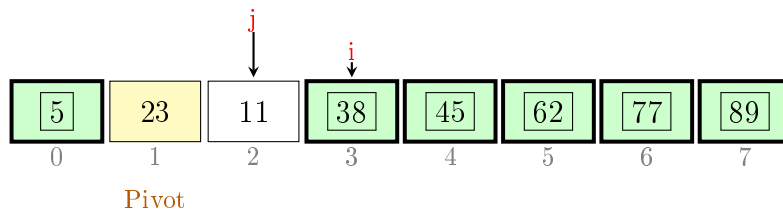
**Schritt 48:**  $\text{array}[2]=11$  Pivot=23  $\rightarrow i$  nach rechts



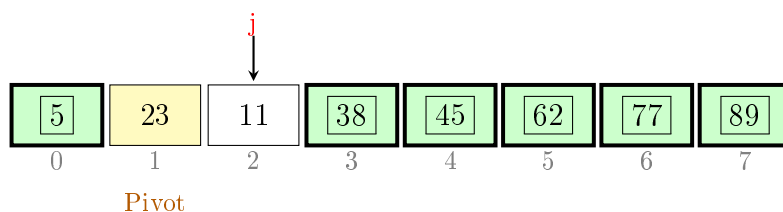
**Schritt 49:**  $\text{array}[2]=11$  Pivot=23  $\rightarrow j$  stoppt



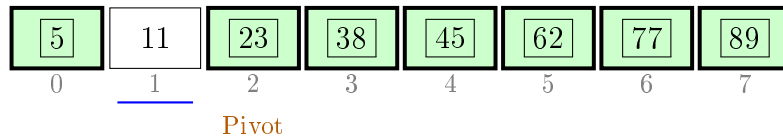
**Schritt 50:**  $i=3$   $j=2 \rightarrow$  Zeiger gekreuzt, Partition beenden



**Schritt 51:** Platziere Pivot: Tausche  $\text{array}[1]=23 \leftrightarrow \text{array}[2]=11$

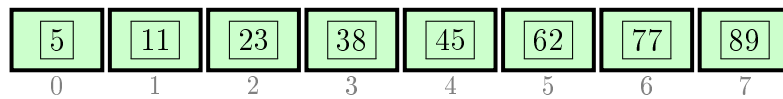


**Schritt 52:** Partition [1..2] fertig: Pivot 23 an Position 2

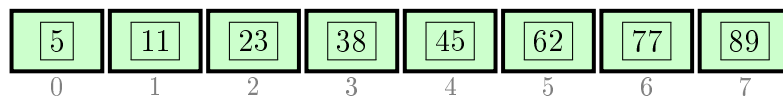


=== Rekursionsebene 4: Basisfall: [1] (Basisfall) ===

**Schritt 53:** quicksort(1, 1):  $l == r$ , einzelnes Element  $a[1]=11$  ist sortiert



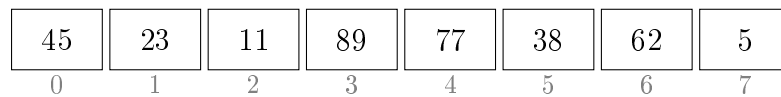
**Schritt 54:** Sortierung abgeschlossen



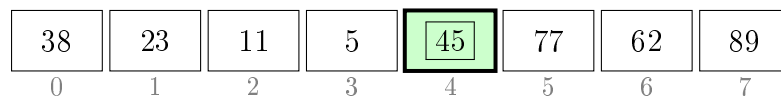
### Übersicht der Rekursionsschritte:

Jede Zeile zeigt den Zustand nach einem Rekursionsschritt. Unterstrichene Teillisten werden im nächsten Schritt parallel verarbeitet. Umrahmte Werte haben ihre endgültige Position erreicht.

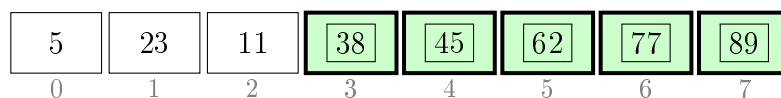
**Zustand 0:** Ausgangszustand



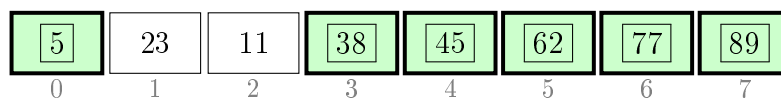
**Zustand 1:** Nach Rekursionsschritt 1



**Zustand 2:** Nach Rekursionsschritt 2



**Zustand 3:** Nach Rekursionsschritt 3



**Zustand 4:** Nach Rekursionsschritt 4

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|   |    |    |    |    |    |    |    |
|---|----|----|----|----|----|----|----|
| 5 | 11 | 23 | 38 | 45 | 62 | 77 | 89 |
| 0 | 1  | 2  | 3  | 4  | 5  | 6  | 7  |

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**Antworten:**

1. Nach erster Partition: **[38, 23, 11, 5, 45, 77, 62, 89]**
2. Nach letzter Partition: **[5, 11, 23, 38, 45, 62, 77, 89]**
3. Anzahl Vergleiche: **22**

**Endergebnis:** [5, 11, 23, 38, 45, 62, 77, 89]

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## Aufgabe 3 – Schwierigkeit: ★★★

Standard-Quicksort mit 10 Elementen

Gegeben sei folgende Liste, die in einem Array abgelegt ist:

**27, 63, 14, 55, 8, 91, 42, 36, 70, 19**

Die Liste soll mit dem Standard-Quicksort-Verfahren (keine Variante) in-situ sortiert werden. Als Pivot-Element soll immer das **erste Element** der zu sortierenden (Teil-)Liste genommen werden.

**Fragen:**

1. Geben Sie den Zustand des Arrays nach der **ersten** Partition an.
2. Geben Sie den Zustand des Arrays nach der **letzten** Partition an.
3. Wie viele Vergleiche werden insgesamt durchgeführt?

### Lösung:

**Schritt-für-Schritt-Lösung:**

**Schritt 0:** Ausgangszustand

|    |    |    |    |   |    |    |    |    |    |
|----|----|----|----|---|----|----|----|----|----|
| 27 | 63 | 14 | 55 | 8 | 91 | 42 | 36 | 70 | 19 |
| 0  | 1  | 2  | 3  | 4 | 5  | 6  | 7  | 8  | 9  |

=== Rekursionsebene 0: Partitioniere [0..9] ===

**Schritt 1:** Partition [0..9]: Pivot = 27 (erstes Element)

|    |    |    |    |   |    |    |    |    |    |
|----|----|----|----|---|----|----|----|----|----|
| 27 | 63 | 14 | 55 | 8 | 91 | 42 | 36 | 70 | 19 |
| 0  | 1  | 2  | 3  | 4 | 5  | 6  | 7  | 8  | 9  |

Pivot

**Schritt 2:** Initialisiere Zeiger: i=1, j=9

|    |    |    |    |   |    |    |    |    |    |
|----|----|----|----|---|----|----|----|----|----|
| 27 | 63 | 14 | 55 | 8 | 91 | 42 | 36 | 70 | 19 |
| 0  | 1  | 2  | 3  | 4 | 5  | 6  | 7  | 8  | 9  |

Pivot

i ↓ j ↓

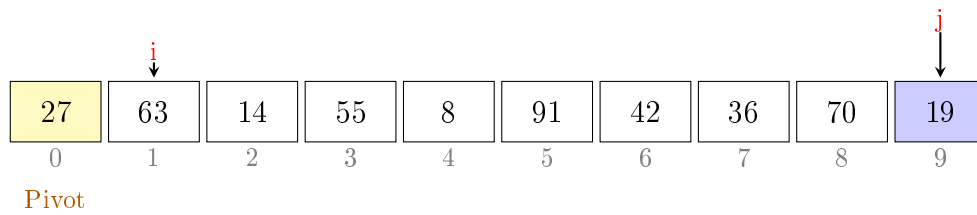
**Schritt 3:** array[1]=63 > Pivot=27 → i stoppt

|    |    |    |    |   |    |    |    |    |    |
|----|----|----|----|---|----|----|----|----|----|
| 27 | 63 | 14 | 55 | 8 | 91 | 42 | 36 | 70 | 19 |
| 0  | 1  | 2  | 3  | 4 | 5  | 6  | 7  | 8  | 9  |

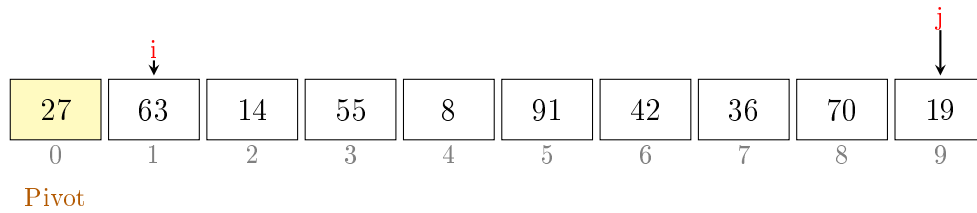
Pivot

i ↓ j ↓

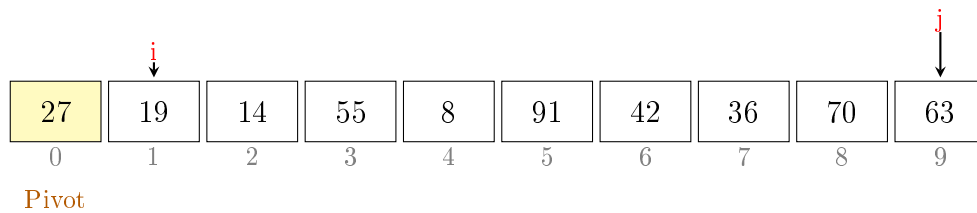
**Schritt 4:** array[9]=19 Pivot=27 → j stoppt



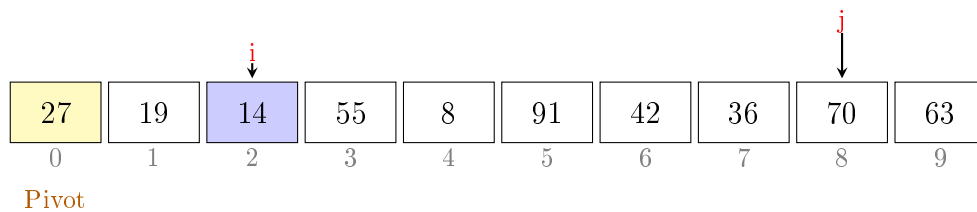
**Schritt 5:** Tausche array[1]=63 ↔ array[9]=19



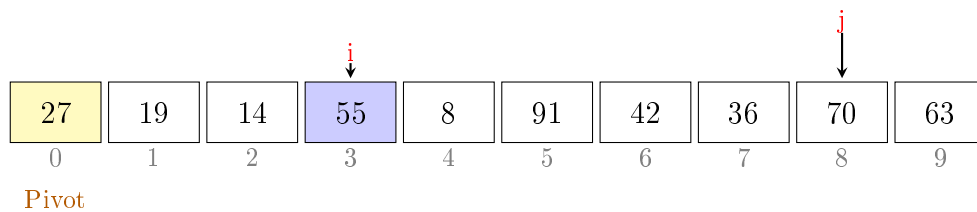
**Schritt 6:** Nach Tausch: array[1]=19, array[9]=63



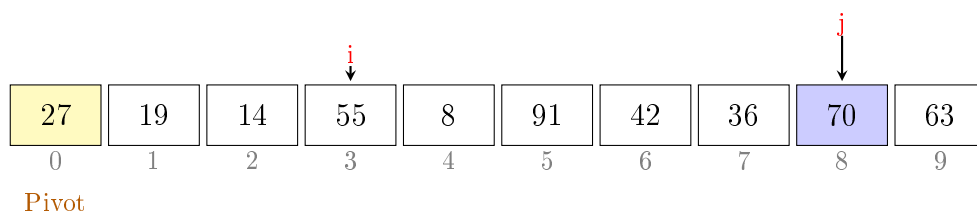
**Schritt 7:** array[2]=14 Pivot=27 → i nach rechts



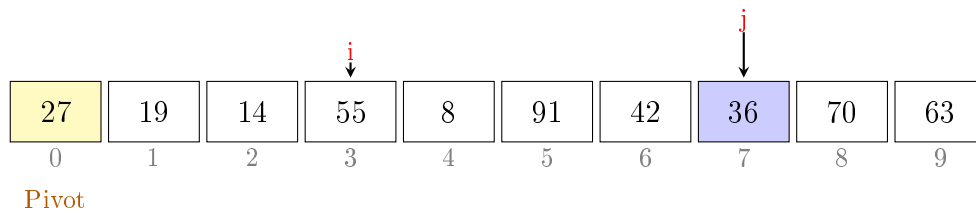
**Schritt 8:** array[3]=55 > Pivot=27 → i stoppt



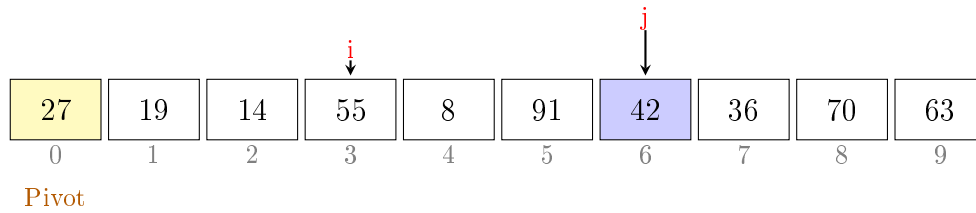
**Schritt 9:** array[8]=70 > Pivot=27 → j nach links



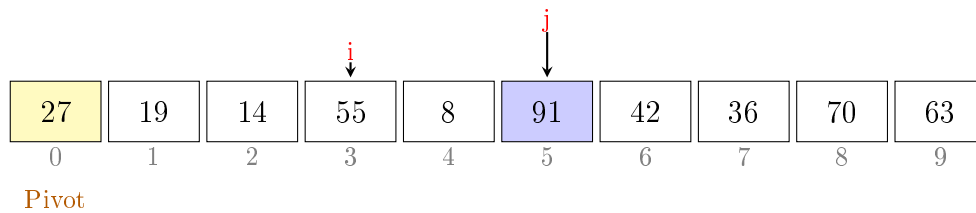
**Schritt 10:**  $\text{array}[7]=36 > \text{Pivot}=27 \rightarrow j$  nach links



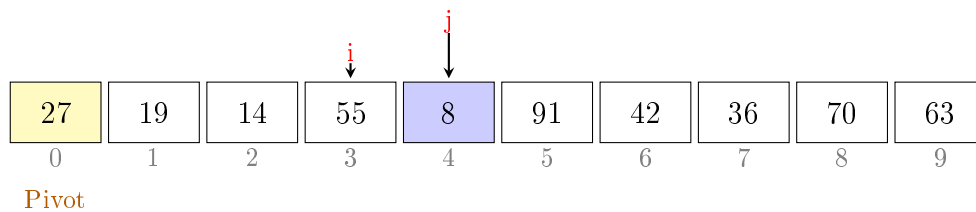
**Schritt 11:**  $\text{array}[6]=42 > \text{Pivot}=27 \rightarrow j$  nach links



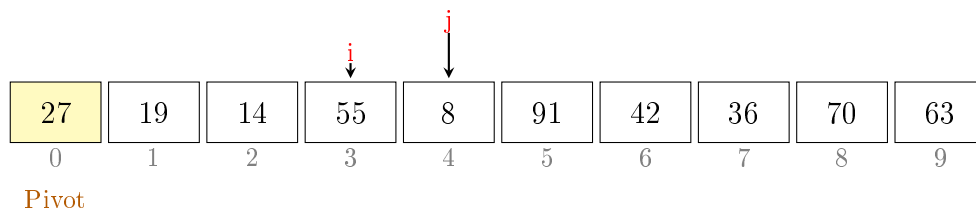
**Schritt 12:**  $\text{array}[5]=91 > \text{Pivot}=27 \rightarrow j$  nach links



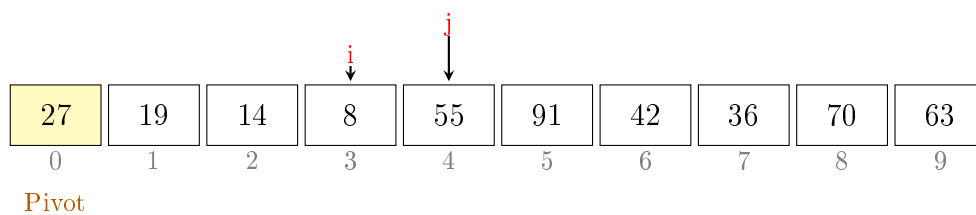
**Schritt 13:**  $\text{array}[4]=8$  Pivot=27  $\rightarrow j$  stoppt



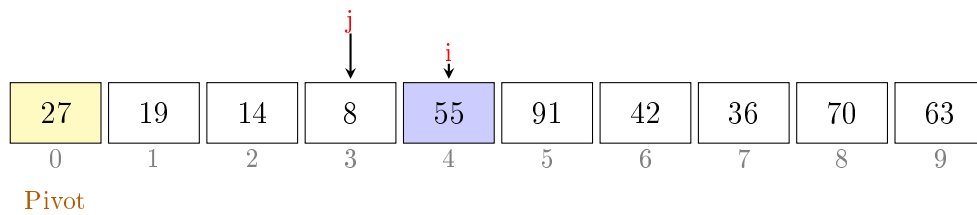
**Schritt 14:** Tausche  $\text{array}[3]=55 \leftrightarrow \text{array}[4]=8$



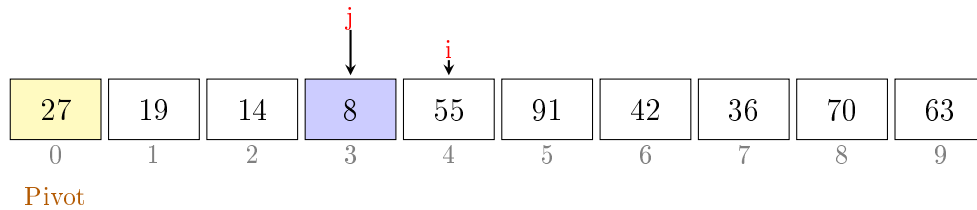
**Schritt 15:** Nach Tausch:  $\text{array}[3]=8$ ,  $\text{array}[4]=55$



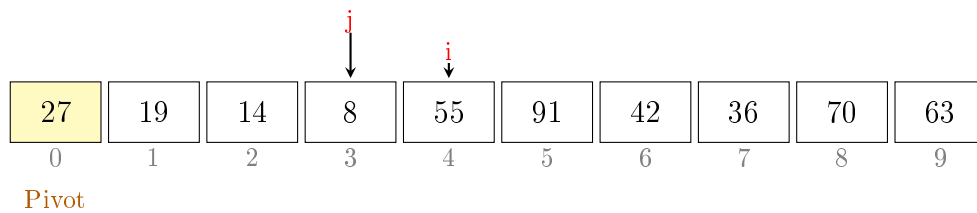
**Schritt 16:**  $\text{array}[4]=55 > \text{Pivot}=27 \rightarrow i$  stoppt



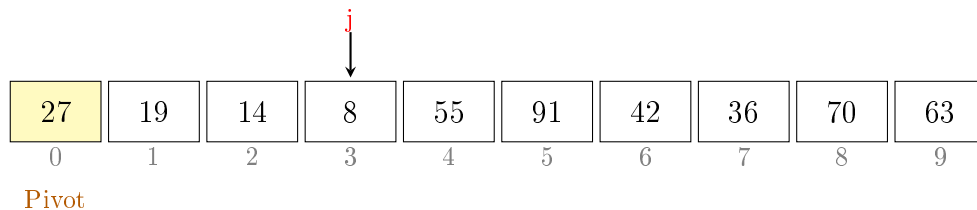
**Schritt 17:**  $\text{array}[3]=8$  Pivot=27  $\rightarrow j$  stoppt



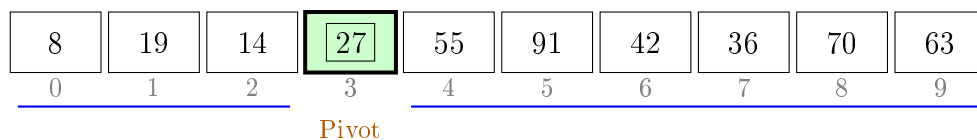
**Schritt 18:**  $i=4$   $j=3 \rightarrow$  Zeiger gekreuzt, Partition beenden



**Schritt 19:** Platziere Pivot: Tausche  $\text{array}[0]=27 \leftrightarrow \text{array}[3]=8$

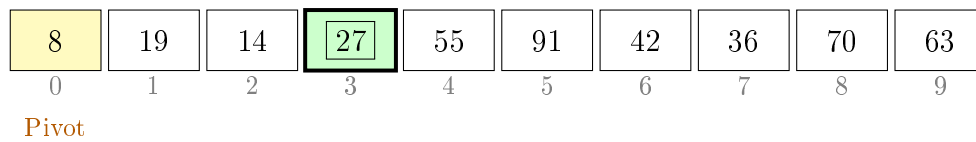


**Schritt 20:** Partition  $[0..9]$  fertig: Pivot 27 an Position 3

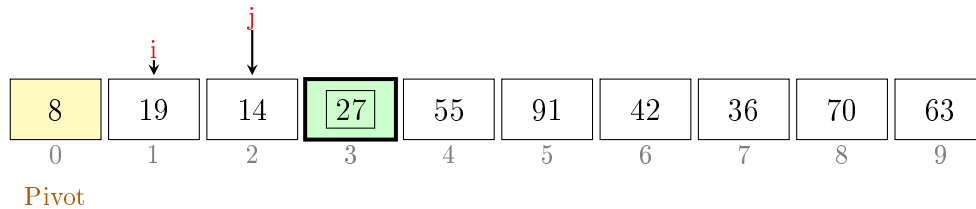


=== Rekursionsebene 1: Partitioniere  $[0..2]$ ,  $[4..9]$  ===

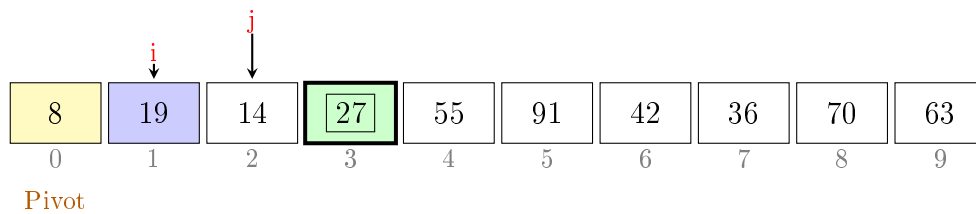
**Schritt 21:** Partition  $[0..2]$ : Pivot = 8 (erstes Element)



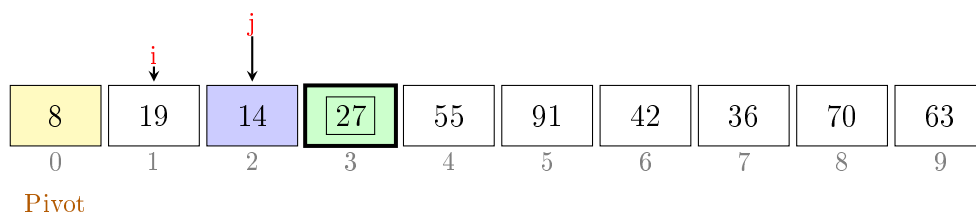
**Schritt 22:** Initialisiere Zeiger:  $i=1$ ,  $j=2$



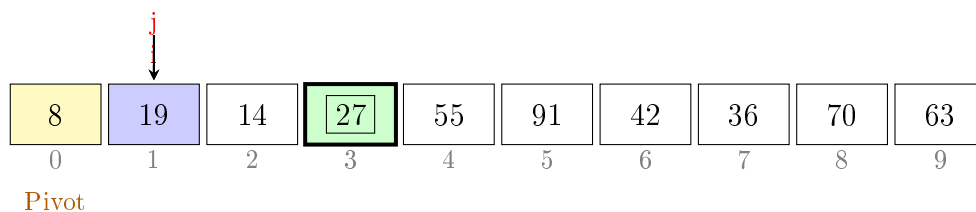
**Schritt 23:**  $\text{array}[1]=19 > \text{Pivot}=8 \rightarrow i$  stoppt



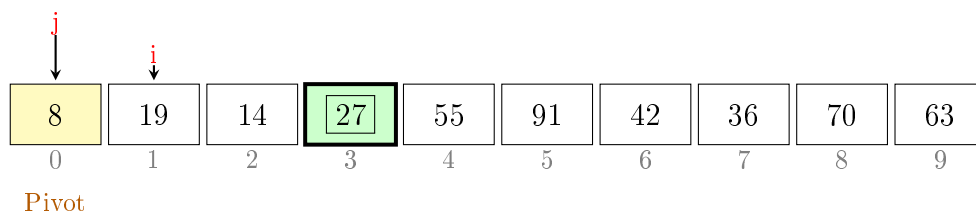
**Schritt 24:**  $\text{array}[2]=14 > \text{Pivot}=8 \rightarrow j$  nach links



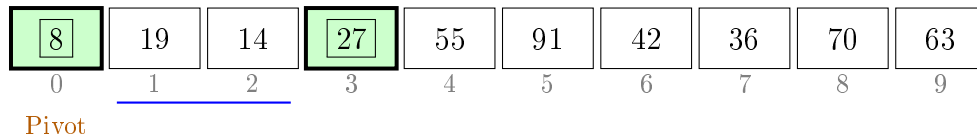
**Schritt 25:**  $\text{array}[1]=19 > \text{Pivot}=8 \rightarrow j$  nach links



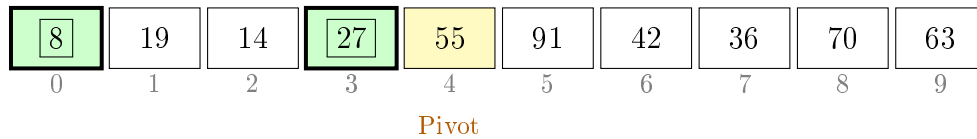
**Schritt 26:**  $i=1$   $j=0 \rightarrow$  Zeiger gekreuzt, Partition beenden



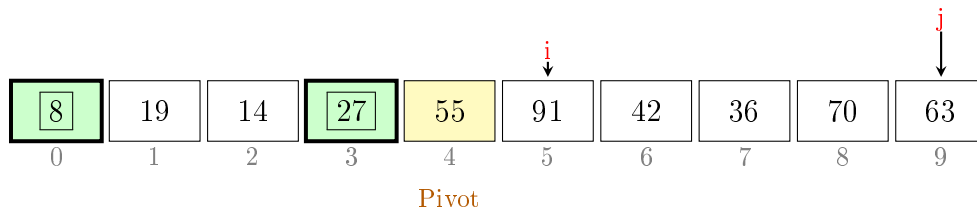
**Schritt 27:** Partition  $[0..2]$  fertig: Pivot 8 an Position 0



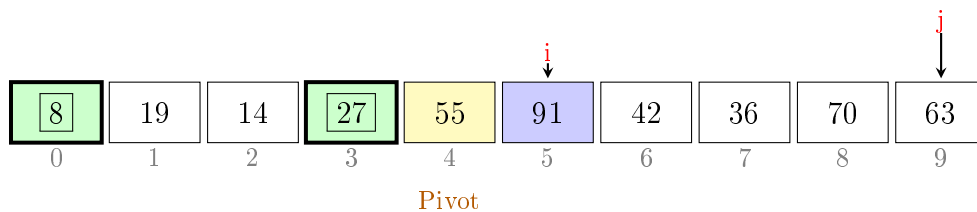
**Schritt 28:** Partition [4..9]: Pivot = 55 (erstes Element)



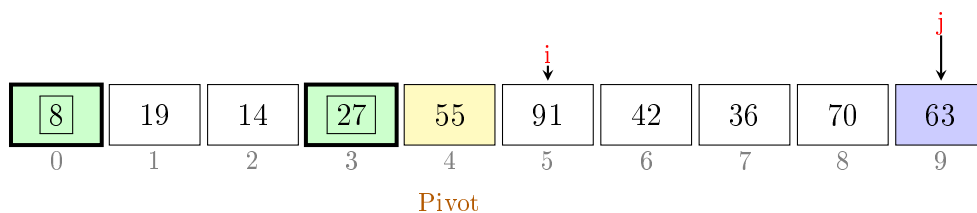
**Schritt 29:** Initialisiere Zeiger: i=5, j=9



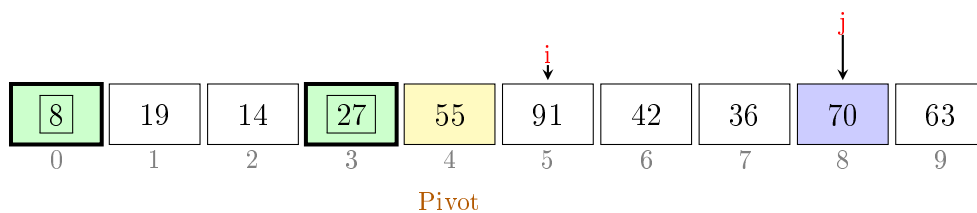
**Schritt 30:** array[5]=91 > Pivot=55 → i stoppt



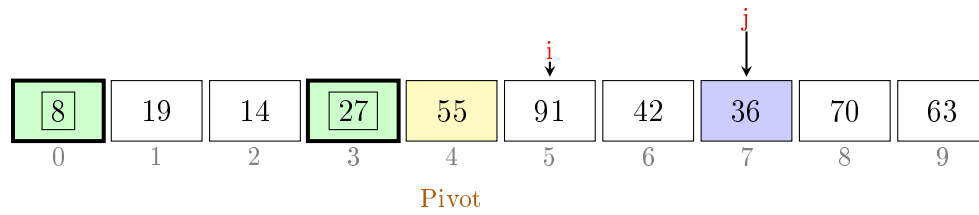
**Schritt 31:** array[9]=63 > Pivot=55 → j nach links



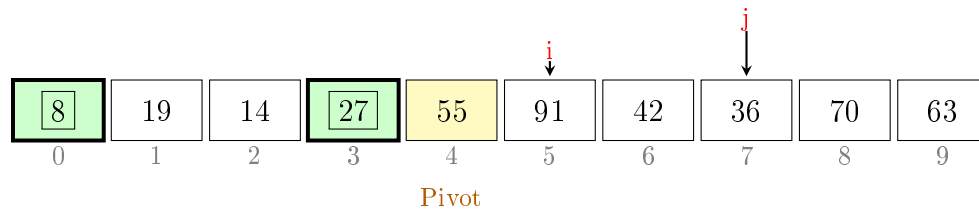
**Schritt 32:** array[8]=70 > Pivot=55 → j nach links



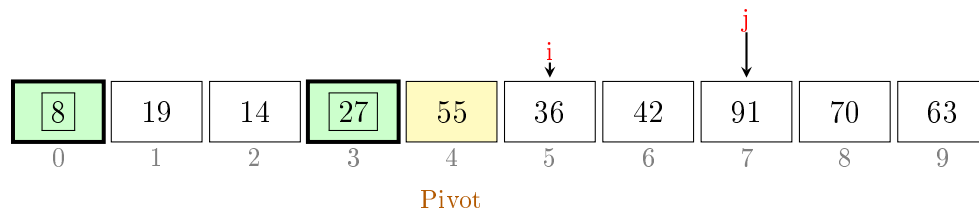
**Schritt 33:** array[7]=36 Pivot=55 → j stoppt



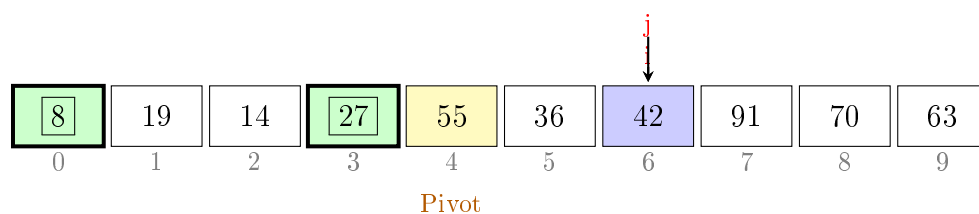
**Schritt 34:** Tausche  $\text{array}[5]=91 \leftrightarrow \text{array}[7]=36$



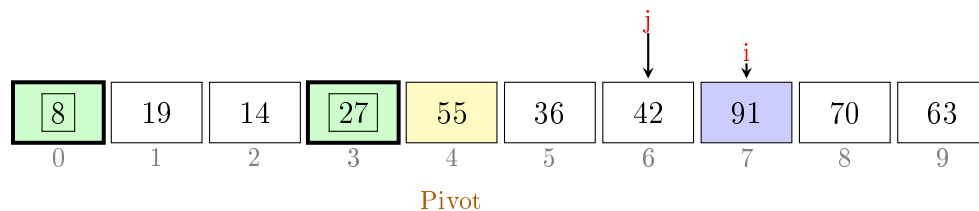
**Schritt 35:** Nach Tausch:  $\text{array}[5]=36$ ,  $\text{array}[7]=91$



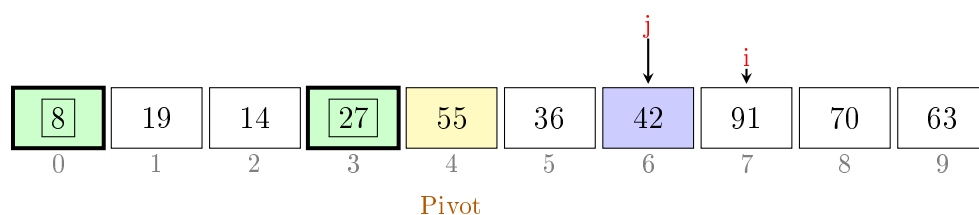
**Schritt 36:**  $\text{array}[6]=42$  Pivot=55  $\rightarrow$  i nach rechts



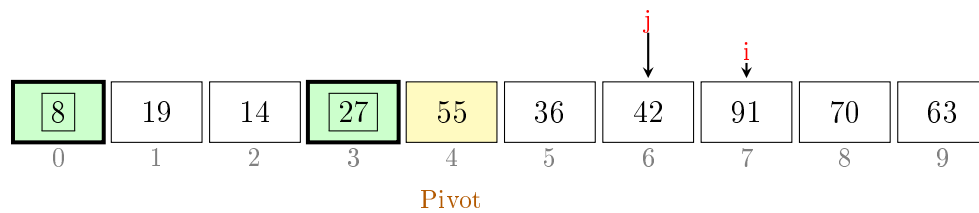
**Schritt 37:**  $\text{array}[7]=91 > \text{Pivot}=55 \rightarrow$  i stoppt



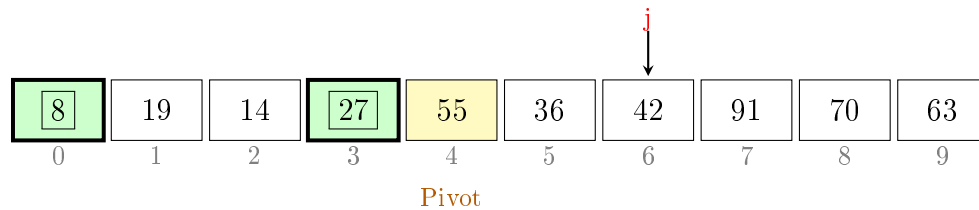
**Schritt 38:**  $\text{array}[6]=42$  Pivot=55  $\rightarrow$  j stoppt



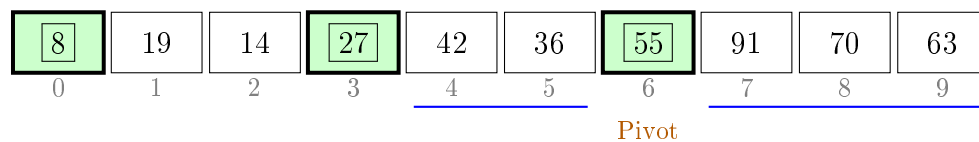
**Schritt 39:**  $i=7$   $j=6 \rightarrow$  Zeiger gekreuzt, Partition beenden



**Schritt 40:** Platziere Pivot: Tausche  $\text{array}[4]=55 \leftrightarrow \text{array}[6]=42$

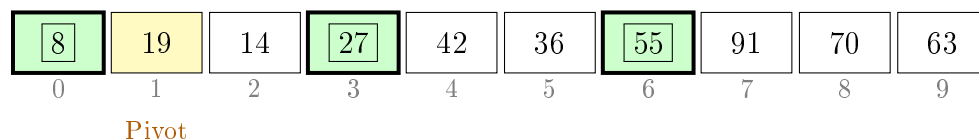


**Schritt 41:** Partition [4..9] fertig: Pivot 55 an Position 6

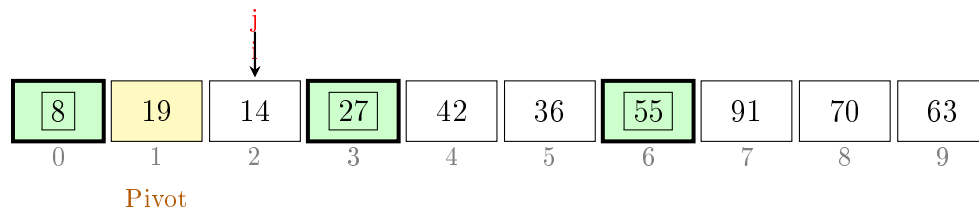


=== Rekursionsebene 2: Partitioniere [1..2], [4..5], [7..9] ===

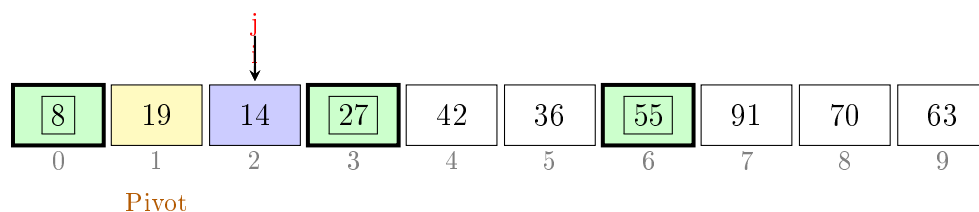
**Schritt 42:** Partition [1..2]: Pivot = 19 (erstes Element)



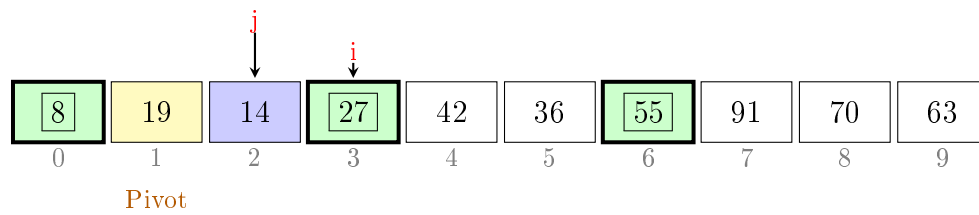
**Schritt 43:** Initialisiere Zeiger:  $i=2$ ,  $j=2$



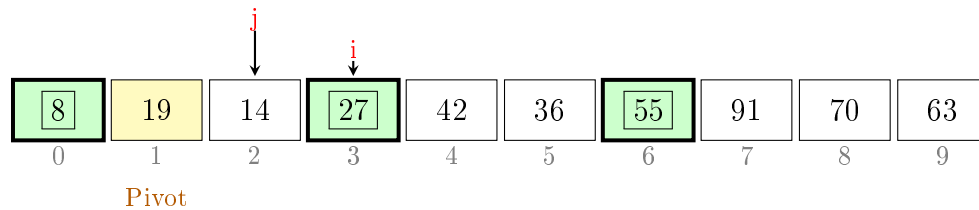
**Schritt 44:**  $\text{array}[2]=14$  Pivot=19  $\rightarrow$  i nach rechts



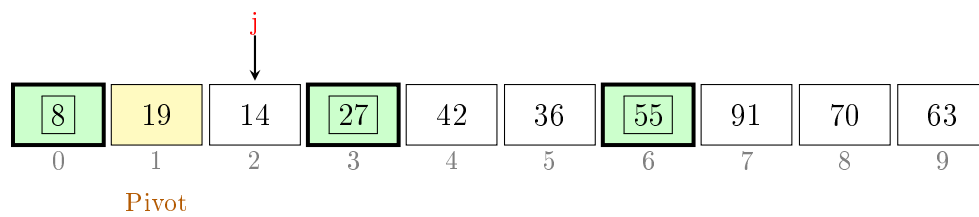
**Schritt 45:**  $\text{array}[2]=14$  Pivot=19  $\rightarrow j$  stoppt



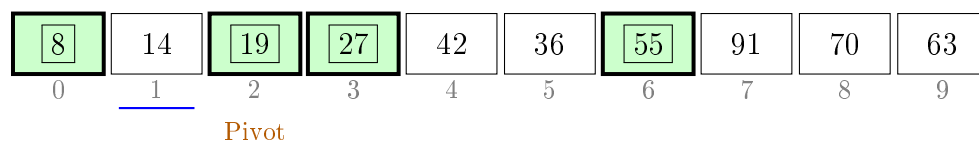
**Schritt 46:**  $i=3$   $j=2$   $\rightarrow$  Zeiger gekreuzt, Partition beenden



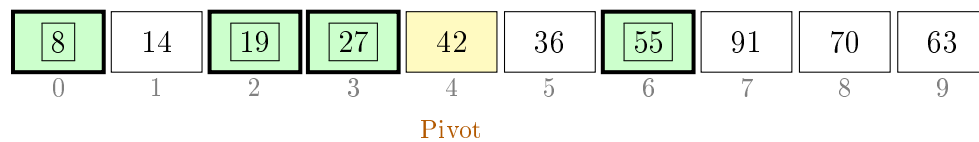
**Schritt 47:** Platziere Pivot: Tausche  $\text{array}[1]=19 \leftrightarrow \text{array}[2]=14$



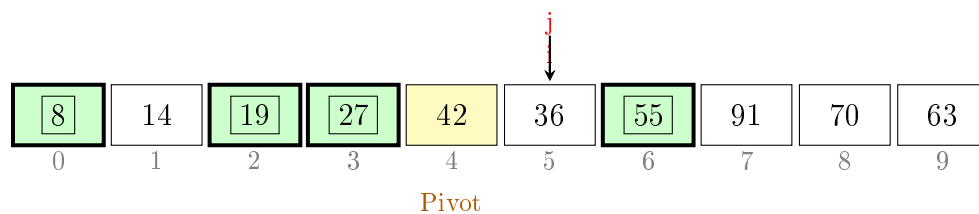
**Schritt 48:** Partition [1..2] fertig: Pivot 19 an Position 2



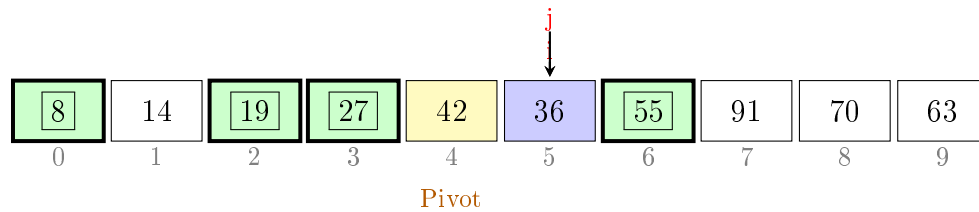
**Schritt 49:** Partition [4..5]: Pivot = 42 (erstes Element)



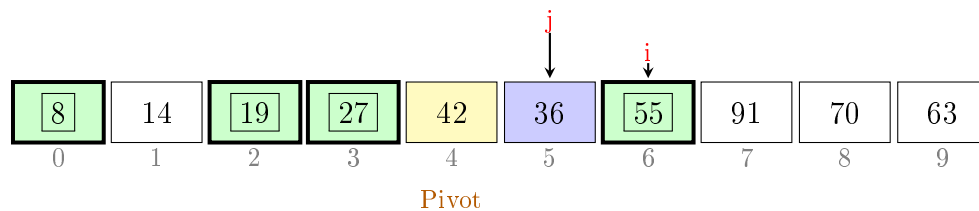
**Schritt 50:** Initialisiere Zeiger:  $i=5$ ,  $j=5$



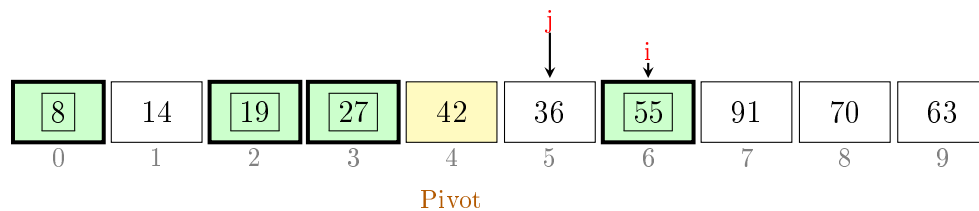
**Schritt 51:**  $\text{array}[5]=36$  Pivot=42  $\rightarrow i$  nach rechts



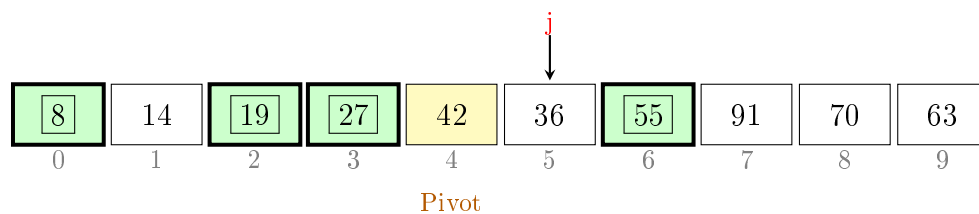
**Schritt 52:**  $\text{array}[5]=36$  Pivot=42  $\rightarrow$  j stoppt



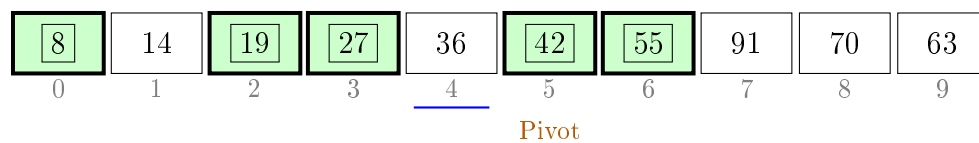
**Schritt 53:**  $i=6$   $j=5$   $\rightarrow$  Zeiger gekreuzt, Partition beenden



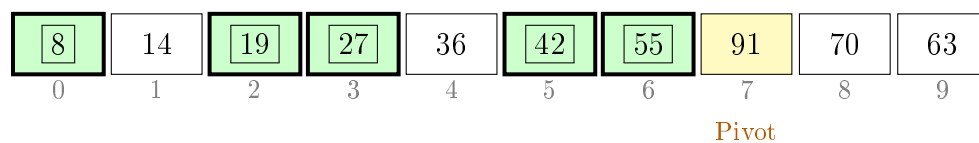
**Schritt 54:** Platziere Pivot: Tausche  $\text{array}[4]=42 \leftrightarrow \text{array}[5]=36$



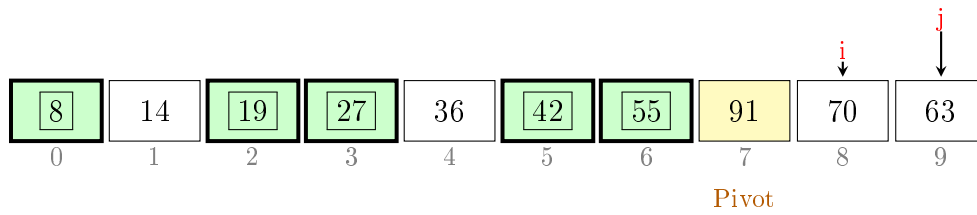
**Schritt 55:** Partition [4..5] fertig: Pivot 42 an Position 5



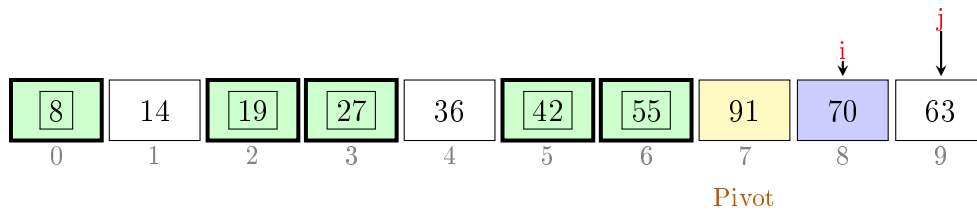
**Schritt 56:** Partition [7..9]: Pivot = 91 (erstes Element)



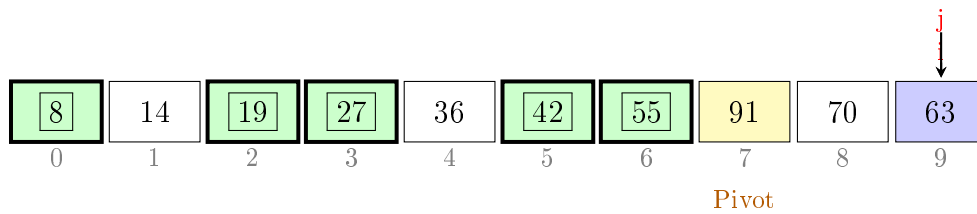
**Schritt 57:** Initialisiere Zeiger:  $i=8$ ,  $j=9$



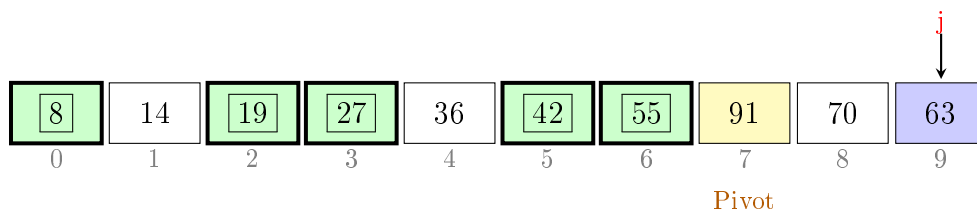
**Schritt 58:**  $\text{array}[8]=70$   $\text{Pivot}=91 \rightarrow i$  nach rechts



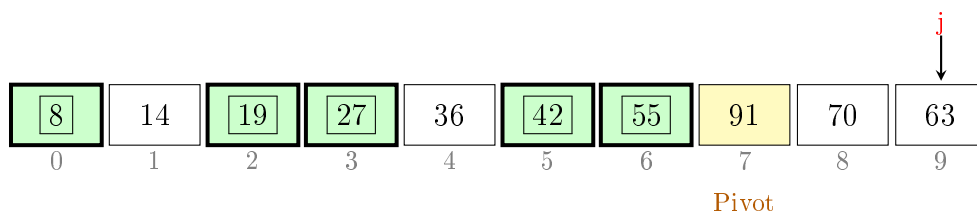
**Schritt 59:**  $\text{array}[9]=63$   $\text{Pivot}=91 \rightarrow i$  nach rechts



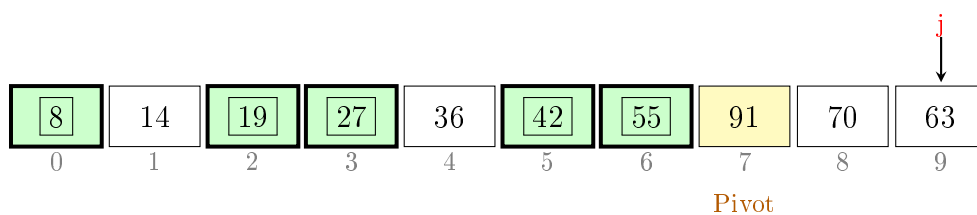
**Schritt 60:**  $\text{array}[9]=63$   $\text{Pivot}=91 \rightarrow j$  stoppt



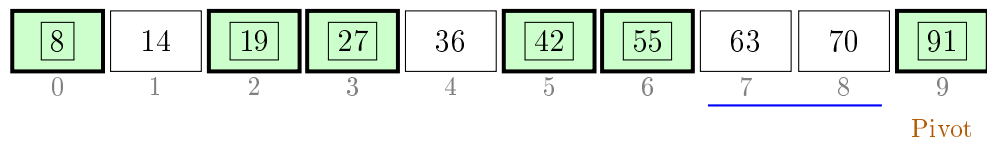
**Schritt 61:**  $i=10$   $j=9 \rightarrow$  Zeiger gekreuzt, Partition beenden



**Schritt 62:** Platziere Pivot: Tausche  $\text{array}[7]=91 \leftrightarrow \text{array}[9]=63$

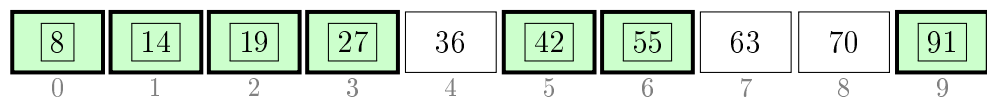


**Schritt 63:** Partition [7..9] fertig: Pivot 91 an Position 9

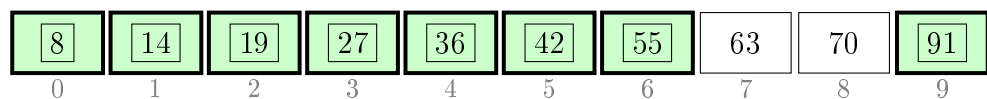


=== Rekursionsebene 3: Partitioniere [7..8]; Basisfall: [1] (Basisfall), [4] (Basisfall) ===

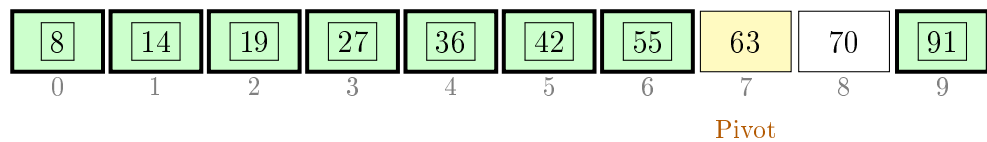
**Schritt 64:** quicksort(1, 1):  $l == r$ , einzelnes Element  $a[1]=14$  ist sortiert



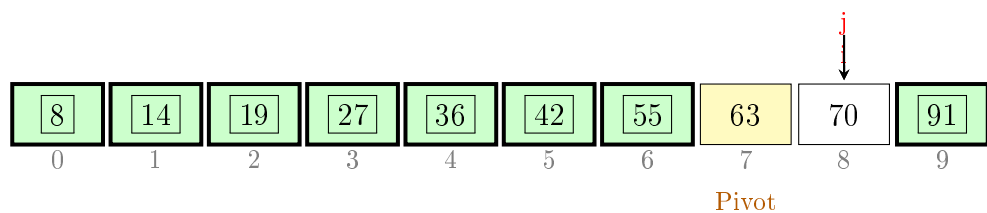
**Schritt 65:** quicksort(4, 4):  $l == r$ , einzelnes Element  $a[4]=36$  ist sortiert



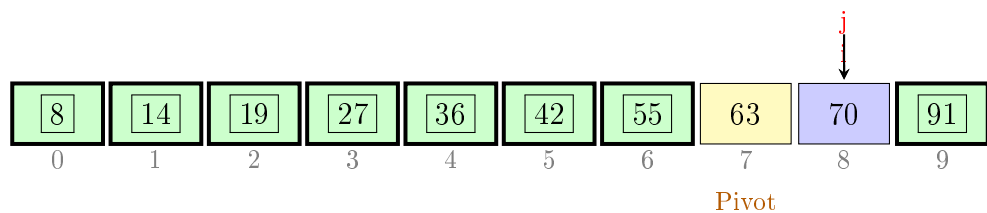
**Schritt 66:** Partition [7..8]: Pivot = 63 (erstes Element)



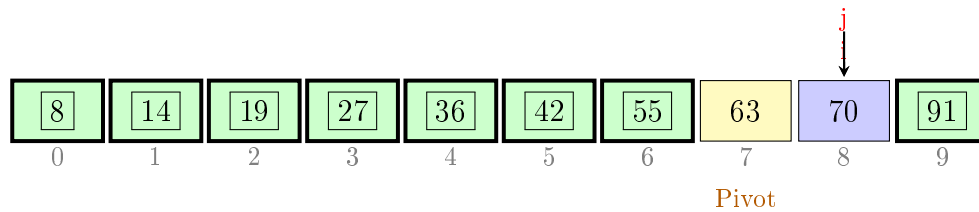
**Schritt 67:** Initialisiere Zeiger:  $i=8, j=8$



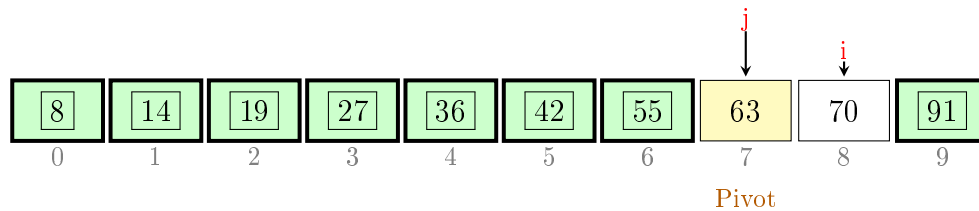
**Schritt 68:**  $array[8]=70 > Pivot=63 \rightarrow i$  stoppt



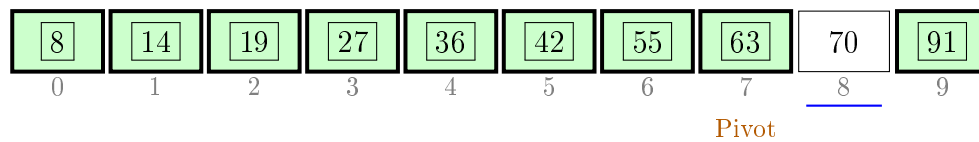
**Schritt 69:**  $array[8]=70 > Pivot=63 \rightarrow j$  nach links



**Schritt 70:**  $i=8$   $j=7 \rightarrow$  Zeiger gekreuzt, Partition beenden

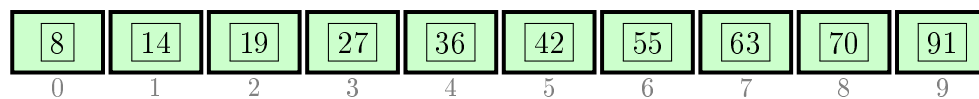


**Schritt 71:** Partition [7..8] fertig: Pivot 63 an Position 7

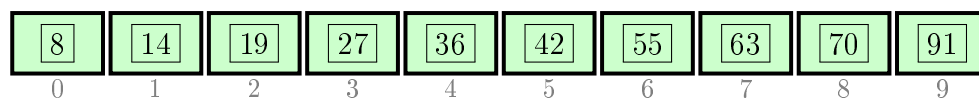


=== Rekursionsebene 4: Basisfall: [8] (Basisfall) ===

**Schritt 72:** quicksort(8, 8):  $l == r$ , einzelnes Element  $a[8]=70$  ist sortiert



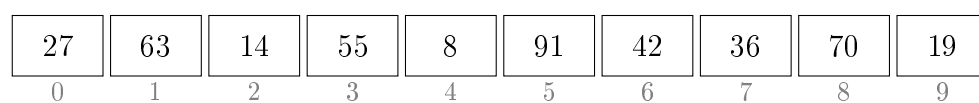
**Schritt 73:** Sortierung abgeschlossen



### Übersicht der Rekursionsschritte:

Jede Zeile zeigt den Zustand nach einem Rekursionsschritt. Unterstrichene Teillisten werden im nächsten Schritt parallel verarbeitet. Umrahmte Werte haben ihre endgültige Position erreicht.

**Zustand 0:** Ausgangszustand



**Zustand 1:** Nach Rekursionsschritt 1

|   |    |    |    |    |    |    |    |    |    |
|---|----|----|----|----|----|----|----|----|----|
| 8 | 19 | 14 | 27 | 55 | 91 | 42 | 36 | 70 | 63 |
| 0 | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  |

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**Zustand 2:** Nach Rekursionsschritt 2

|   |    |    |    |    |    |    |    |    |    |
|---|----|----|----|----|----|----|----|----|----|
| 8 | 19 | 14 | 27 | 42 | 36 | 55 | 91 | 70 | 63 |
| 0 | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  |

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**Zustand 3:** Nach Rekursionsschritt 3

|   |    |    |    |    |    |    |    |    |    |
|---|----|----|----|----|----|----|----|----|----|
| 8 | 14 | 19 | 27 | 36 | 42 | 55 | 63 | 70 | 91 |
| 0 | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  |

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**Zustand 4:** Nach Rekursionsschritt 4

|   |    |    |    |    |    |    |    |    |    |
|---|----|----|----|----|----|----|----|----|----|
| 8 | 14 | 19 | 27 | 36 | 42 | 55 | 63 | 70 | 91 |
| 0 | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  |

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**Antworten:**

1. Nach erster Partition: [8, 19, 14, 27, 55, 91, 42, 36, 70, 63]
2. Nach letzter Partition: [8, 14, 19, 27, 36, 42, 55, 63, 70, 91]
3. Anzahl Vergleiche: **30**

**Endergebnis:** [8, 14, 19, 27, 36, 42, 55, 63, 70, 91]

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## Aufgabe 4 – Schwierigkeit: ★★★★★

*Standard-Quicksort mit 12 Elementen*

Gegeben sei folgende Liste, die in einem Array abgelegt ist:

**34, 78, 12, 45, 89, 23, 67, 9, 56, 41, 95, 31**

Die Liste soll mit dem Standard-Quicksort-Verfahren (keine Variante) in-situ sortiert werden. Als Pivot-Element soll immer das **erste Element** der zu sortierenden (Teil-)Liste genommen werden.

**Fragen:**

1. Geben Sie den Zustand des Arrays nach der **ersten** Partition an.
2. Geben Sie den Zustand des Arrays nach der **letzten** Partition an.
3. Wie viele Vergleiche werden insgesamt durchgeführt?

### Lösung:

**Schritt-für-Schritt-Lösung:**

**Schritt 0:** Ausgangszustand

|    |    |    |    |    |    |    |   |    |    |    |    |
|----|----|----|----|----|----|----|---|----|----|----|----|
| 34 | 78 | 12 | 45 | 89 | 23 | 67 | 9 | 56 | 41 | 95 | 31 |
| 0  | 1  | 2  | 3  | 4  | 5  | 6  | 7 | 8  | 9  | 10 | 11 |

=== Rekursionsebene 0: Partitioniere [0..11] ===

**Schritt 1:** Partition [0..11]: Pivot = 34 (erstes Element)

|    |    |    |    |    |    |    |   |    |    |    |    |
|----|----|----|----|----|----|----|---|----|----|----|----|
| 34 | 78 | 12 | 45 | 89 | 23 | 67 | 9 | 56 | 41 | 95 | 31 |
| 0  | 1  | 2  | 3  | 4  | 5  | 6  | 7 | 8  | 9  | 10 | 11 |

Pivot

**Schritt 2:** Initialisiere Zeiger: i=1, j=11

|    |    |    |    |    |    |    |   |    |    |    |    |
|----|----|----|----|----|----|----|---|----|----|----|----|
| 34 | 78 | 12 | 45 | 89 | 23 | 67 | 9 | 56 | 41 | 95 | 31 |
| 0  | 1  | 2  | 3  | 4  | 5  | 6  | 7 | 8  | 9  | 10 | 11 |

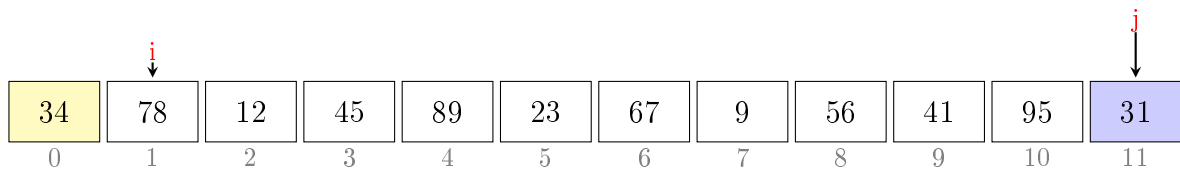
Pivot

**Schritt 3:** array[1]=78 > Pivot=34 → i stoppt

|    |    |    |    |    |    |    |   |    |    |    |    |
|----|----|----|----|----|----|----|---|----|----|----|----|
| 34 | 78 | 12 | 45 | 89 | 23 | 67 | 9 | 56 | 41 | 95 | 31 |
| 0  | 1  | 2  | 3  | 4  | 5  | 6  | 7 | 8  | 9  | 10 | 11 |

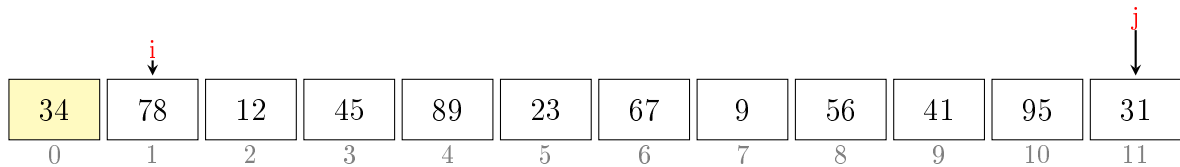
Pivot

**Schritt 4:**  $\text{array}[11]=31$  Pivot=34  $\rightarrow j$  stoppt



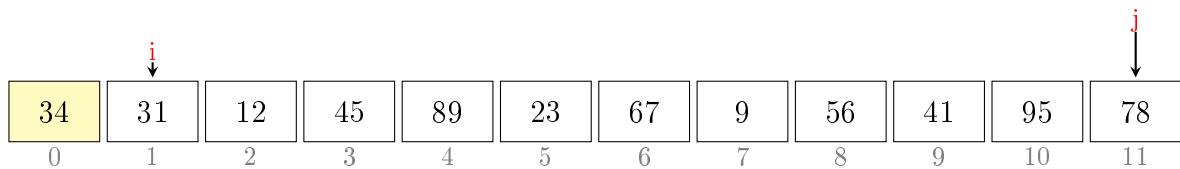
Pivot

**Schritt 5:** Tausche  $\text{array}[1]=78 \leftrightarrow \text{array}[11]=31$



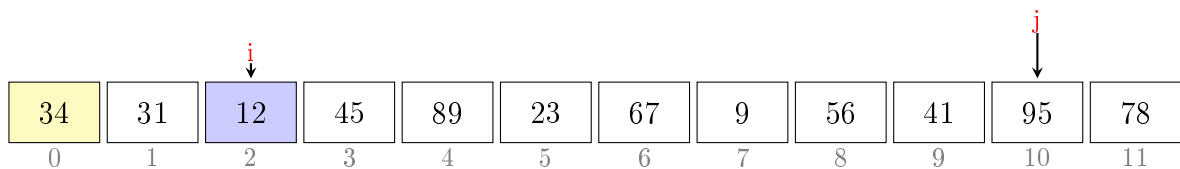
Pivot

**Schritt 6:** Nach Tausch:  $\text{array}[1]=31$ ,  $\text{array}[11]=78$



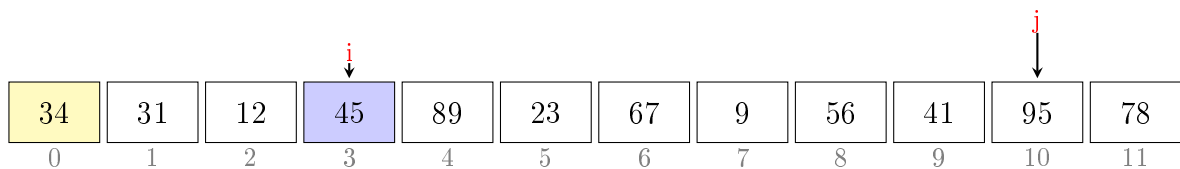
Pivot

**Schritt 7:**  $\text{array}[2]=12$  Pivot=34  $\rightarrow i$  nach rechts



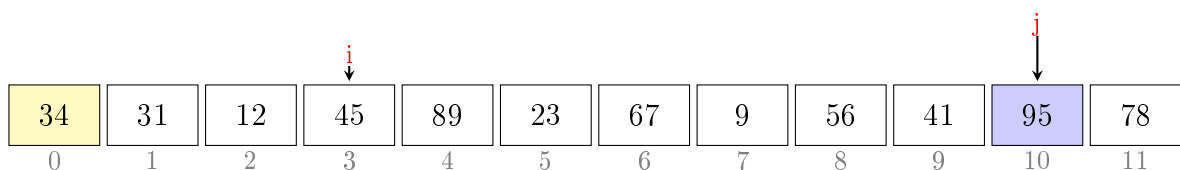
Pivot

**Schritt 8:**  $\text{array}[3]=45 > \text{Pivot}=34 \rightarrow i$  stoppt



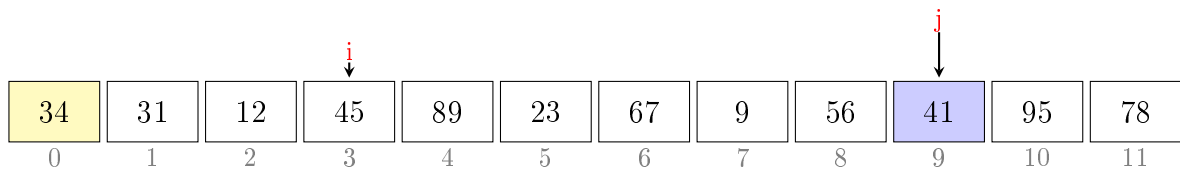
Pivot

**Schritt 9:**  $\text{array}[10]=95 > \text{Pivot}=34 \rightarrow j$  nach links



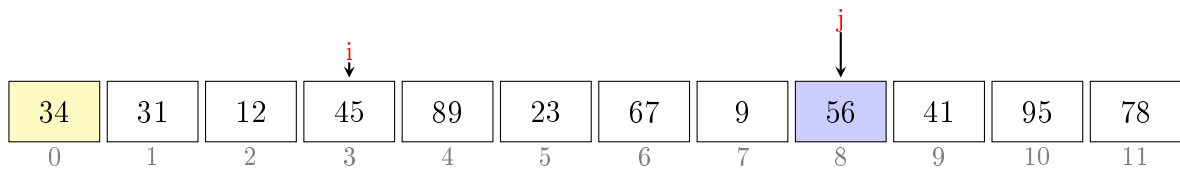
Pivot

**Schritt 10:**  $\text{array}[9]=41 > \text{Pivot}=34 \rightarrow j$  nach links



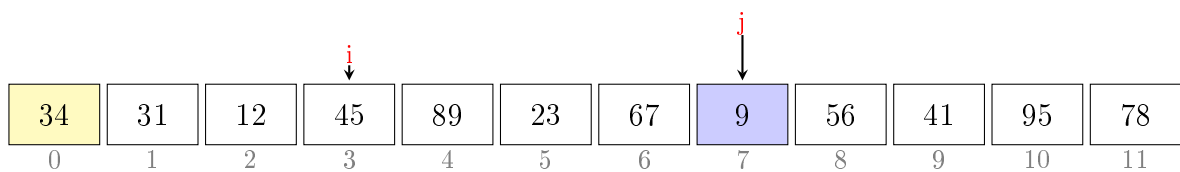
Pivot

**Schritt 11:**  $\text{array}[8]=56 > \text{Pivot}=34 \rightarrow j$  nach links



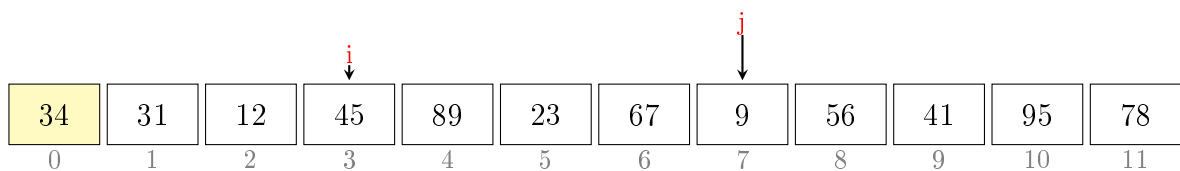
Pivot

**Schritt 12:**  $\text{array}[7]=9$  Pivot=34  $\rightarrow j$  stoppt



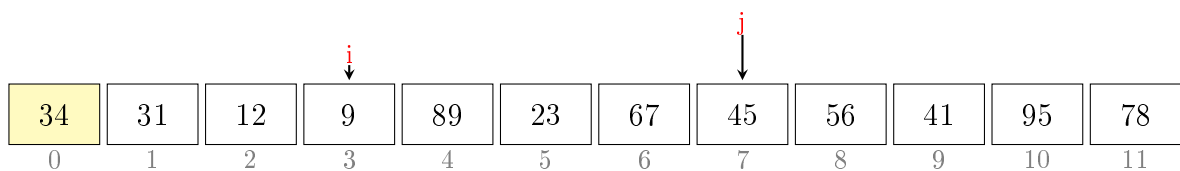
Pivot

**Schritt 13:** Tausche  $\text{array}[3]=45 \leftrightarrow \text{array}[7]=9$



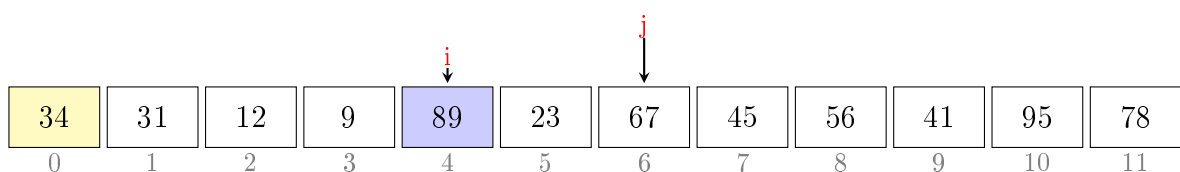
Pivot

**Schritt 14:** Nach Tausch:  $\text{array}[3]=9$ ,  $\text{array}[7]=45$



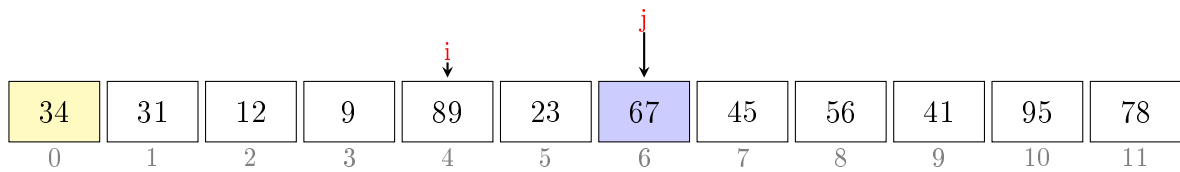
Pivot

**Schritt 15:**  $\text{array}[4]=89 > \text{Pivot}=34 \rightarrow i$  stoppt



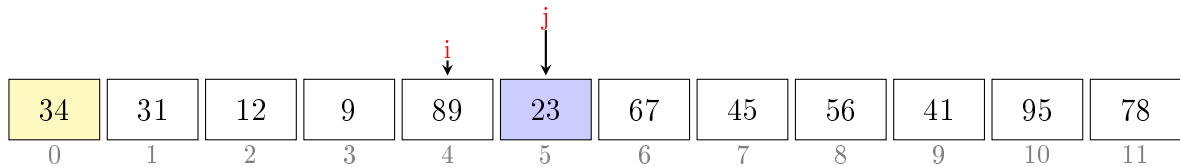
Pivot

**Schritt 16:**  $\text{array}[6]=67 > \text{Pivot}=34 \rightarrow j$  nach links



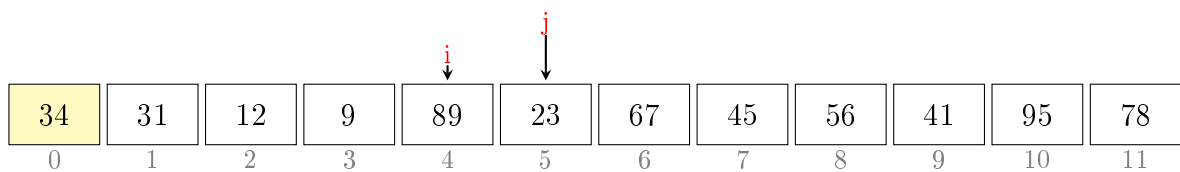
Pivot

**Schritt 17:**  $\text{array}[5]=23$  Pivot=34  $\rightarrow j$  stoppt



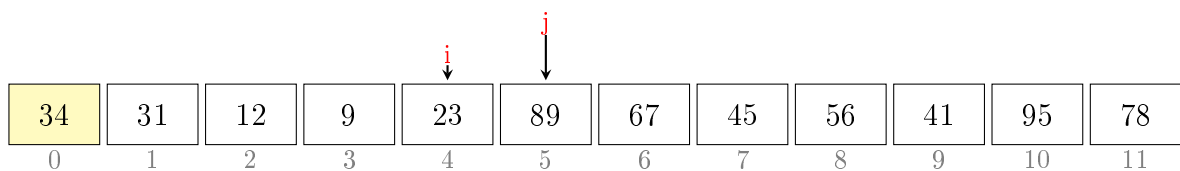
Pivot

**Schritt 18:** Tausche  $\text{array}[4]=89 \leftrightarrow \text{array}[5]=23$



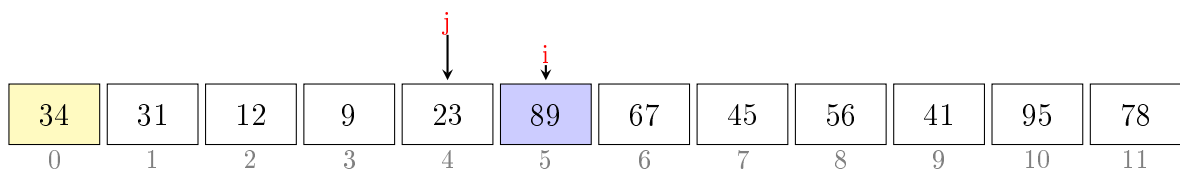
Pivot

**Schritt 19:** Nach Tausch:  $\text{array}[4]=23$ ,  $\text{array}[5]=89$



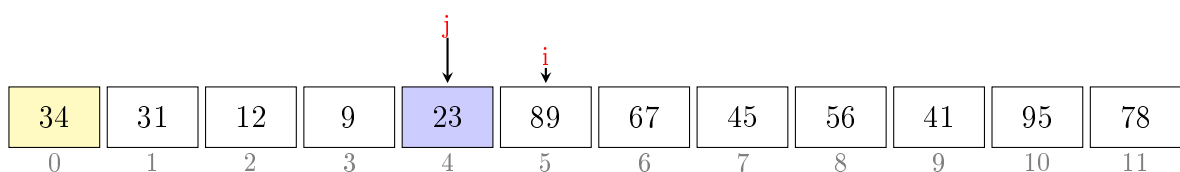
Pivot

**Schritt 20:**  $\text{array}[5]=89 > \text{Pivot}=34 \rightarrow i$  stoppt



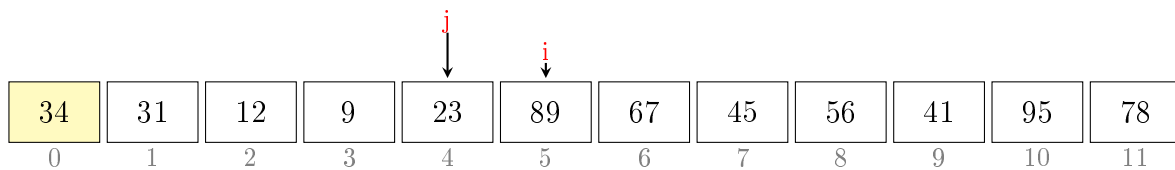
Pivot

**Schritt 21:**  $\text{array}[4]=23$  Pivot=34  $\rightarrow j$  stoppt



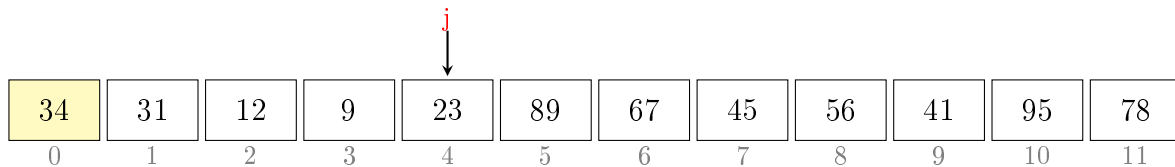
Pivot

**Schritt 22:**  $i=5$   $j=4 \rightarrow$  Zeiger gekreuzt, Partition beenden



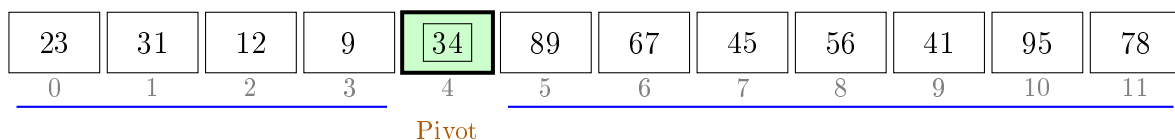
Pivot

**Schritt 23:** Platziere Pivot: Tausche  $\text{array}[0]=34 \leftrightarrow \text{array}[4]=23$



Pivot

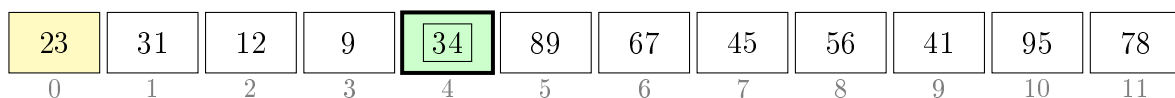
**Schritt 24:** Partition  $[0..11]$  fertig: Pivot 34 an Position 4



Pivot

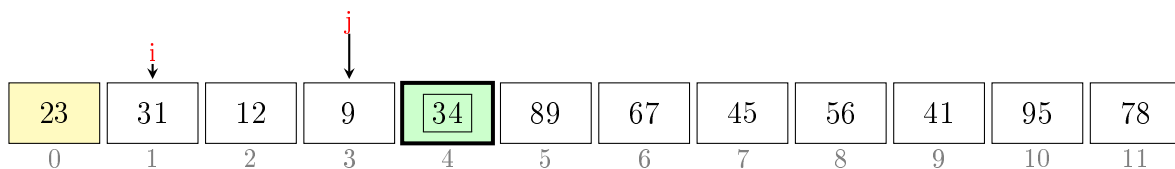
=== Rekursionsebene 1: Partitioniere  $[0..3]$ ,  $[5..11]$  ===

**Schritt 25:** Partition  $[0..3]$ : Pivot = 23 (erstes Element)



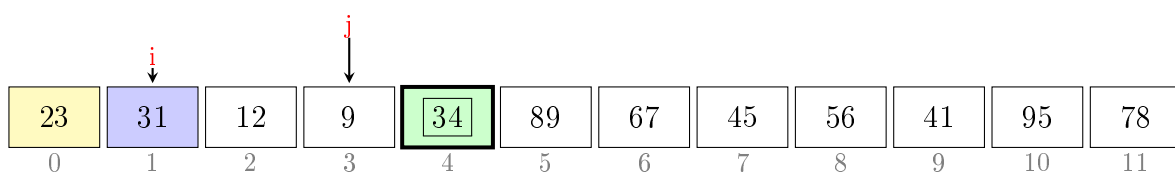
Pivot

**Schritt 26:** Initialisiere Zeiger:  $i=1$ ,  $j=3$



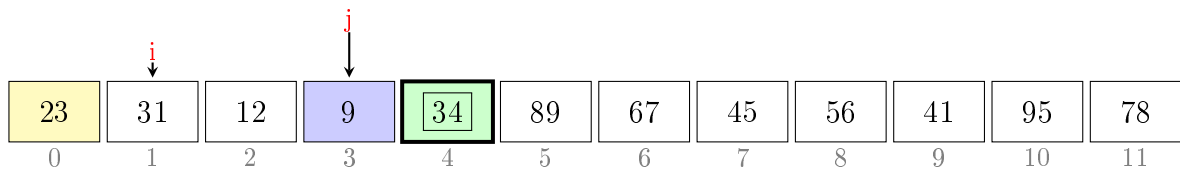
Pivot

**Schritt 27:**  $\text{array}[1]=31 > \text{Pivot}=23 \rightarrow i$  stoppt



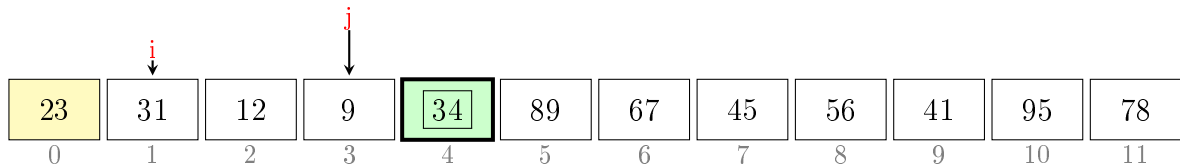
Pivot

**Schritt 28:**  $\text{array}[3]=9$  Pivot=23  $\rightarrow j$  stoppt



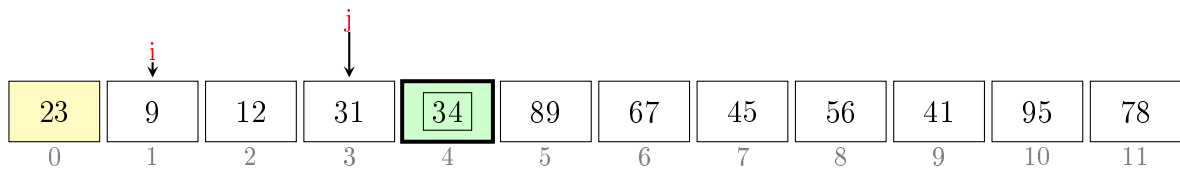
Pivot

**Schritt 29:** Tausche  $\text{array}[1]=31 \leftrightarrow \text{array}[3]=9$



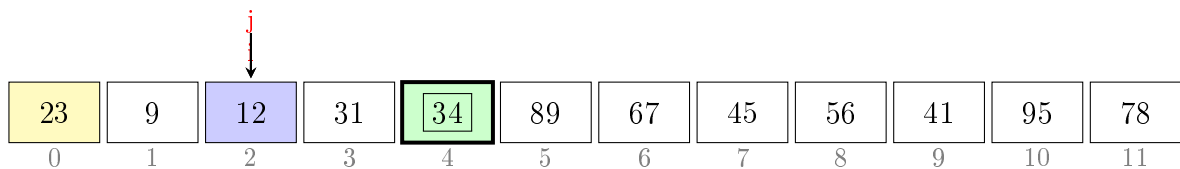
Pivot

**Schritt 30:** Nach Tausch:  $\text{array}[1]=9$ ,  $\text{array}[3]=31$



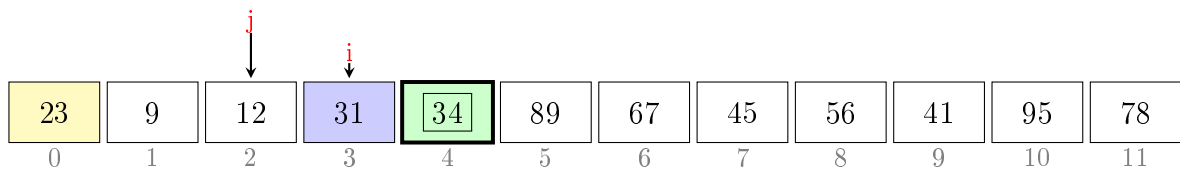
Pivot

**Schritt 31:**  $\text{array}[2]=12$  Pivot=23  $\rightarrow i$  nach rechts



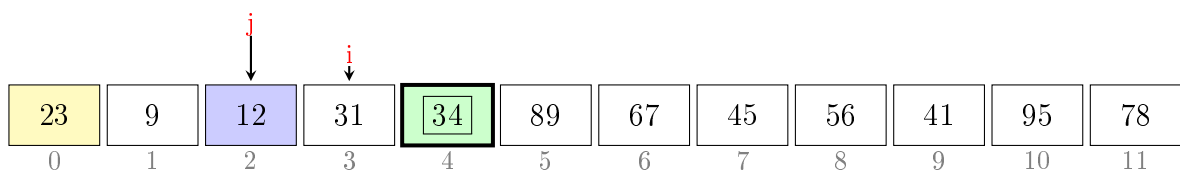
Pivot

**Schritt 32:**  $\text{array}[3]=31 > \text{Pivot}=23 \rightarrow i$  stoppt



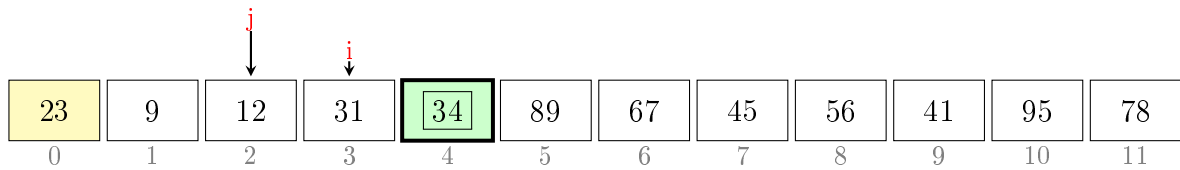
Pivot

**Schritt 33:**  $\text{array}[2]=12$  Pivot=23  $\rightarrow j$  stoppt

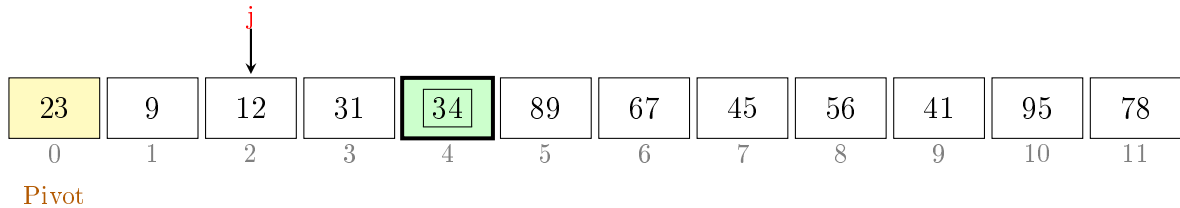


Pivot

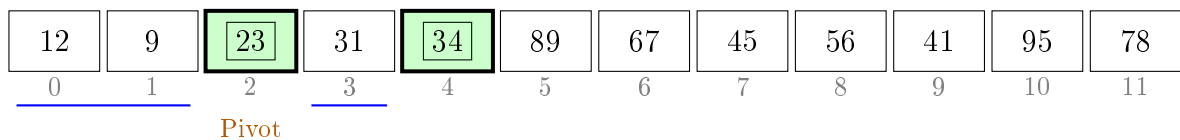
**Schritt 34:**  $i=3$   $j=2 \rightarrow$  Zeiger gekreuzt, Partition beenden



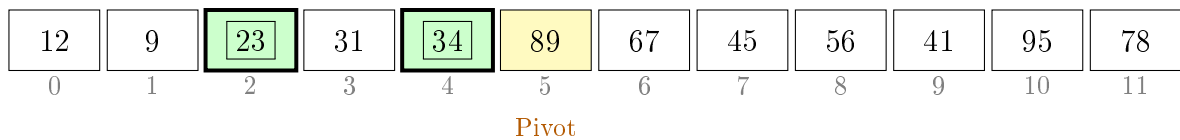
**Schritt 35:** Platziere Pivot: Tausche  $\text{array}[0]=23 \leftrightarrow \text{array}[2]=12$



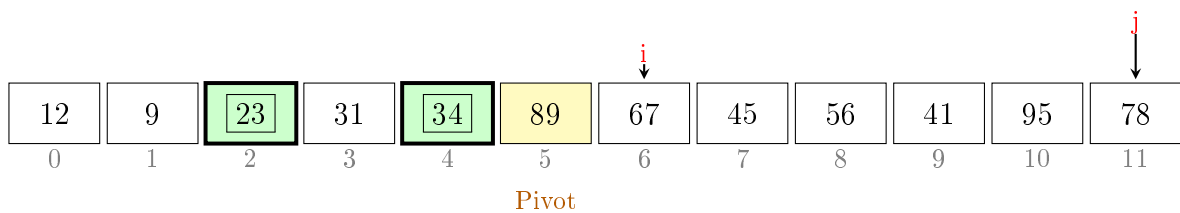
**Schritt 36:** Partition  $[0..3]$  fertig: Pivot 23 an Position 2



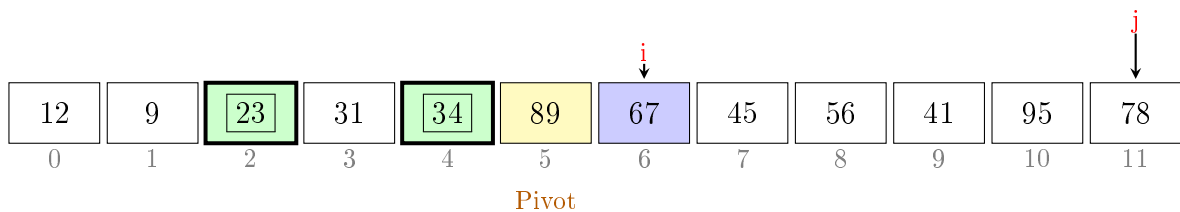
**Schritt 37:** Partition  $[5..11]$ : Pivot = 89 (erstes Element)



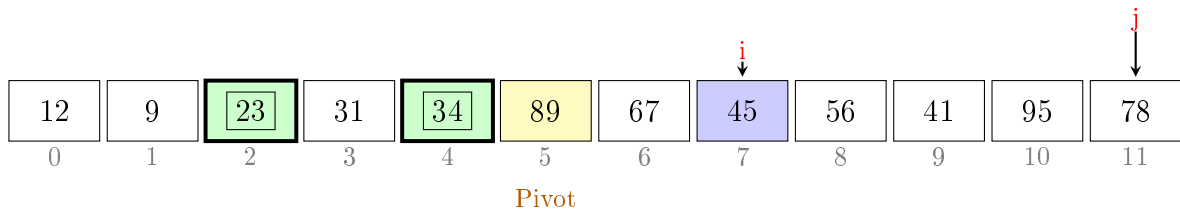
**Schritt 38:** Initialisiere Zeiger:  $i=6, j=11$



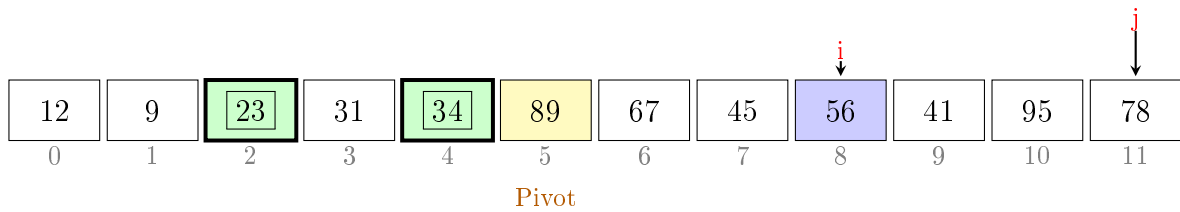
**Schritt 39:**  $\text{array}[6]=67$  Pivot=89  $\rightarrow i$  nach rechts



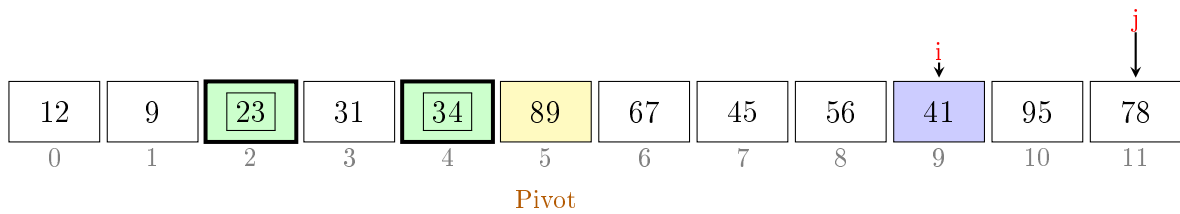
**Schritt 40:**  $\text{array}[7]=45$  Pivot=89  $\rightarrow i$  nach rechts



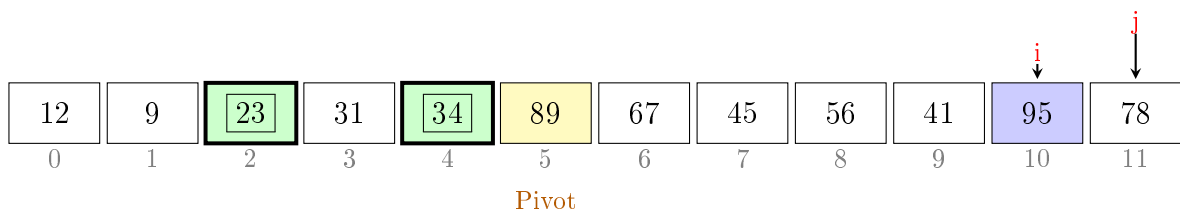
**Schritt 41:** array[8]=56 Pivot=89 → i nach rechts



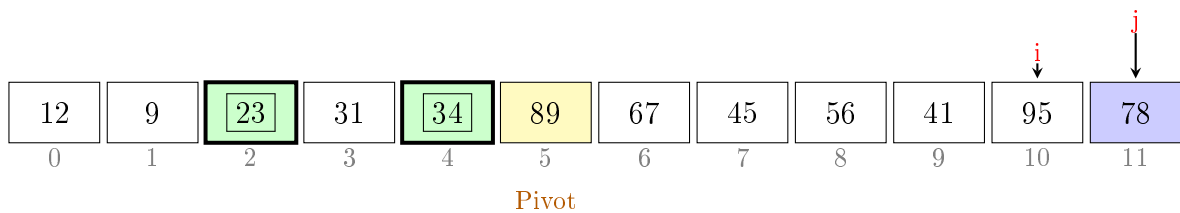
**Schritt 42:** array[9]=41 Pivot=89 → i nach rechts



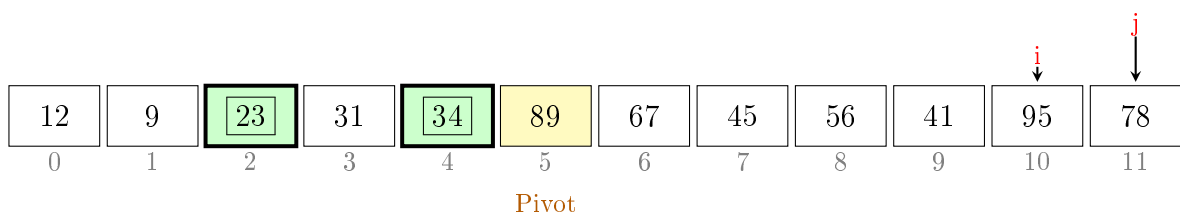
**Schritt 43:** array[10]=95 > Pivot=89 → i stoppt



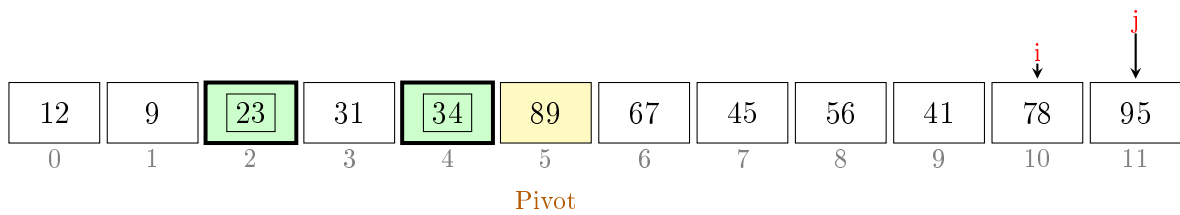
**Schritt 44:** array[11]=78 Pivot=89 → j stoppt



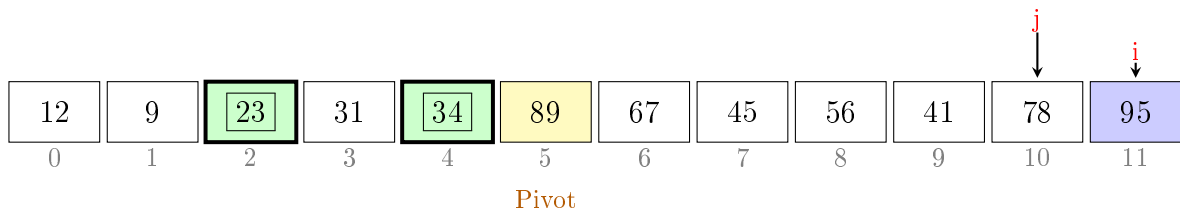
**Schritt 45:** Tausche array[10]=95 ↔ array[11]=78



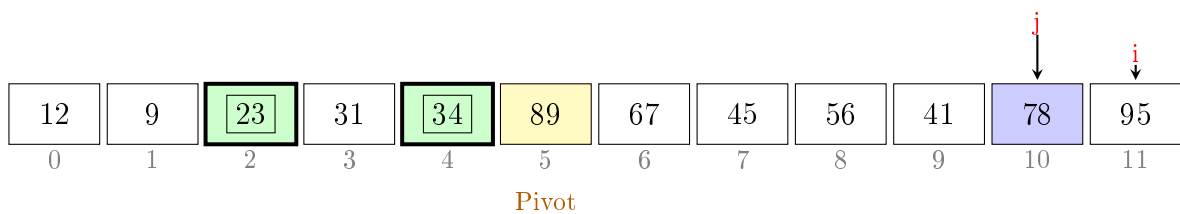
**Schritt 46:** Nach Tausch:  $\text{array}[10]=78$ ,  $\text{array}[11]=95$



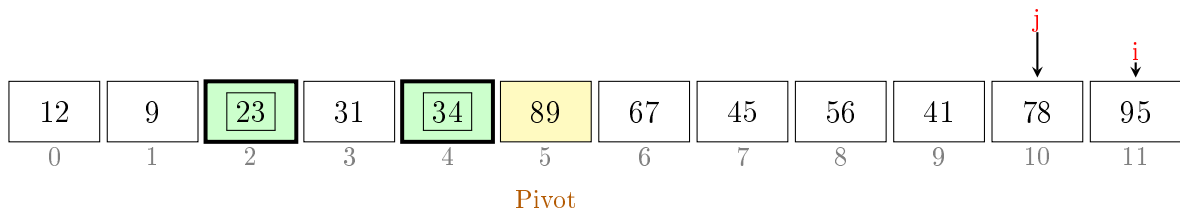
**Schritt 47:**  $\text{array}[11]=95 > \text{Pivot}=89 \rightarrow i$  stoppt



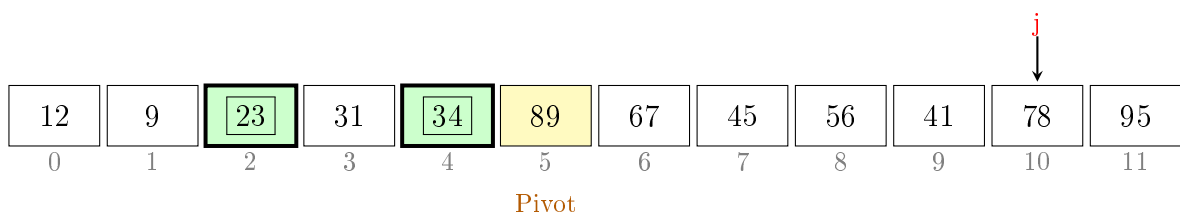
**Schritt 48:**  $\text{array}[10]=78$  Pivot=89  $\rightarrow j$  stoppt



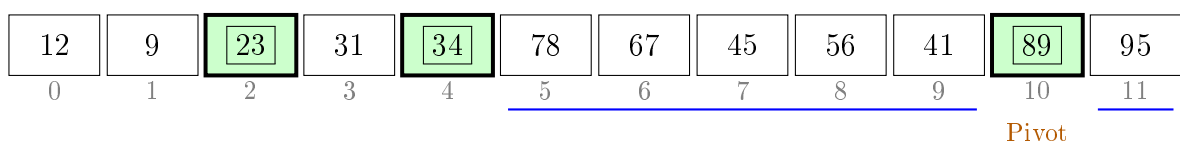
**Schritt 49:**  $i=11$   $j=10 \rightarrow$  Zeiger gekreuzt, Partition beenden



**Schritt 50:** Platziere Pivot: Tausche  $\text{array}[5]=89 \leftrightarrow \text{array}[10]=78$



**Schritt 51:** Partition [5..11] fertig: Pivot 89 an Position 10



=== Rekursionsebene 2: Partitioniere [0..1], [5..9]; Basisfall: [3] (Basisfall), [11] (Basisfall) ===

**Schritt 52:** Partition [0..1]: Pivot = 12 (erstes Element)

|    |   |    |    |    |    |    |    |    |    |    |    |
|----|---|----|----|----|----|----|----|----|----|----|----|
| 12 | 9 | 23 | 31 | 34 | 78 | 67 | 45 | 56 | 41 | 89 | 95 |
| 0  | 1 | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  | 10 | 11 |

Pivot

**Schritt 53:** Initialisiere Zeiger: i=1, j=1

|    |   |    |    |    |    |    |    |    |    |    |    |
|----|---|----|----|----|----|----|----|----|----|----|----|
| 12 | 9 | 23 | 31 | 34 | 78 | 67 | 45 | 56 | 41 | 89 | 95 |
| 0  | 1 | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  | 10 | 11 |

Pivot

**Schritt 54:** array[1]=9 Pivot=12 → i nach rechts

|    |   |    |    |    |    |    |    |    |    |    |    |
|----|---|----|----|----|----|----|----|----|----|----|----|
| 12 | 9 | 23 | 31 | 34 | 78 | 67 | 45 | 56 | 41 | 89 | 95 |
| 0  | 1 | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  | 10 | 11 |

Pivot

**Schritt 55:** array[1]=9 Pivot=12 → j stoppt

|    |   |    |    |    |    |    |    |    |    |    |    |
|----|---|----|----|----|----|----|----|----|----|----|----|
| 12 | 9 | 23 | 31 | 34 | 78 | 67 | 45 | 56 | 41 | 89 | 95 |
| 0  | 1 | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  | 10 | 11 |

Pivot

**Schritt 56:** i=2 j=1 → Zeiger gekreuzt, Partition beenden

|    |   |    |    |    |    |    |    |    |    |    |    |
|----|---|----|----|----|----|----|----|----|----|----|----|
| 12 | 9 | 23 | 31 | 34 | 78 | 67 | 45 | 56 | 41 | 89 | 95 |
| 0  | 1 | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  | 10 | 11 |

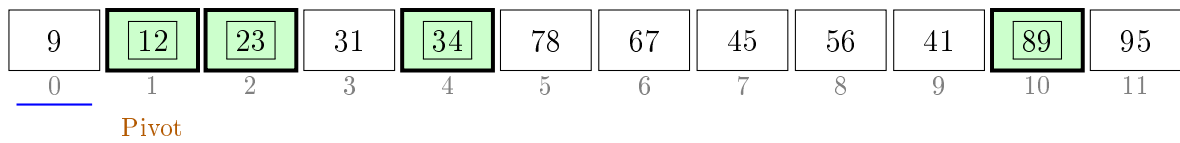
Pivot

**Schritt 57:** Platziere Pivot: Tausche array[0]=12 ↔ array[1]=9

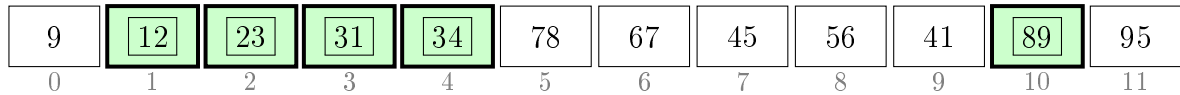
|    |   |    |    |    |    |    |    |    |    |    |    |
|----|---|----|----|----|----|----|----|----|----|----|----|
| 12 | 9 | 23 | 31 | 34 | 78 | 67 | 45 | 56 | 41 | 89 | 95 |
| 0  | 1 | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  | 10 | 11 |

Pivot

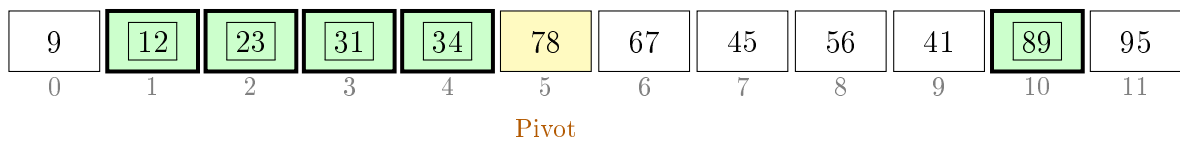
**Schritt 58:** Partition [0..1] fertig: Pivot 12 an Position 1



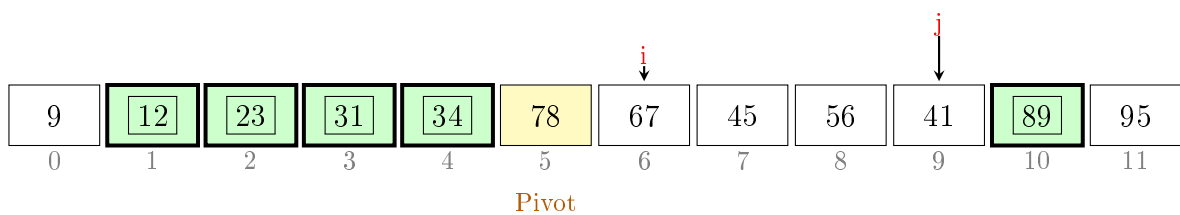
**Schritt 59:** quicksort(3, 3):  $l == r$ , einzelnes Element  $a[3]=31$  ist sortiert



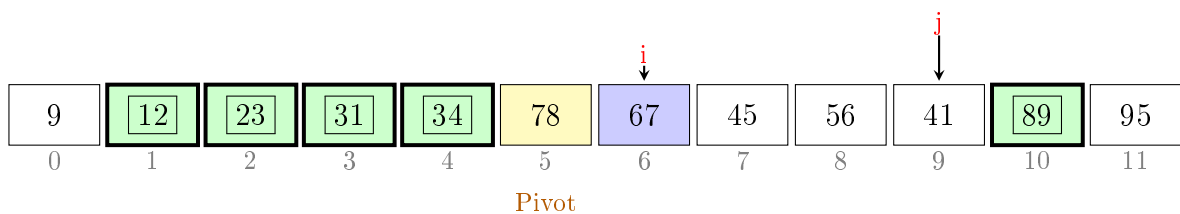
**Schritt 60:** Partition [5..9]: Pivot = 78 (erstes Element)



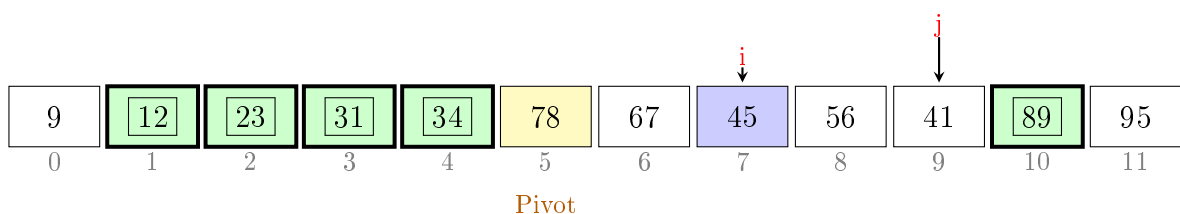
**Schritt 61:** Initialisiere Zeiger:  $i=6, j=9$



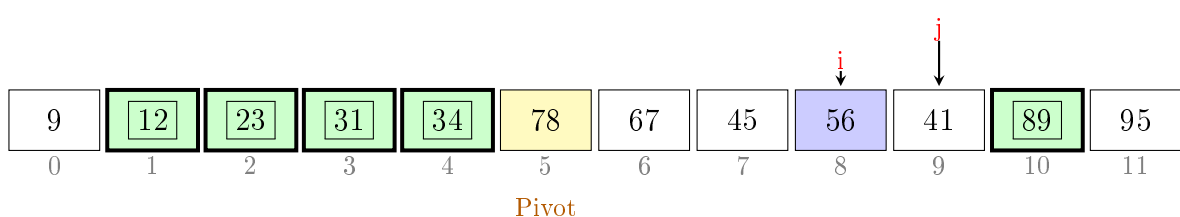
**Schritt 62:**  $array[6]=67$  Pivot=78  $\rightarrow i$  nach rechts



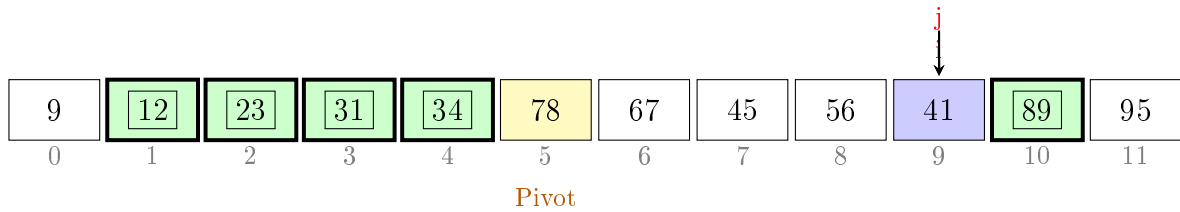
**Schritt 63:**  $array[7]=45$  Pivot=78  $\rightarrow i$  nach rechts



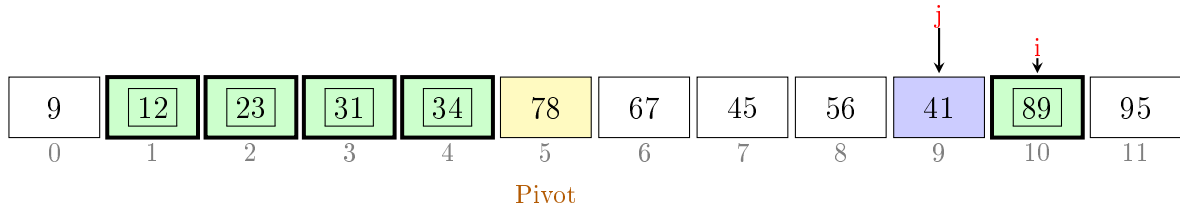
**Schritt 64:**  $array[8]=56$  Pivot=78  $\rightarrow i$  nach rechts



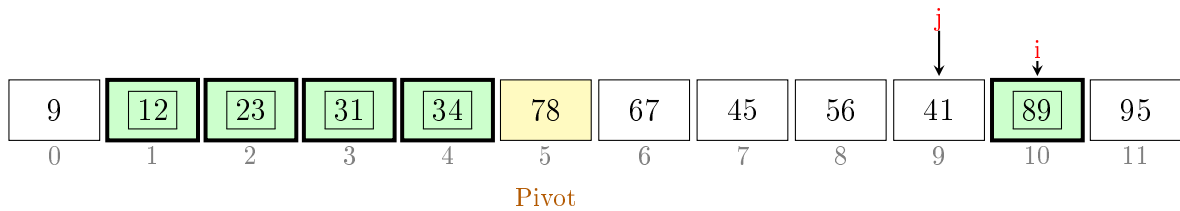
**Schritt 65:** array[9]=41 Pivot=78  $\rightarrow$  i nach rechts



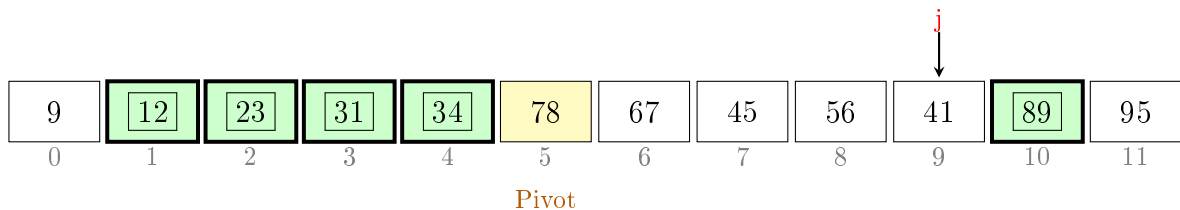
**Schritt 66:** array[9]=41 Pivot=78  $\rightarrow$  j stoppt



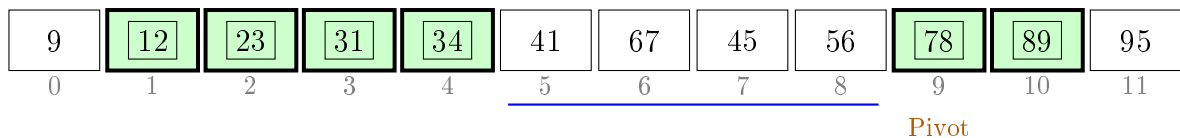
**Schritt 67:** i=10 j=9  $\rightarrow$  Zeiger gekreuzt, Partition beenden



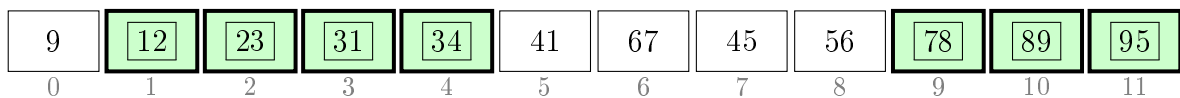
**Schritt 68:** Platziere Pivot: Tausche array[5]=78  $\leftrightarrow$  array[9]=41



**Schritt 69:** Partition [5..9] fertig: Pivot 78 an Position 9

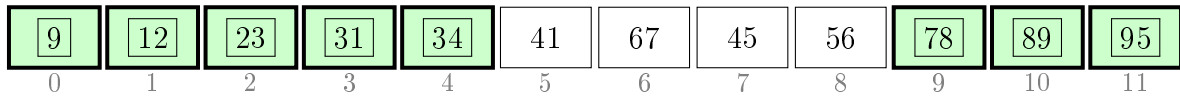


**Schritt 70:** quicksort(11, 11): l == r, einzelnes Element a[11]=95 ist sortiert

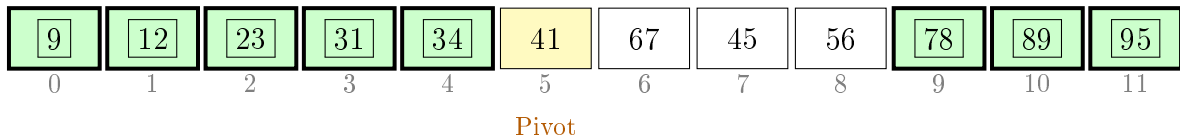


=== Rekursionsebene 3: Partitioniere [5..8]; Basisfall: [0] (Basisfall) ===

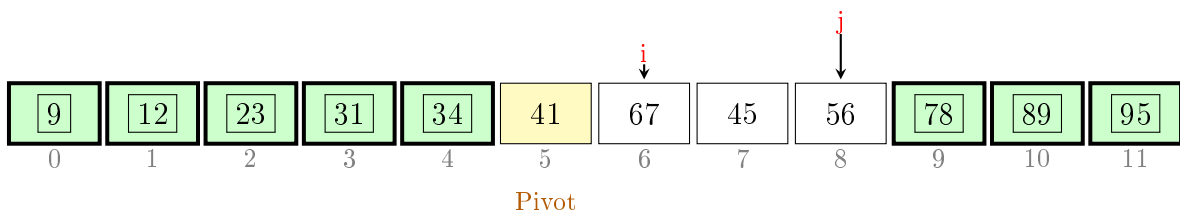
**Schritt 71:** quicksort(0, 0):  $l == r$ , einzelnes Element  $a[0]=9$  ist sortiert



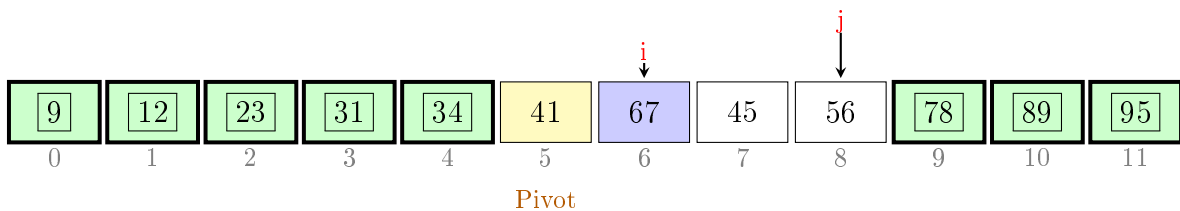
**Schritt 72:** Partition [5..8]: Pivot = 41 (erstes Element)



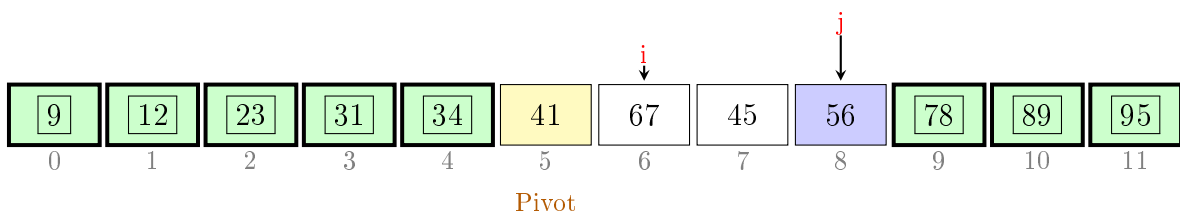
**Schritt 73:** Initialisiere Zeiger:  $i=6, j=8$



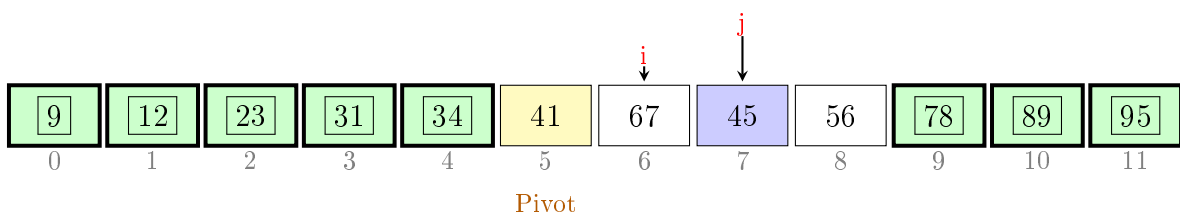
**Schritt 74:**  $\text{array}[6]=67 > \text{Pivot}=41 \rightarrow i$  stoppt



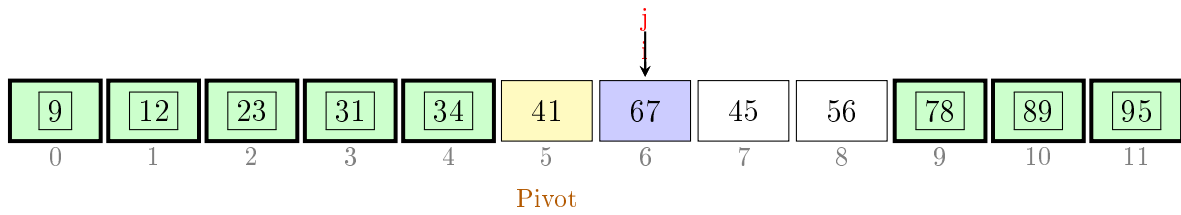
**Schritt 75:**  $\text{array}[8]=56 > \text{Pivot}=41 \rightarrow j$  nach links



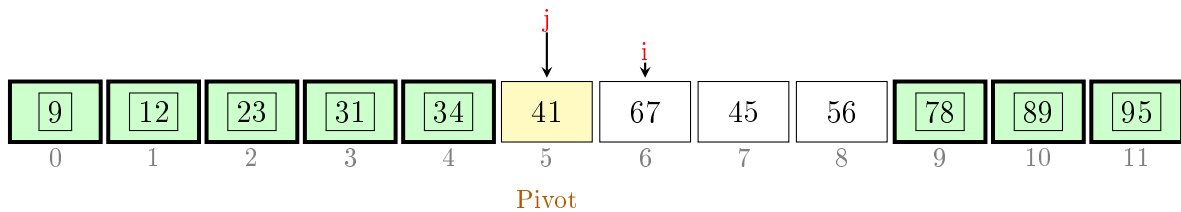
**Schritt 76:**  $\text{array}[7]=45 > \text{Pivot}=41 \rightarrow j$  nach links



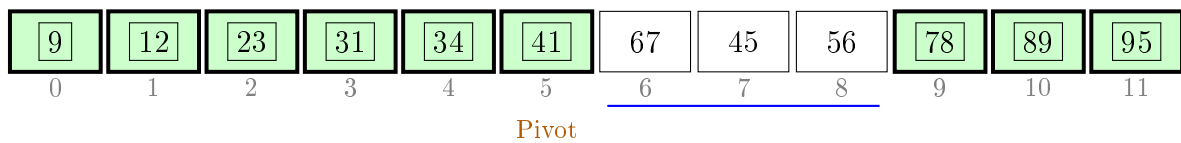
**Schritt 77:**  $\text{array}[6]=67 > \text{Pivot}=41 \rightarrow j$  nach links



**Schritt 78:**  $i=6$   $j=5 \rightarrow$  Zeiger gekreuzt, Partition beenden

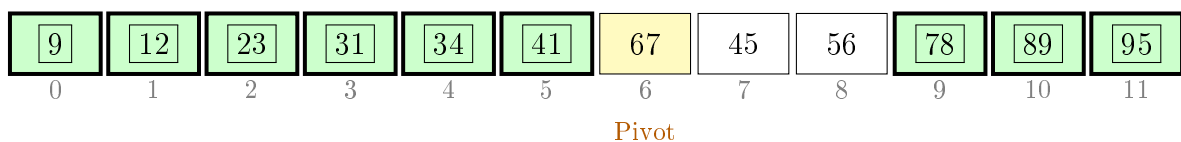


**Schritt 79:** Partition [5..8] fertig: Pivot 41 an Position 5

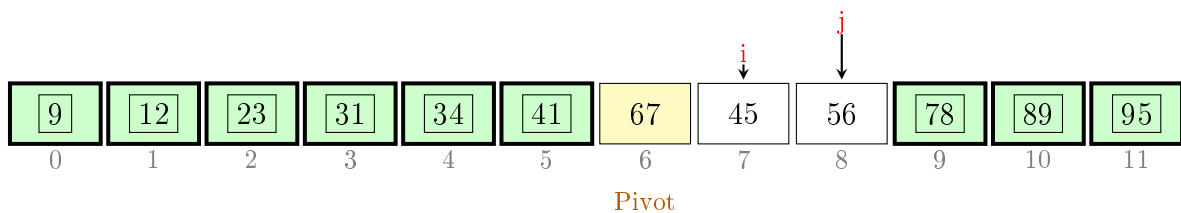


=== Rekursionsebene 4: Partitioniere [6..8] ===

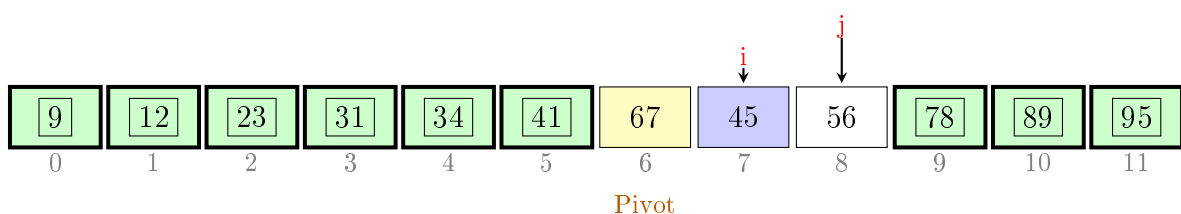
**Schritt 80:** Partition [6..8]: Pivot = 67 (erstes Element)



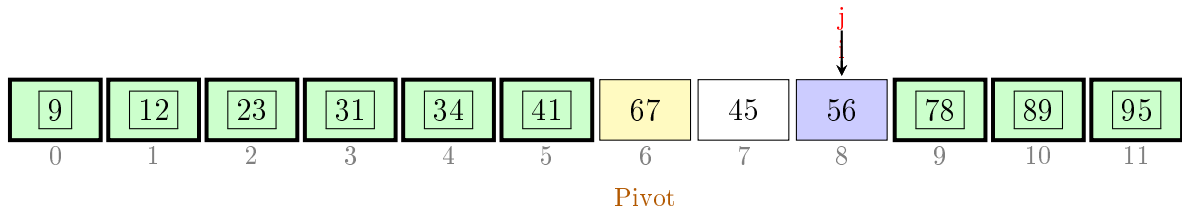
**Schritt 81:** Initialisiere Zeiger:  $i=7$ ,  $j=8$



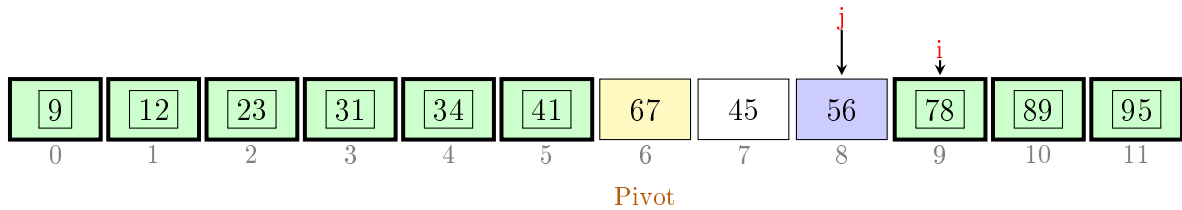
**Schritt 82:**  $\text{array}[7]=45$  Pivot=67  $\rightarrow$   $i$  nach rechts



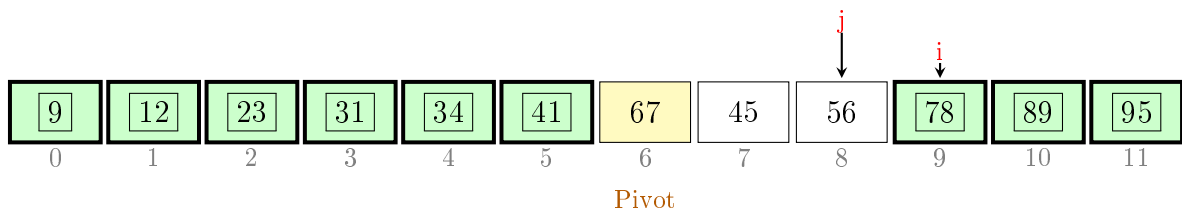
**Schritt 83:** array[8]=56 Pivot=67 → i nach rechts



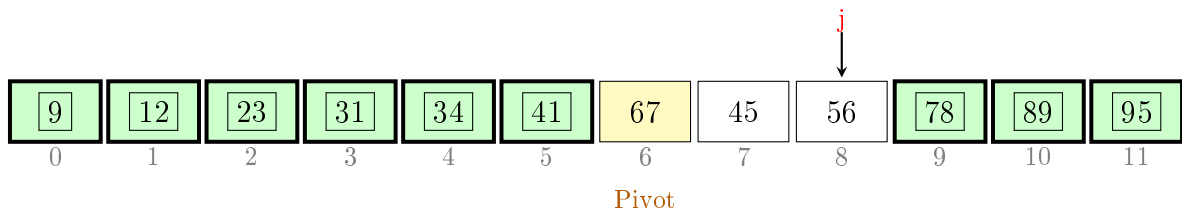
**Schritt 84:** array[8]=56 Pivot=67 → j stoppt



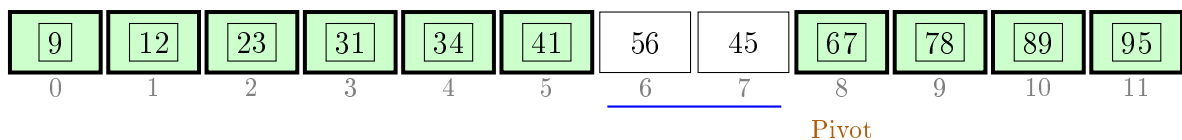
**Schritt 85:** i=9 j=8 → Zeiger gekreuzt, Partition beenden



**Schritt 86:** Platziere Pivot: Tausche array[6]=67 ↔ array[8]=56

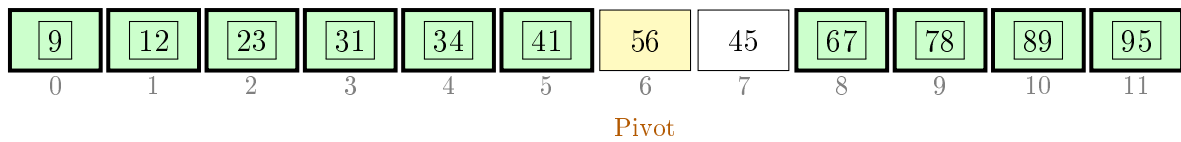


**Schritt 87:** Partition [6..8] fertig: Pivot 67 an Position 8

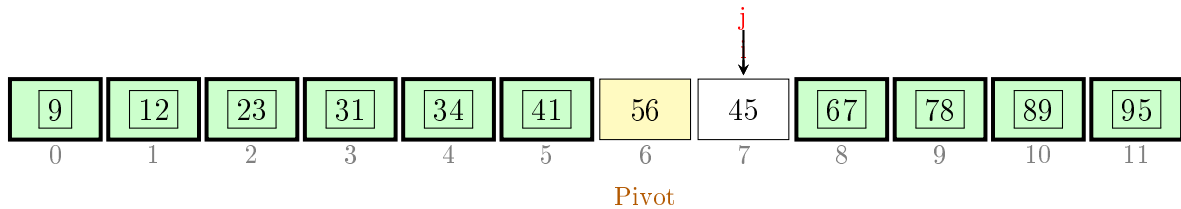


=== Rekursionsebene 5: Partitioniere [6..7] ===

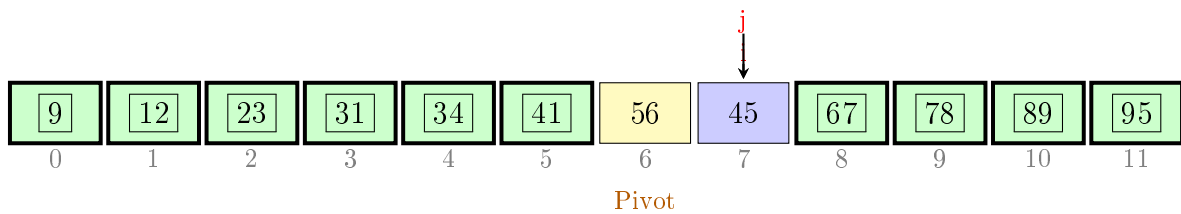
**Schritt 88:** Partition [6..7]: Pivot = 56 (erstes Element)



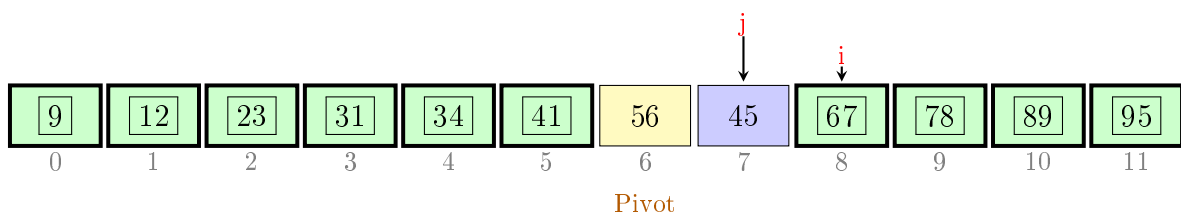
**Schritt 89:** Initialisiere Zeiger:  $i=7, j=7$



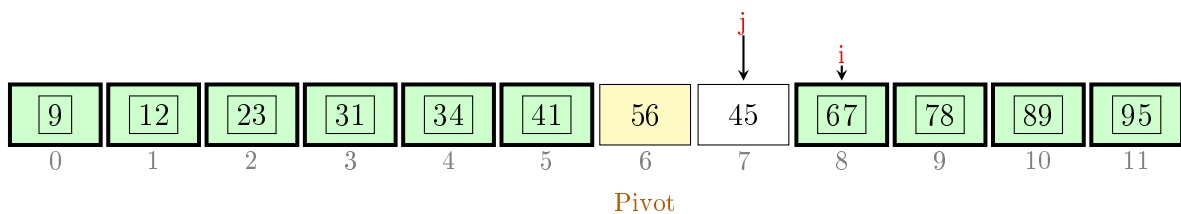
**Schritt 90:**  $\text{array}[7]=45$  Pivot=56  $\rightarrow i$  nach rechts



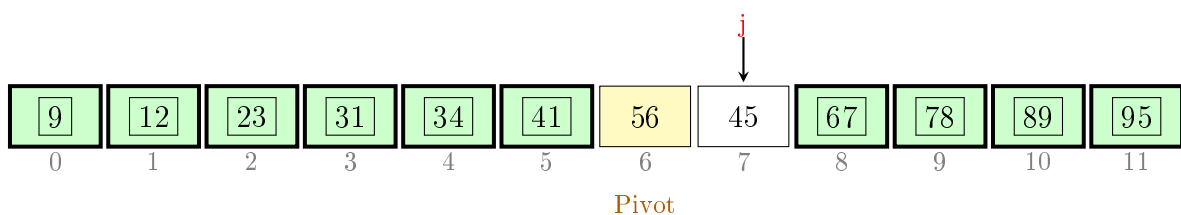
**Schritt 91:**  $\text{array}[7]=45$  Pivot=56  $\rightarrow j$  stoppt



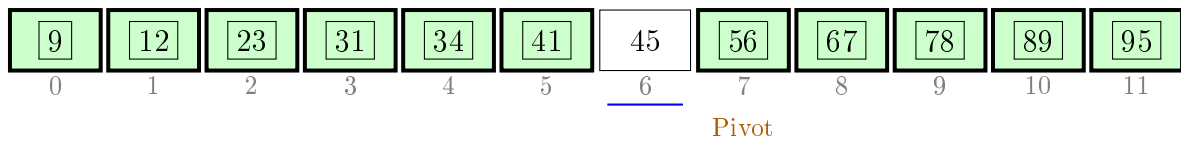
**Schritt 92:**  $i=8, j=7 \rightarrow$  Zeiger gekreuzt, Partition beenden



**Schritt 93:** Platziere Pivot: Tausche  $\text{array}[6]=56 \leftrightarrow \text{array}[7]=45$

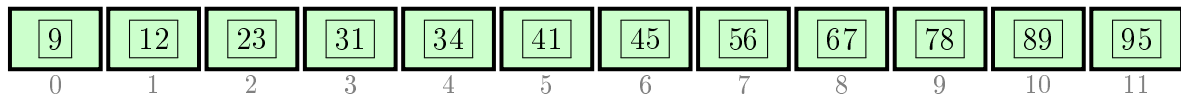


**Schritt 94:** Partition [6..7] fertig: Pivot 56 an Position 7

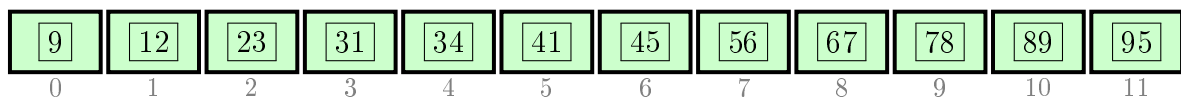


=== Rekursionsebene 6: Basisfall: [6] (Basisfall) ===

**Schritt 95:** quicksort(6, 6):  $l == r$ , einzelnes Element  $a[6]=45$  ist sortiert



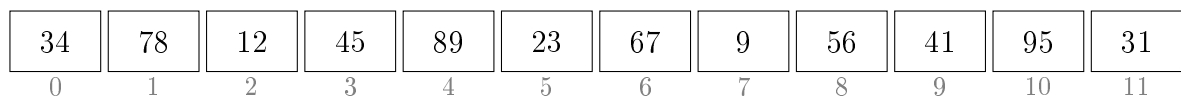
**Schritt 96:** Sortierung abgeschlossen



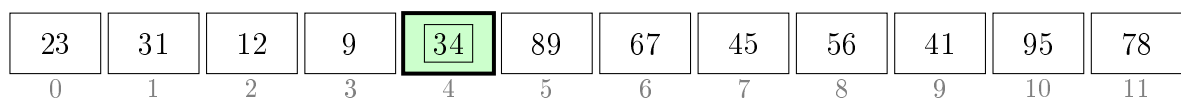
### Übersicht der Rekursionsschritte:

Jede Zeile zeigt den Zustand nach einem Rekursionsschritt. Unterstrichene Teillisten werden im nächsten Schritt parallel verarbeitet. Umrahmte Werte haben ihre endgültige Position erreicht.

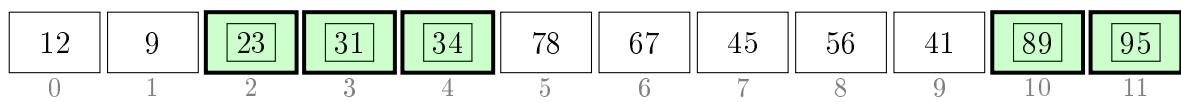
**Zustand 0:** Ausgangszustand



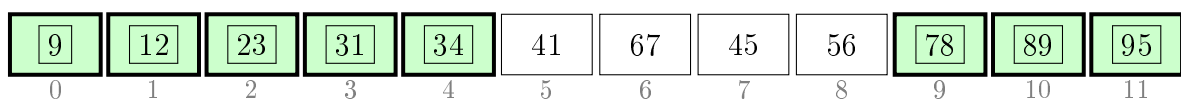
**Zustand 1:** Nach Rekursionsschritt 1



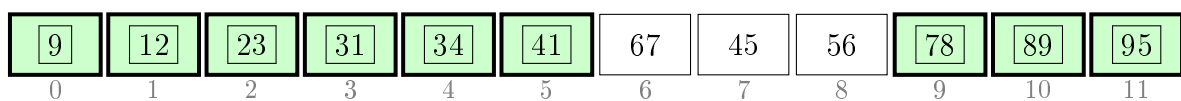
**Zustand 2:** Nach Rekursionsschritt 2



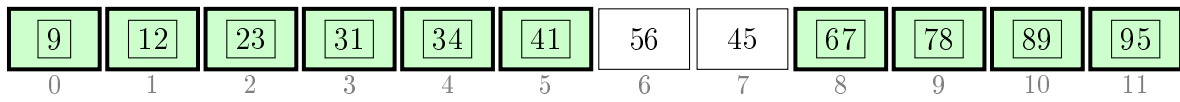
**Zustand 3:** Nach Rekursionsschritt 3



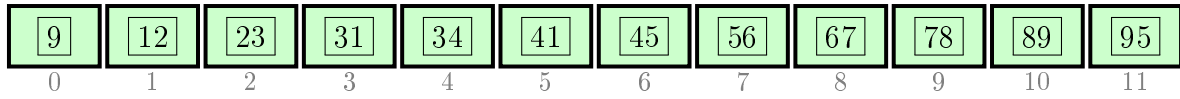
**Zustand 4:** Nach Rekursionsschritt 4



**Zustand 5:** Nach Rekursionsschritt 5



**Zustand 6:** Nach Rekursionsschritt 6



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**Antworten:**

1. Nach erster Partition: [23, 31, 12, 9, 34, 89, 67, 45, 56, 41, 95, 78]
2. Nach letzter Partition: [9, 12, 23, 31, 34, 41, 45, 56, 67, 78, 89, 95]
3. Anzahl Vergleiche: **42**

**Endergebnis:** [9, 12, 23, 31, 34, 41, 45, 56, 67, 78, 89, 95]

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## Aufgabe 5 – Schwierigkeit: ★★★

*Quicksort mit tiefer Rekursion*

Gegeben sei folgende Liste, die in einem Array abgelegt ist:

**1, 18, 20, 81, 59, 73, 40, 16**

Die Liste soll mit dem Standard-Quicksort-Verfahren (keine Variante) in-situ sortiert werden. Als Pivot-Element soll immer das **erste Element** der zu sortierenden (Teil-)Liste genommen werden.

**Fragen:**

1. Geben Sie den Zustand des Arrays nach der **ersten** Partition an.
2. Geben Sie den Zustand des Arrays nach der **letzten** Partition an.
3. Wie viele Vergleiche werden insgesamt durchgeführt?

---

### Lösung:

**Schritt-für-Schritt-Lösung:**

**Schritt 0:** Ausgangszustand

|   |    |    |    |    |    |    |    |
|---|----|----|----|----|----|----|----|
| 1 | 18 | 20 | 81 | 59 | 73 | 40 | 16 |
| 0 | 1  | 2  | 3  | 4  | 5  | 6  | 7  |

=== Rekursionsebene 0: Partitioniere [0..7] ===

**Schritt 1:** Partition [0..7]: Pivot = 1 (erstes Element)

|   |    |    |    |    |    |    |    |
|---|----|----|----|----|----|----|----|
| 1 | 18 | 20 | 81 | 59 | 73 | 40 | 16 |
| 0 | 1  | 2  | 3  | 4  | 5  | 6  | 7  |

Pivot

**Schritt 2:** Initialisiere Zeiger: i=1, j=7

|   |    |    |    |    |    |    |    |
|---|----|----|----|----|----|----|----|
| 1 | 18 | 20 | 81 | 59 | 73 | 40 | 16 |
| 0 | 1  | 2  | 3  | 4  | 5  | 6  | 7  |

Pivot

i ↓ j ↓

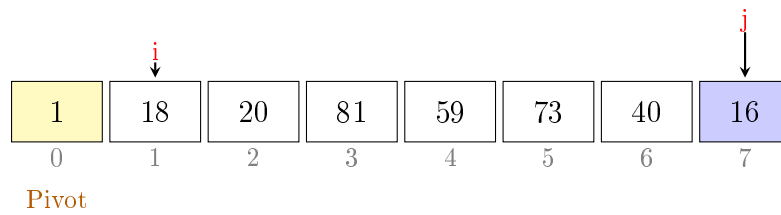
**Schritt 3:** array[1]=18 > Pivot=1 → i stoppt

|   |    |    |    |    |    |    |    |
|---|----|----|----|----|----|----|----|
| 1 | 18 | 20 | 81 | 59 | 73 | 40 | 16 |
| 0 | 1  | 2  | 3  | 4  | 5  | 6  | 7  |

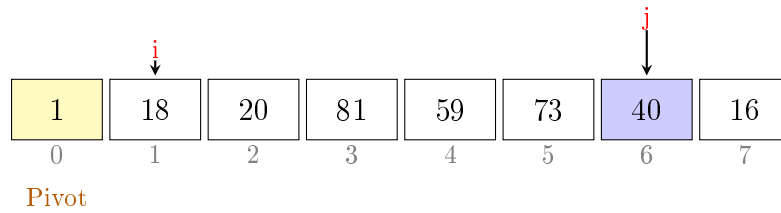
Pivot

i ↓ j ↓

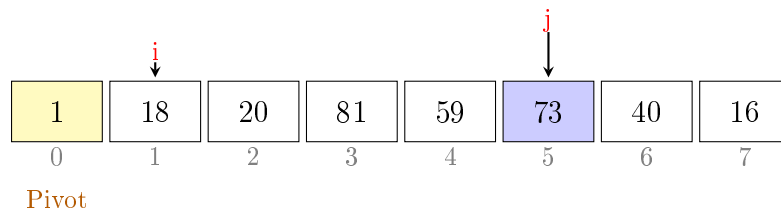
**Schritt 4:**  $\text{array}[7]=16 > \text{Pivot}=1 \rightarrow j$  nach links



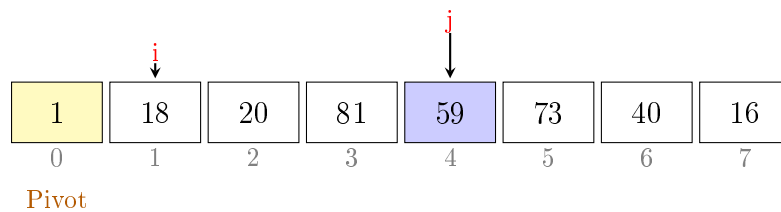
**Schritt 5:**  $\text{array}[6]=40 > \text{Pivot}=1 \rightarrow j$  nach links



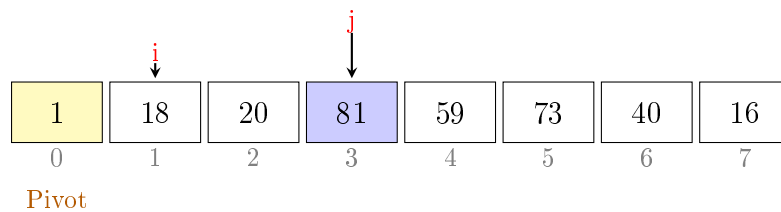
**Schritt 6:**  $\text{array}[5]=73 > \text{Pivot}=1 \rightarrow j$  nach links



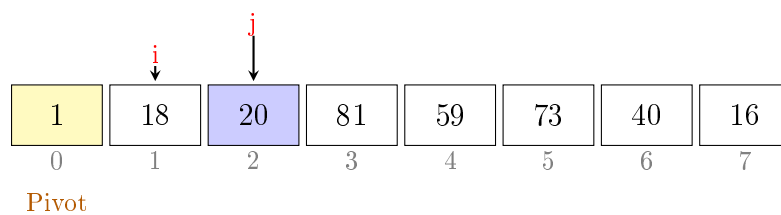
**Schritt 7:**  $\text{array}[4]=59 > \text{Pivot}=1 \rightarrow j$  nach links



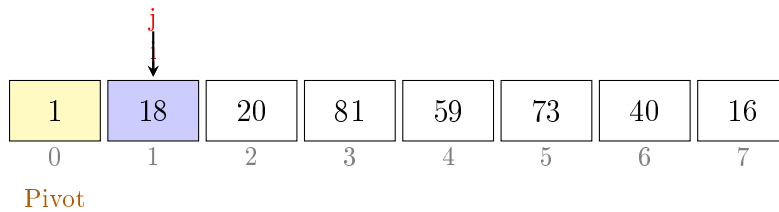
**Schritt 8:**  $\text{array}[3]=81 > \text{Pivot}=1 \rightarrow j$  nach links



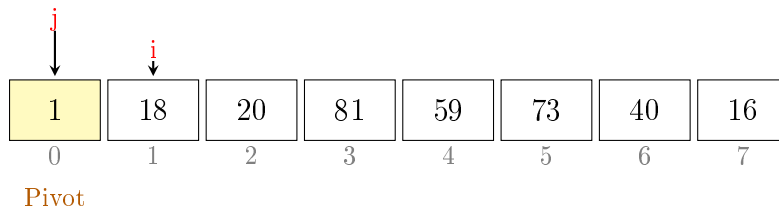
**Schritt 9:**  $\text{array}[2]=20 > \text{Pivot}=1 \rightarrow j$  nach links



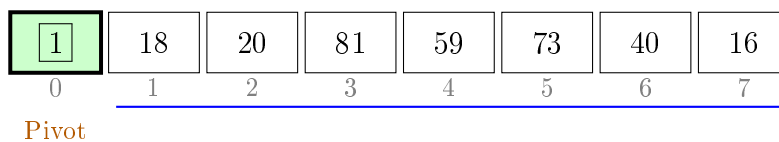
**Schritt 10:**  $\text{array}[1]=18 > \text{Pivot}=1 \rightarrow j$  nach links



**Schritt 11:**  $i=1 \ j=0 \rightarrow$  Zeiger gekreuzt, Partition beenden

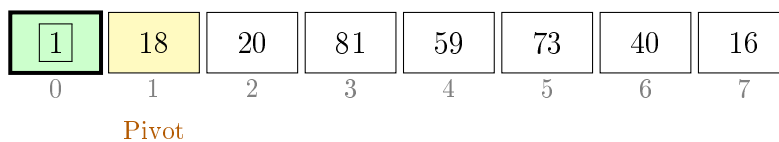


**Schritt 12:** Partition  $[0..7]$  fertig: Pivot 1 an Position 0

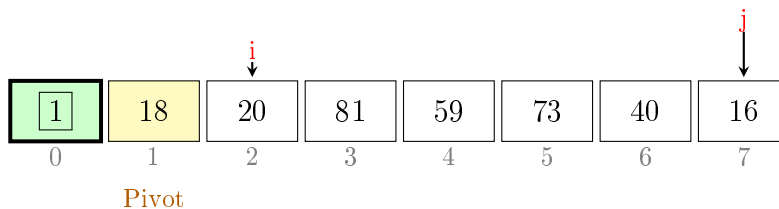


=== Rekursionsebene 1: Partitioniere  $[1..7]$  ===

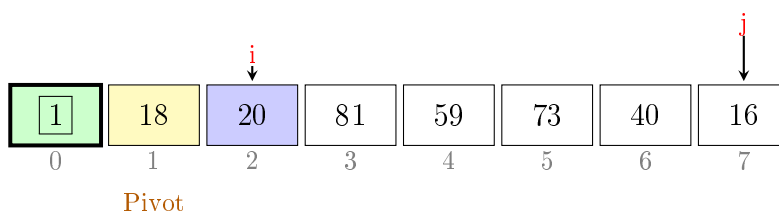
**Schritt 13:** Partition  $[1..7]$ : Pivot = 18 (erstes Element)



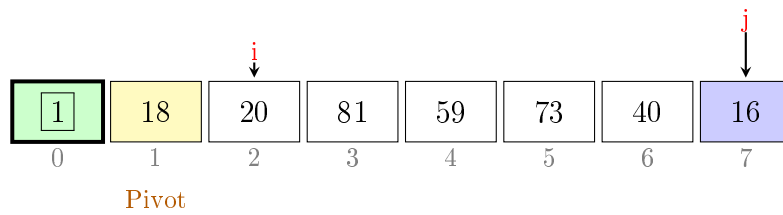
**Schritt 14:** Initialisiere Zeiger:  $i=2, j=7$



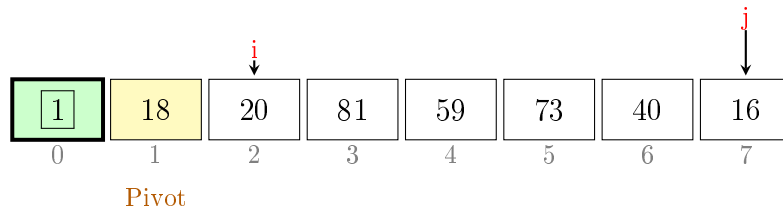
**Schritt 15:**  $\text{array}[2]=20 > \text{Pivot}=18 \rightarrow i$  stoppt



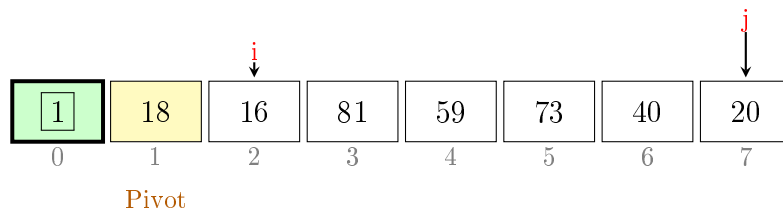
**Schritt 16:**  $\text{array}[7]=16$  Pivot=18  $\rightarrow$  j stoppt



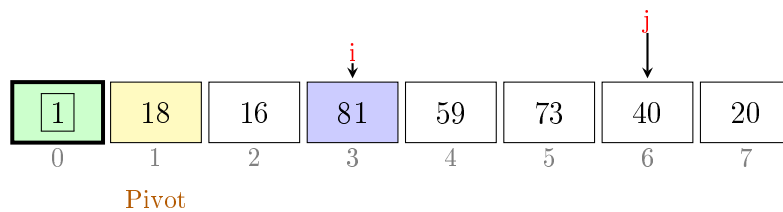
**Schritt 17:** Tausche  $\text{array}[2]=20 \leftrightarrow \text{array}[7]=16$



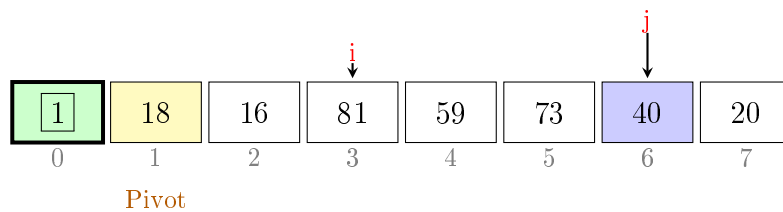
**Schritt 18:** Nach Tausch:  $\text{array}[2]=16$ ,  $\text{array}[7]=20$



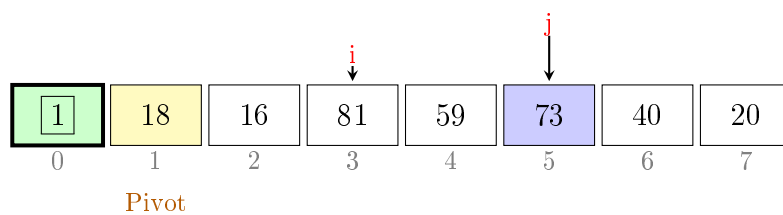
**Schritt 19:**  $\text{array}[3]=81 > \text{Pivot}=18 \rightarrow$  i stoppt



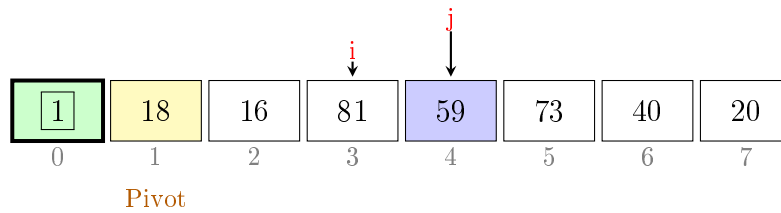
**Schritt 20:**  $\text{array}[6]=40 > \text{Pivot}=18 \rightarrow$  j nach links



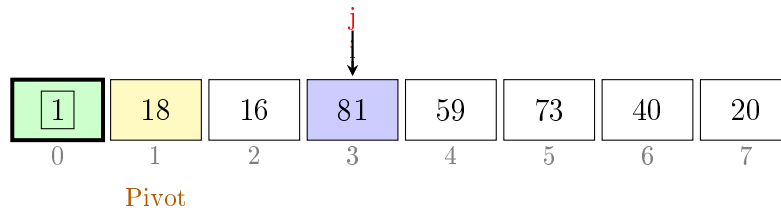
**Schritt 21:**  $\text{array}[5]=73 > \text{Pivot}=18 \rightarrow$  j nach links



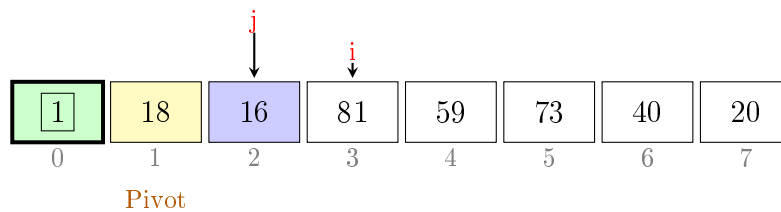
**Schritt 22:**  $\text{array}[4]=59 > \text{Pivot}=18 \rightarrow j$  nach links



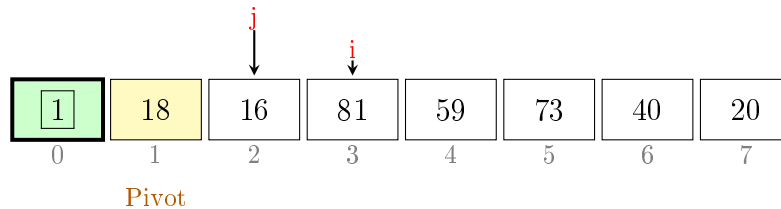
**Schritt 23:**  $\text{array}[3]=81 > \text{Pivot}=18 \rightarrow j$  nach links



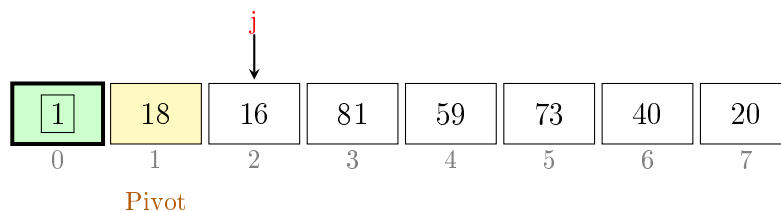
**Schritt 24:**  $\text{array}[2]=16 \leq \text{Pivot}=18 \rightarrow j$  stoppt



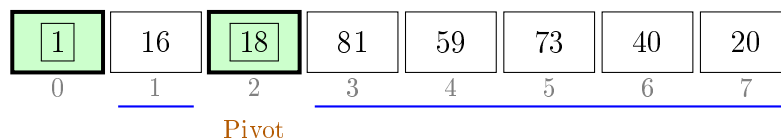
**Schritt 25:**  $i=3 \ j=2 \rightarrow$  Zeiger gekreuzt, Partition beenden



**Schritt 26:** Platziere Pivot: Tausche  $\text{array}[1]=18 \leftrightarrow \text{array}[2]=16$

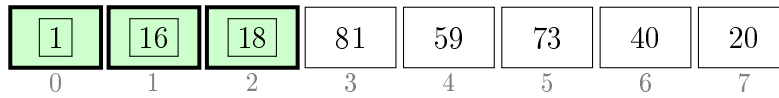


**Schritt 27:** Partition [1..7] fertig: Pivot 18 an Position 2

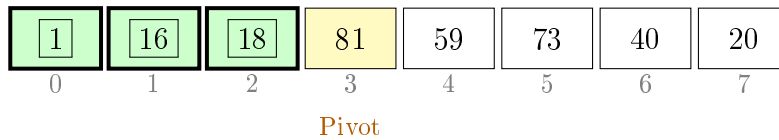


=== Rekursionsebene 2: Partitioniere [3..7]; Basisfall: [1] (Basisfall) ===

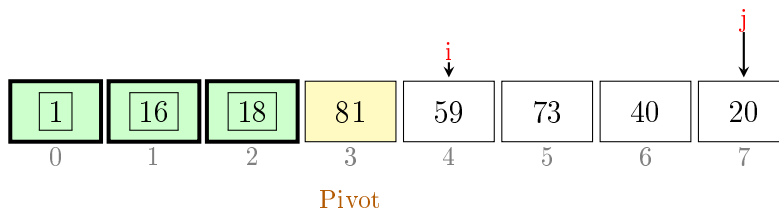
**Schritt 28:** quicksort(1, 1):  $l == r$ , einzelnes Element  $a[1]=16$  ist sortiert



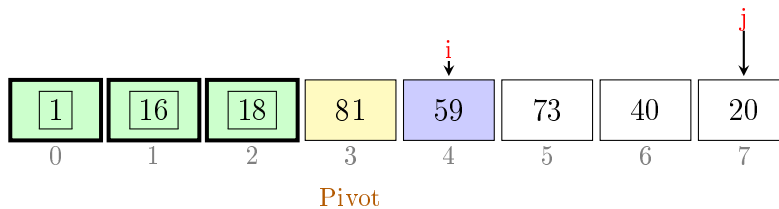
**Schritt 29:** Partition [3..7]: Pivot = 81 (erstes Element)



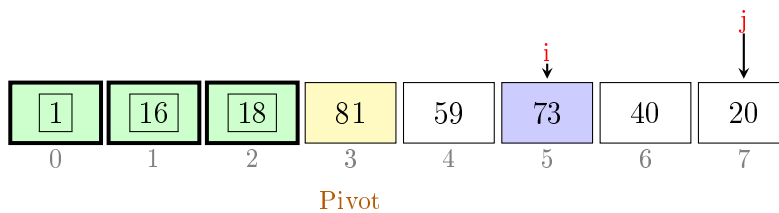
**Schritt 30:** Initialisiere Zeiger:  $i=4$ ,  $j=7$



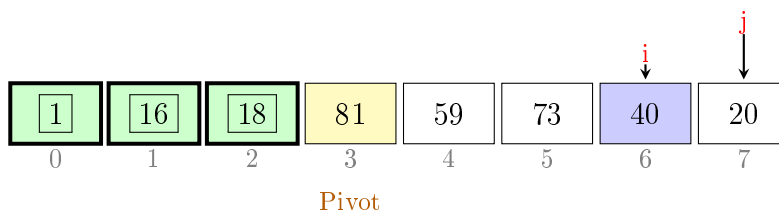
**Schritt 31:**  $array[4]=59$  Pivot=81  $\rightarrow i$  nach rechts



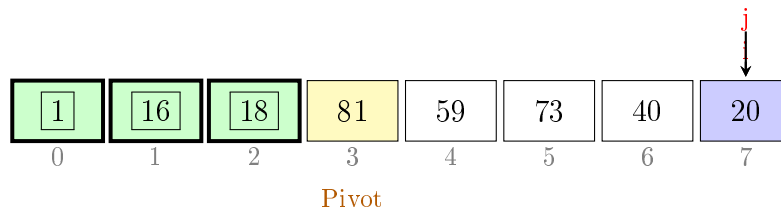
**Schritt 32:**  $array[5]=73$  Pivot=81  $\rightarrow i$  nach rechts



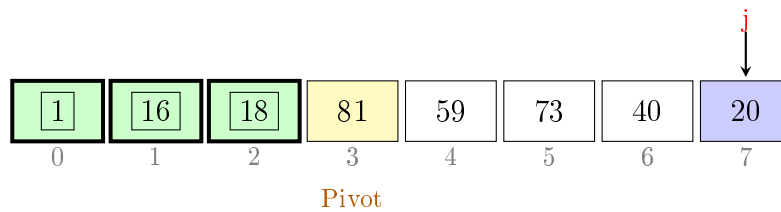
**Schritt 33:**  $array[6]=40$  Pivot=81  $\rightarrow i$  nach rechts



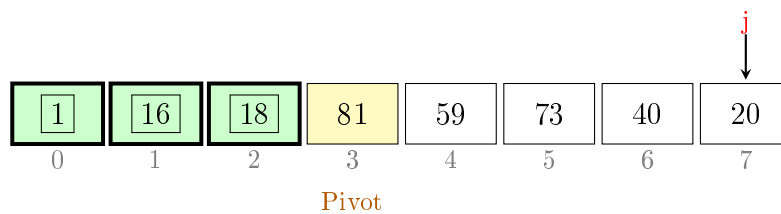
**Schritt 34:**  $array[7]=20$  Pivot=81  $\rightarrow i$  nach rechts



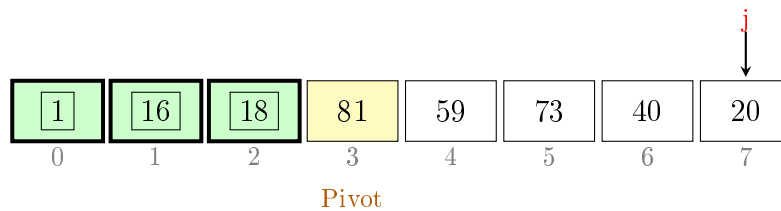
**Schritt 35:**  $\text{array}[7]=20$  Pivot=81  $\rightarrow$  j stoppt



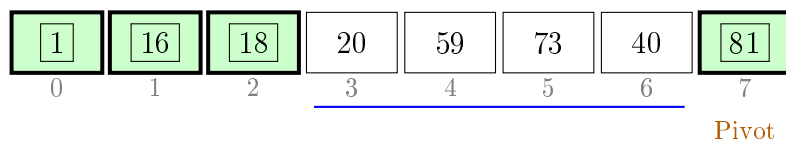
**Schritt 36:**  $i=8$   $j=7 \rightarrow$  Zeiger gekreuzt, Partition beenden



**Schritt 37:** Platziere Pivot: Tausche  $\text{array}[3]=81 \leftrightarrow \text{array}[7]=20$

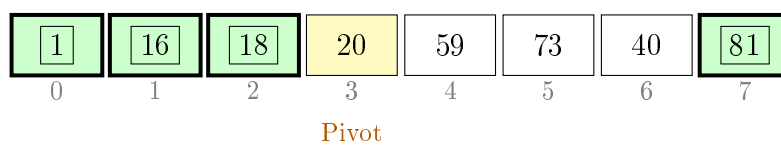


**Schritt 38:** Partition [3..7] fertig: Pivot 81 an Position 7

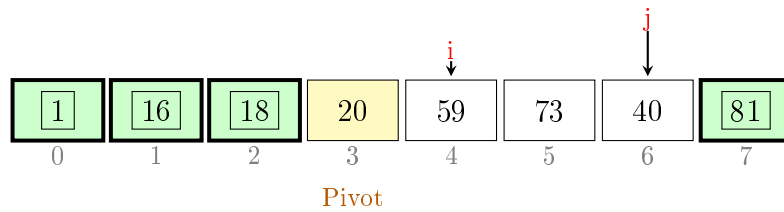


=== Rekursionsebene 3: Partitioniere [3..6] ===

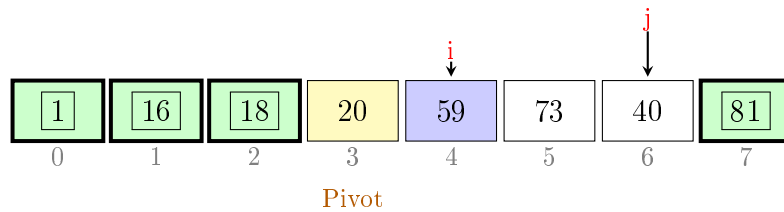
**Schritt 39:** Partition [3..6]: Pivot = 20 (erstes Element)



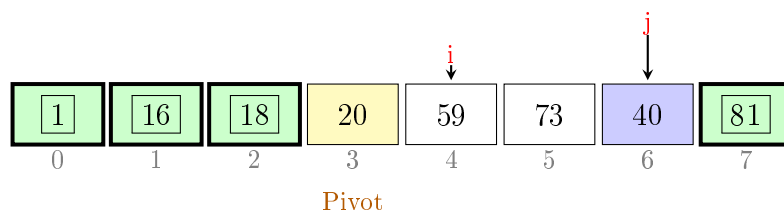
**Schritt 40:** Initialisiere Zeiger:  $i=4$ ,  $j=6$



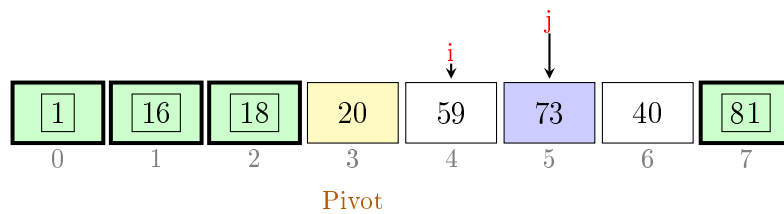
**Schritt 41:**  $\text{array}[4]=59 > \text{Pivot}=20 \rightarrow i$  stoppt



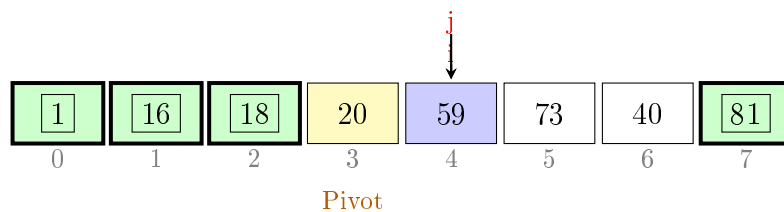
**Schritt 42:**  $\text{array}[6]=40 > \text{Pivot}=20 \rightarrow j$  nach links



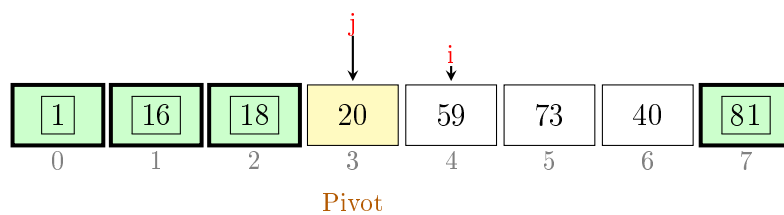
**Schritt 43:**  $\text{array}[5]=73 > \text{Pivot}=20 \rightarrow j$  nach links



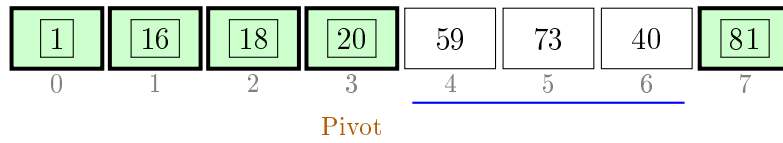
**Schritt 44:**  $\text{array}[4]=59 > \text{Pivot}=20 \rightarrow j$  nach links



**Schritt 45:**  $i=4$   $j=3 \rightarrow$  Zeiger gekreuzt, Partition beenden

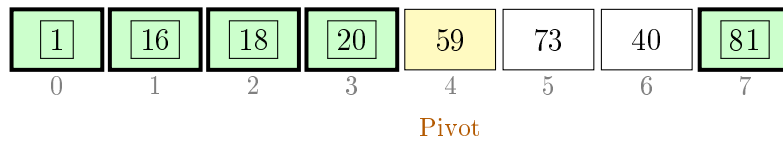


**Schritt 46:** Partition [3..6] fertig: Pivot 20 an Position 3

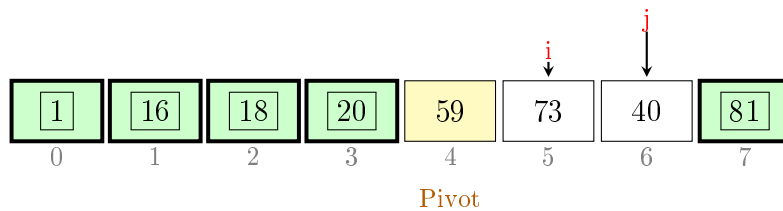


=== Rekursionsebene 4: Partitioniere [4..6] ===

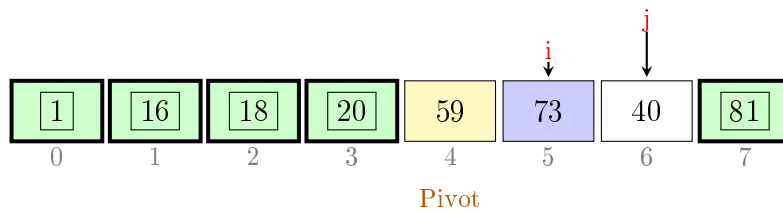
**Schritt 47:** Partition [4..6]: Pivot = 59 (erstes Element)



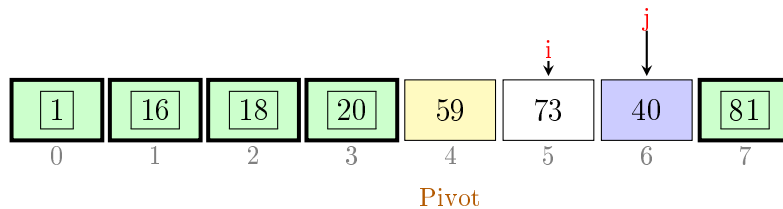
**Schritt 48:** Initialisiere Zeiger: i=5, j=6



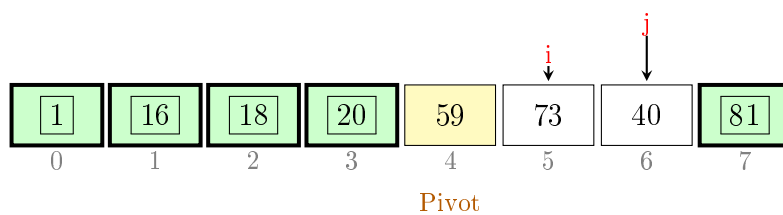
**Schritt 49:** array[5]=73 > Pivot=59 → i stoppt



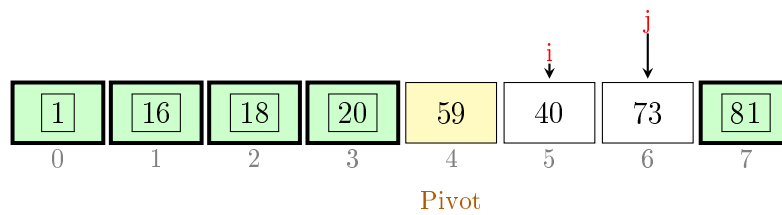
**Schritt 50:** array[6]=40 Pivot=59 → j stoppt



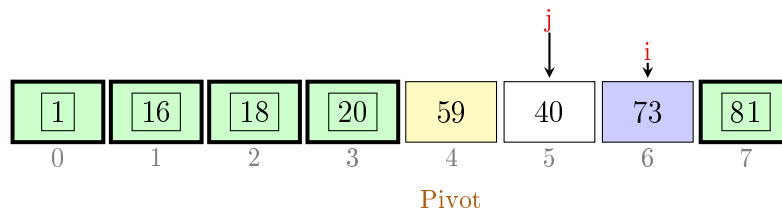
**Schritt 51:** Tausche array[5]=73 ↔ array[6]=40



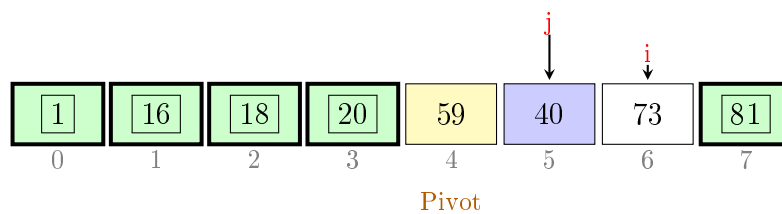
**Schritt 52:** Nach Tausch:  $\text{array}[5]=40$ ,  $\text{array}[6]=73$



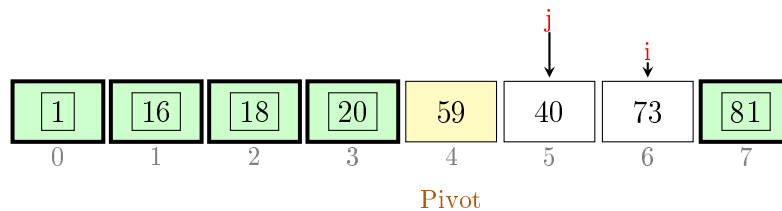
**Schritt 53:**  $\text{array}[6]=73 > \text{Pivot}=59 \rightarrow i$  stoppt



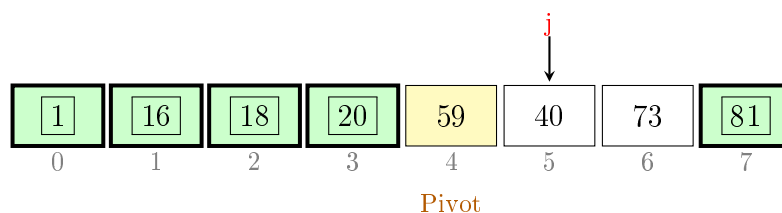
**Schritt 54:**  $\text{array}[5]=40$  Pivot=59  $\rightarrow j$  stoppt



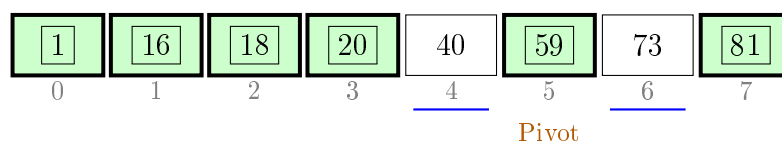
**Schritt 55:**  $i=6$   $j=5 \rightarrow$  Zeiger gekreuzt, Partition beenden



**Schritt 56:** Platziere Pivot: Tausche  $\text{array}[4]=59 \leftrightarrow \text{array}[5]=40$

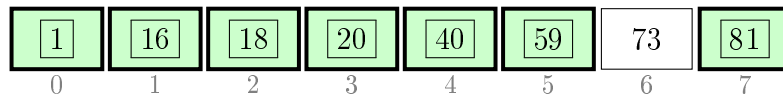


**Schritt 57:** Partition [4..6] fertig: Pivot 59 an Position 5

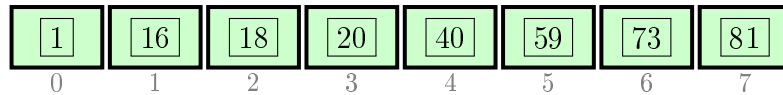


=== Rekursionsebene 5: Basisfall: [4] (Basisfall), [6] (Basisfall) ===

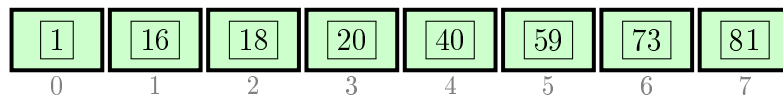
**Schritt 58:** quicksort(4, 4):  $l == r$ , einzelnes Element  $a[4]=40$  ist sortiert



**Schritt 59:** quicksort(6, 6):  $l == r$ , einzelnes Element  $a[6]=73$  ist sortiert



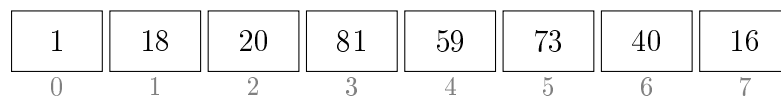
**Schritt 60:** Sortierung abgeschlossen



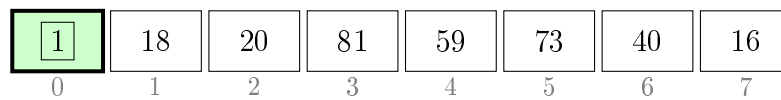
### Übersicht der Rekursionsschritte:

Jede Zeile zeigt den Zustand nach einem Rekursionsschritt. Unterstrichene Teillisten werden im nächsten Schritt parallel verarbeitet. Umrahmte Werte haben ihre endgültige Position erreicht.

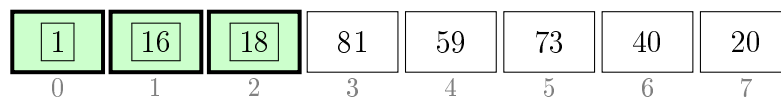
**Zustand 0:** Ausgangszustand



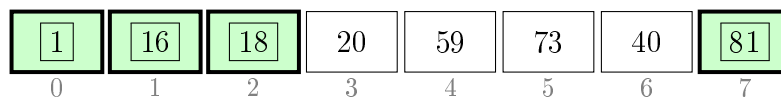
**Zustand 1:** Nach Rekursionsschritt 1



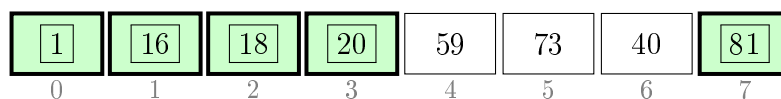
**Zustand 2:** Nach Rekursionsschritt 2



**Zustand 3:** Nach Rekursionsschritt 3



**Zustand 4:** Nach Rekursionsschritt 4



**Zustand 5:** Nach Rekursionsschritt 5

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|   |    |    |    |    |    |    |    |
|---|----|----|----|----|----|----|----|
| 1 | 16 | 18 | 20 | 40 | 59 | 73 | 81 |
| 0 | 1  | 2  | 3  | 4  | 5  | 6  | 7  |

---

**Antworten:**

1. Nach erster Partition: [1, 18, 20, 81, 59, 73, 40, 16]
2. Nach letzter Partition: [1, 16, 18, 20, 40, 59, 73, 81]
3. Anzahl Vergleiche: **29**

**Endergebnis:** [1, 16, 18, 20, 40, 59, 73, 81]

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## Aufgabe 6 – Schwierigkeit: ★★★★★

Quicksort mit 10 Elementen (anspruchsvoll)

Gegeben sei folgende Liste, die in einem Array abgelegt ist:

**28, 10, 76, 52, 98, 18, 96, 85, 86, 70**

Die Liste soll mit dem Standard-Quicksort-Verfahren (keine Variante) in-situ sortiert werden. Als Pivot-Element soll immer das **erste Element** der zu sortierenden (Teil-)Liste genommen werden.

**Fragen:**

1. Geben Sie den Zustand des Arrays nach der **ersten** Partition an.
2. Geben Sie den Zustand des Arrays nach der **letzten** Partition an.
3. Wie viele Vergleiche werden insgesamt durchgeführt?

### Lösung:

**Schritt-für-Schritt-Lösung:**

**Schritt 0:** Ausgangszustand

|    |    |    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|----|----|
| 28 | 10 | 76 | 52 | 98 | 18 | 96 | 85 | 86 | 70 |
| 0  | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  |

=== Rekursionsebene 0: Partitioniere [0..9] ===

**Schritt 1:** Partition [0..9]: Pivot = 28 (erstes Element)

|    |    |    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|----|----|
| 28 | 10 | 76 | 52 | 98 | 18 | 96 | 85 | 86 | 70 |
| 0  | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  |

Pivot

**Schritt 2:** Initialisiere Zeiger: i=1, j=9

|    |    |    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|----|----|
| 28 | 10 | 76 | 52 | 98 | 18 | 96 | 85 | 86 | 70 |
| 0  | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  |

Pivot

i ↓ j ↓

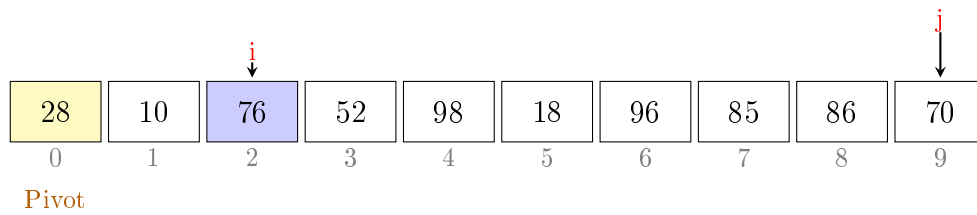
**Schritt 3:** array[1]=10 Pivot=28 → i nach rechts

|    |    |    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|----|----|
| 28 | 10 | 76 | 52 | 98 | 18 | 96 | 85 | 86 | 70 |
| 0  | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  |

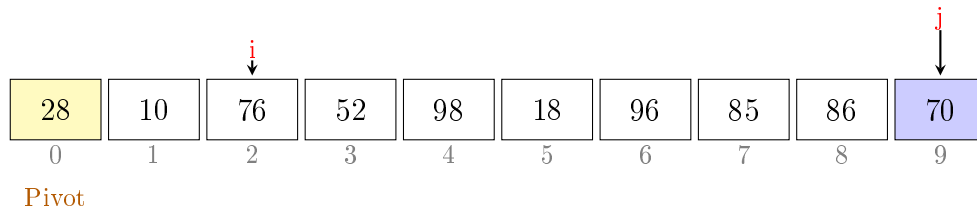
Pivot

i ↓ j ↓

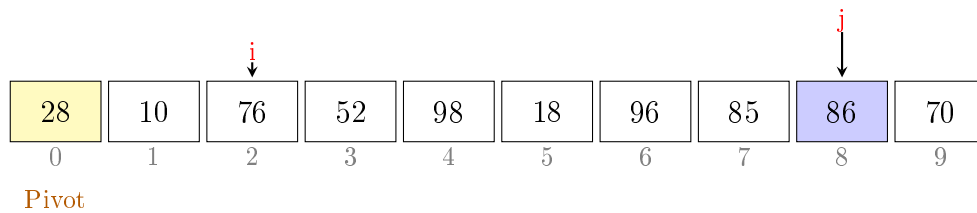
**Schritt 4:**  $\text{array}[2]=76 > \text{Pivot}=28 \rightarrow i$  stoppt



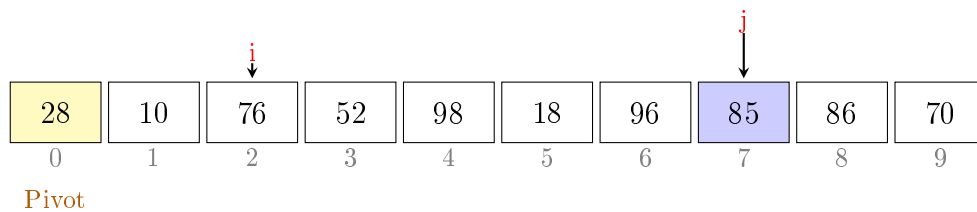
**Schritt 5:**  $\text{array}[9]=70 > \text{Pivot}=28 \rightarrow j$  nach links



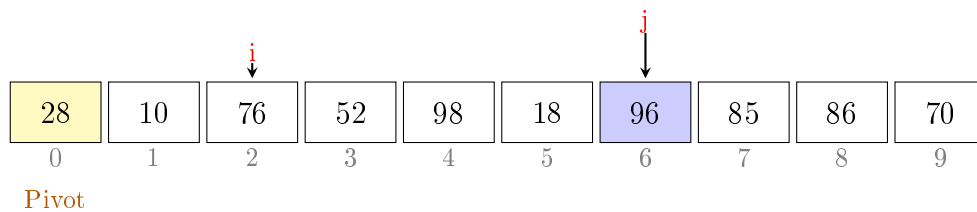
**Schritt 6:**  $\text{array}[8]=86 > \text{Pivot}=28 \rightarrow j$  nach links



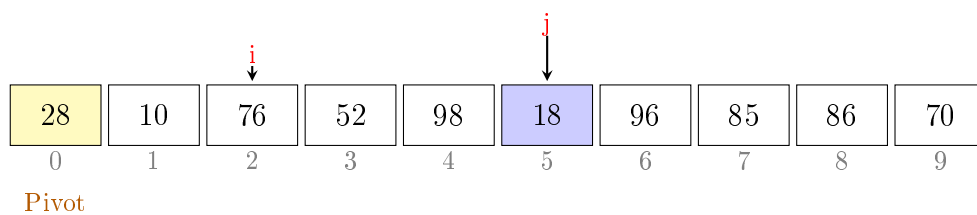
**Schritt 7:**  $\text{array}[7]=85 > \text{Pivot}=28 \rightarrow j$  nach links



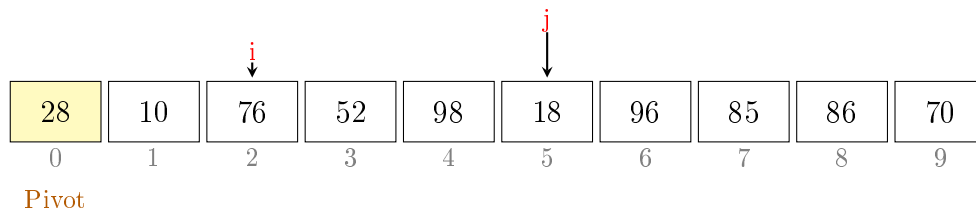
**Schritt 8:**  $\text{array}[6]=96 > \text{Pivot}=28 \rightarrow j$  nach links



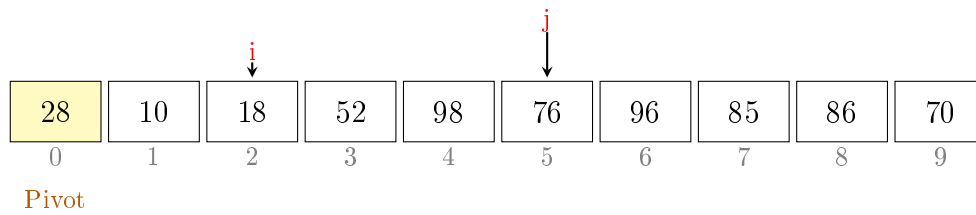
**Schritt 9:**  $\text{array}[5]=18$  Pivot=28  $\rightarrow j$  stoppt



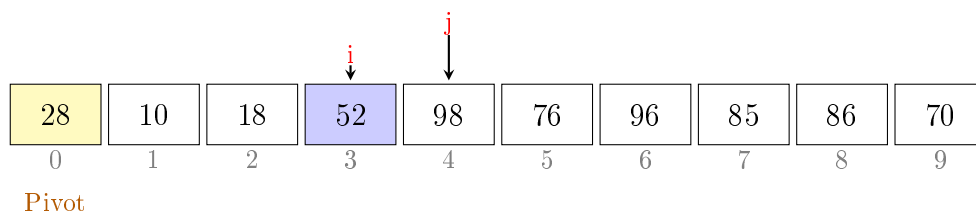
**Schritt 10:** Tausche  $\text{array}[2]=76 \leftrightarrow \text{array}[5]=18$



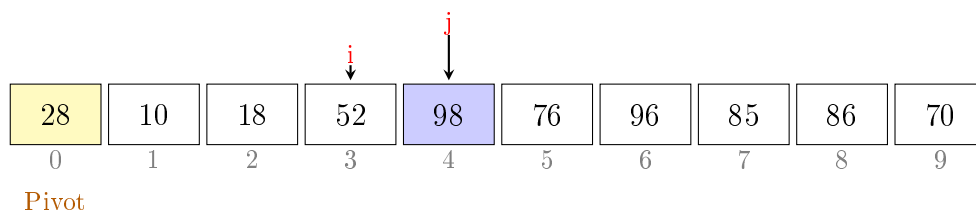
**Schritt 11:** Nach Tausch:  $\text{array}[2]=18$ ,  $\text{array}[5]=76$



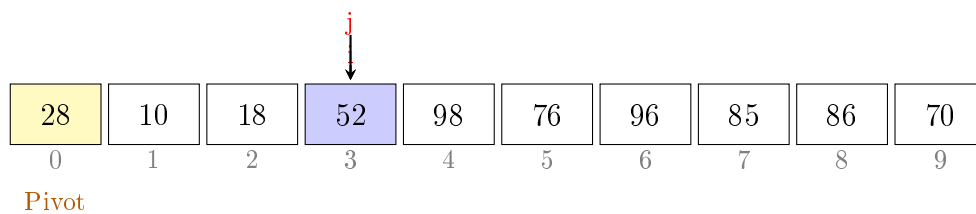
**Schritt 12:**  $\text{array}[3]=52 > \text{Pivot}=28 \rightarrow i$  stoppt



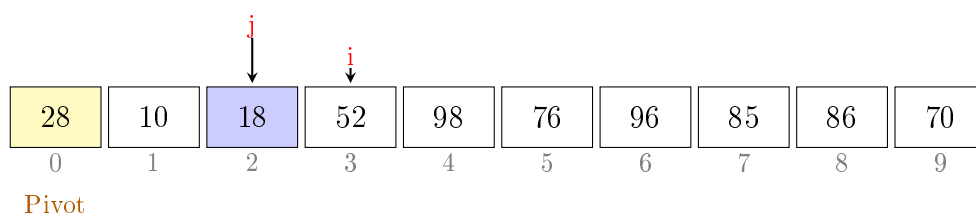
**Schritt 13:**  $\text{array}[4]=98 > \text{Pivot}=28 \rightarrow j$  nach links



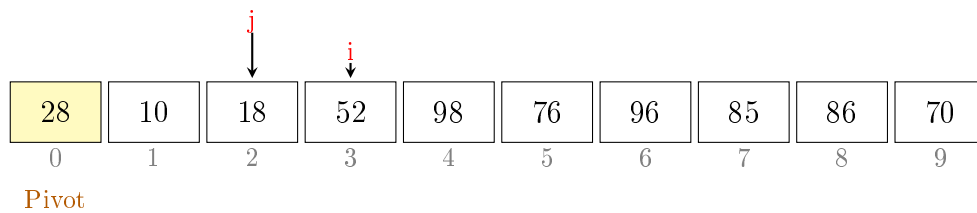
**Schritt 14:**  $\text{array}[3]=52 > \text{Pivot}=28 \rightarrow j$  nach links



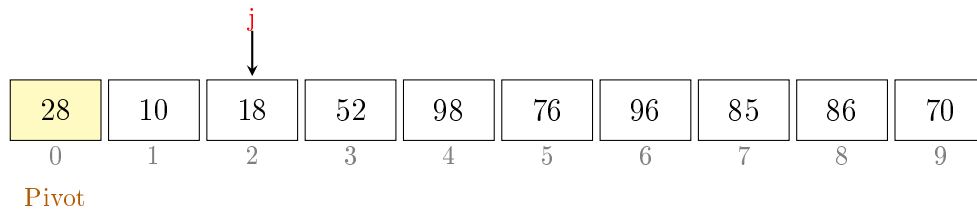
**Schritt 15:**  $\text{array}[2]=18$  Pivot=28  $\rightarrow j$  stoppt



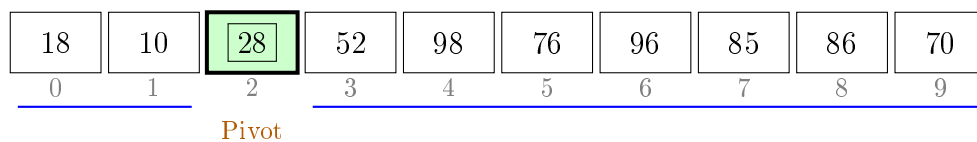
**Schritt 16:**  $i=3$   $j=2 \rightarrow$  Zeiger gekreuzt, Partition beenden



**Schritt 17:** Platziere Pivot: Tausche  $\text{array}[0]=28 \leftrightarrow \text{array}[2]=18$

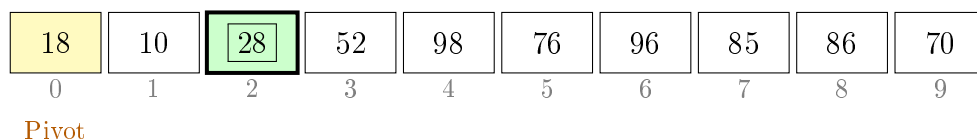


**Schritt 18:** Partition  $[0..9]$  fertig: Pivot 28 an Position 2

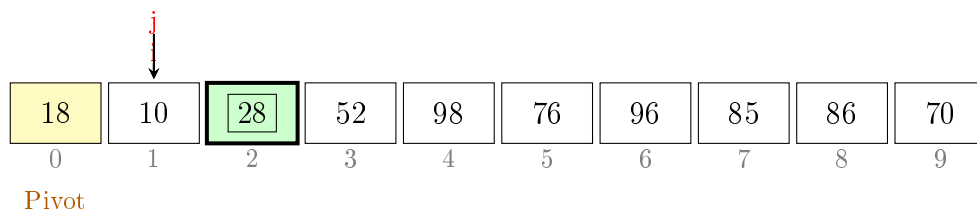


=== Rekursionsebene 1: Partitioniere  $[0..1]$ ,  $[3..9]$  ===

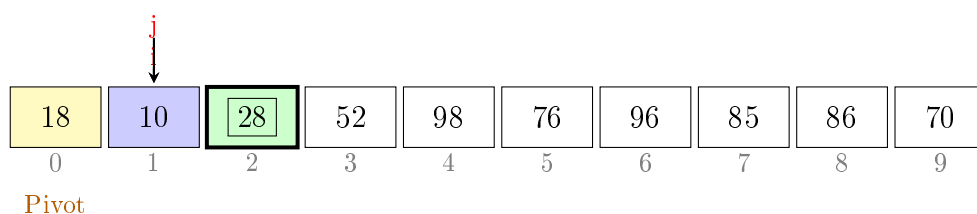
**Schritt 19:** Partition  $[0..1]$ : Pivot = 18 (erstes Element)



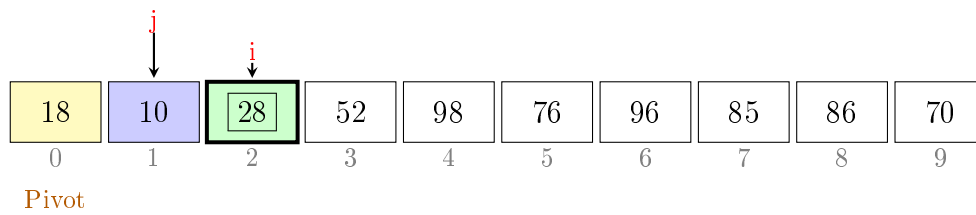
**Schritt 20:** Initialisiere Zeiger:  $i=1$ ,  $j=1$



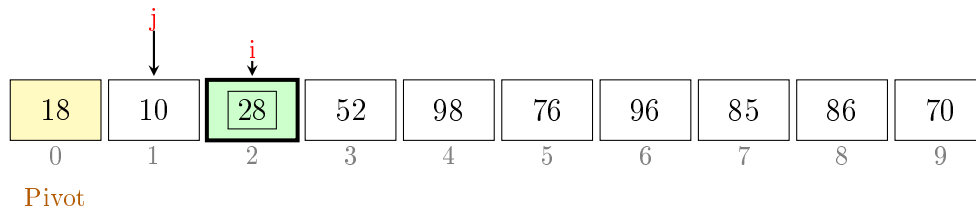
**Schritt 21:**  $\text{array}[1]=10$  Pivot=18  $\rightarrow$  i nach rechts



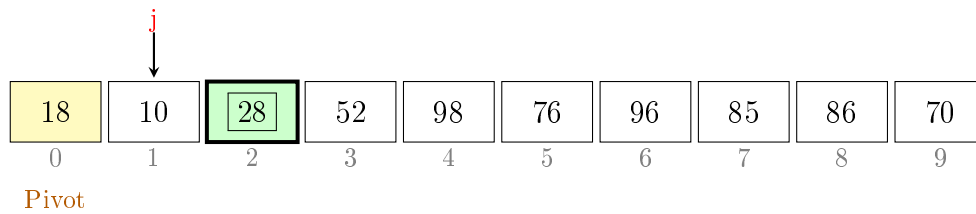
**Schritt 22:**  $\text{array}[1]=10$  Pivot=18  $\rightarrow j$  stoppt



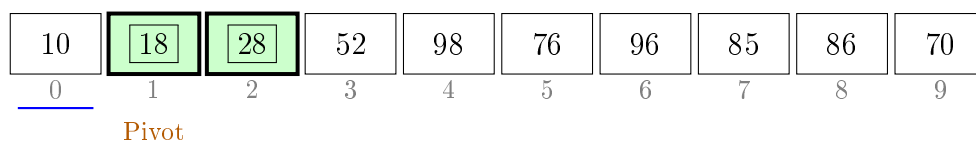
**Schritt 23:**  $i=2$   $j=1$   $\rightarrow$  Zeiger gekreuzt, Partition beenden



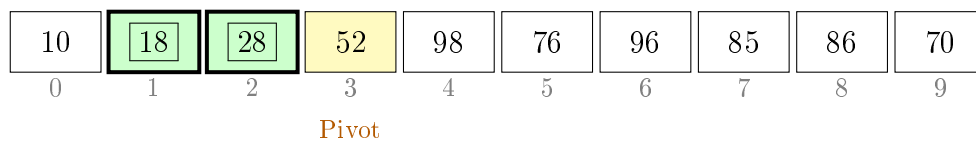
**Schritt 24:** Platziere Pivot: Tausche  $\text{array}[0]=18 \leftrightarrow \text{array}[1]=10$



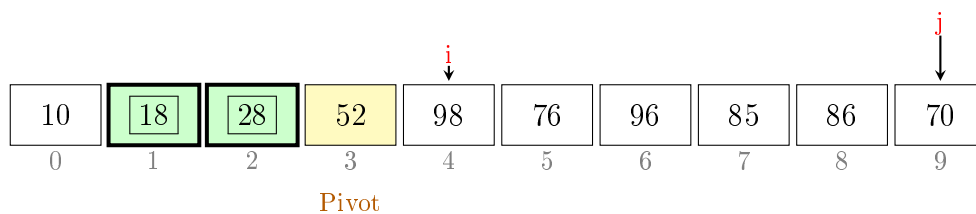
**Schritt 25:** Partition  $[0..1]$  fertig: Pivot 18 an Position 1



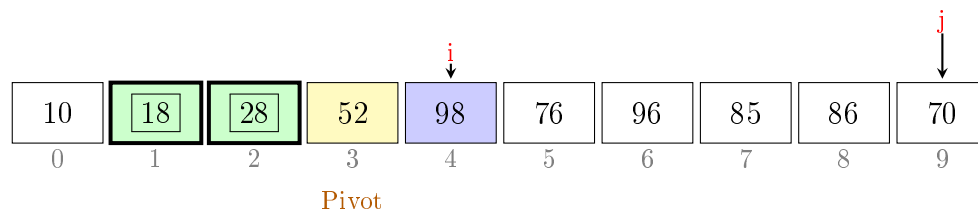
**Schritt 26:** Partition  $[3..9]$ : Pivot = 52 (erstes Element)



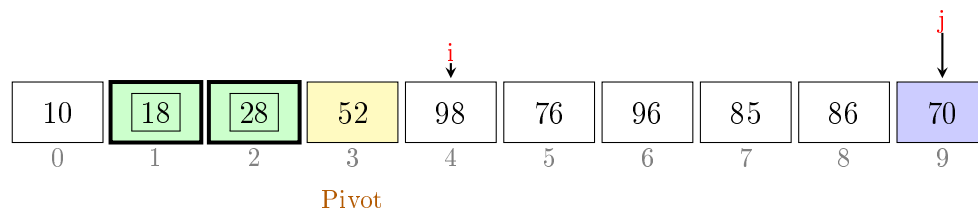
**Schritt 27:** Initialisiere Zeiger:  $i=4$ ,  $j=9$



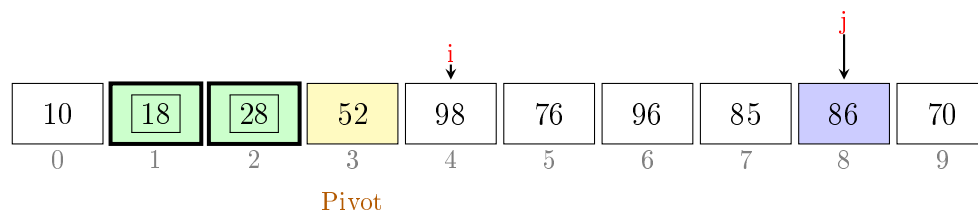
**Schritt 28:**  $\text{array}[4]=98 > \text{Pivot}=52 \rightarrow i$  stoppt



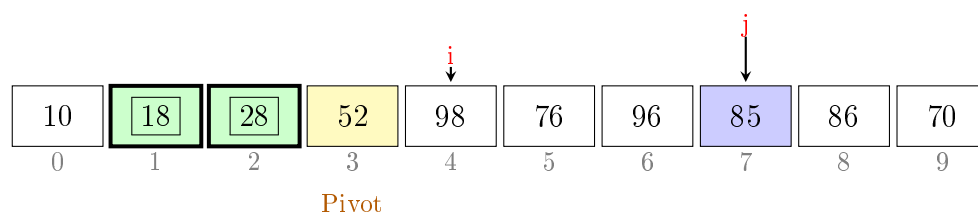
**Schritt 29:**  $\text{array}[9]=70 > \text{Pivot}=52 \rightarrow j$  nach links



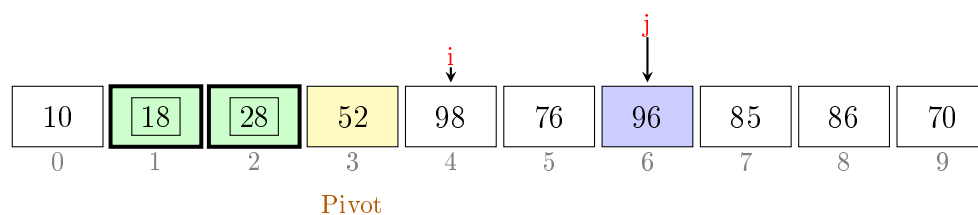
**Schritt 30:**  $\text{array}[8]=86 > \text{Pivot}=52 \rightarrow j$  nach links



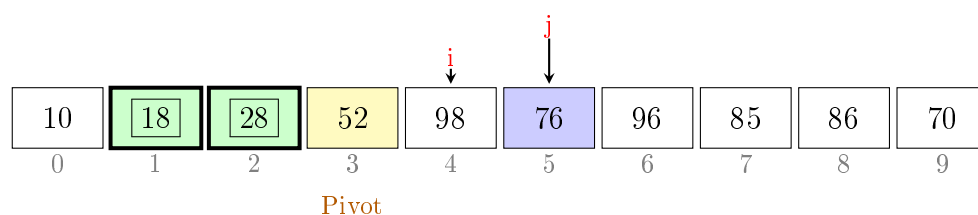
**Schritt 31:**  $\text{array}[7]=85 > \text{Pivot}=52 \rightarrow j$  nach links



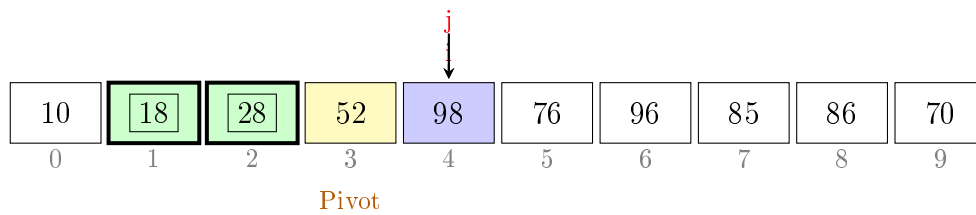
**Schritt 32:**  $\text{array}[6]=96 > \text{Pivot}=52 \rightarrow j$  nach links



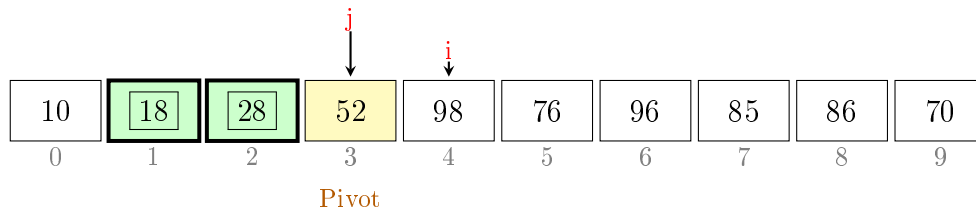
**Schritt 33:**  $\text{array}[5]=76 > \text{Pivot}=52 \rightarrow j$  nach links



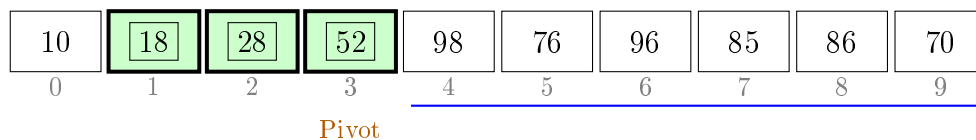
**Schritt 34:**  $\text{array}[4]=98 > \text{Pivot}=52 \rightarrow j$  nach links



**Schritt 35:**  $i=4 \ j=3 \rightarrow$  Zeiger gekreuzt, Partition beenden

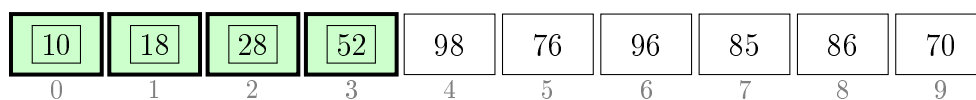


**Schritt 36:** Partition [3..9] fertig: Pivot 52 an Position 3

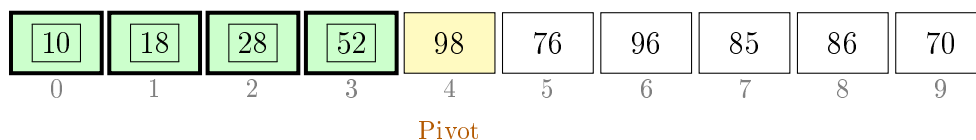


=== Rekursionsebene 2: Partitioniere [4..9]; Basisfall: [0] (Basisfall) ===

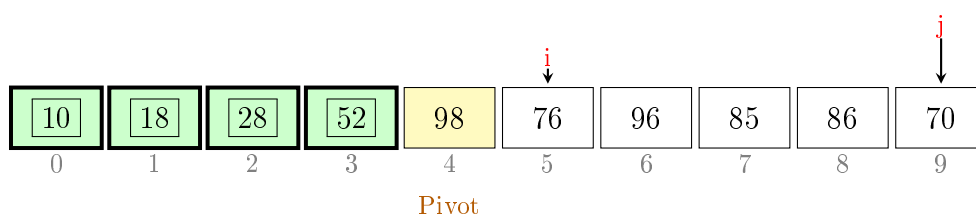
**Schritt 37:** quicksort(0, 0):  $l == r$ , einzelnes Element  $a[0]=10$  ist sortiert



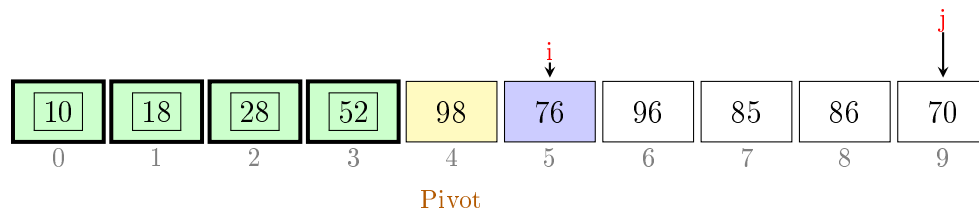
**Schritt 38:** Partition [4..9]: Pivot = 98 (erstes Element)



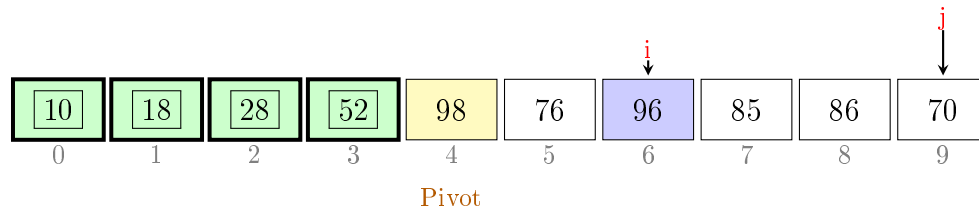
**Schritt 39:** Initialisiere Zeiger:  $i=5, j=9$



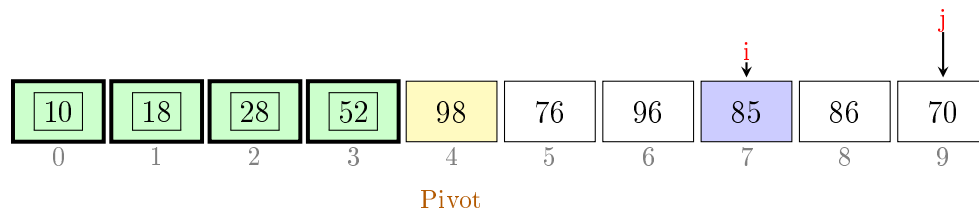
**Schritt 40:** array[5]=76 Pivot=98 → i nach rechts



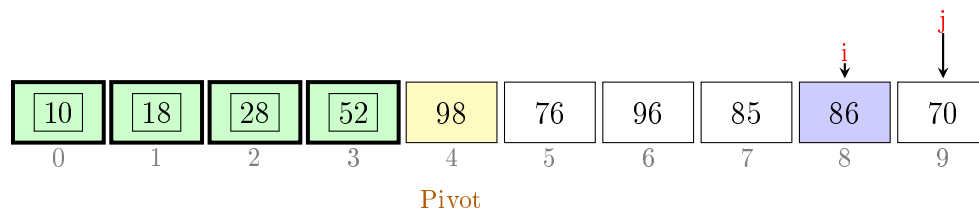
**Schritt 41:** array[6]=96 Pivot=98 → i nach rechts



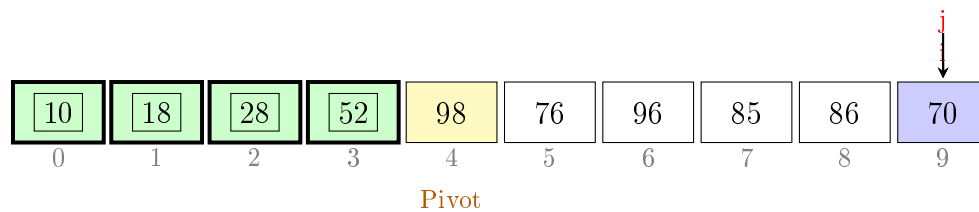
**Schritt 42:** array[7]=85 Pivot=98 → i nach rechts



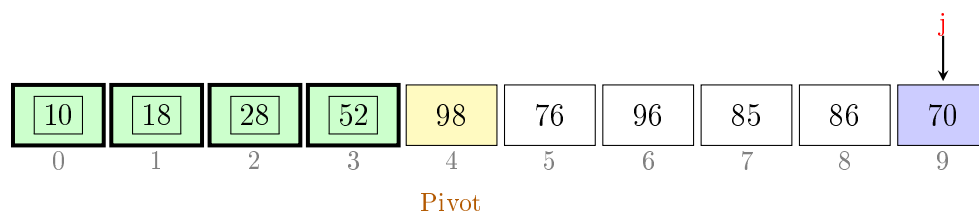
**Schritt 43:** array[8]=86 Pivot=98 → i nach rechts



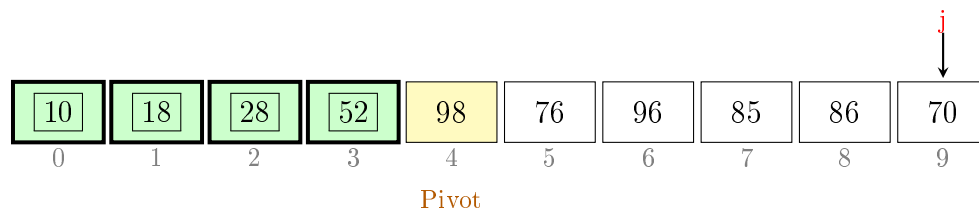
**Schritt 44:** array[9]=70 Pivot=98 → i nach rechts



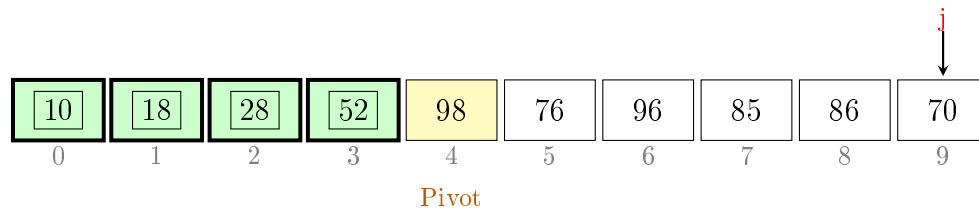
**Schritt 45:** array[9]=70 Pivot=98 → j stoppt



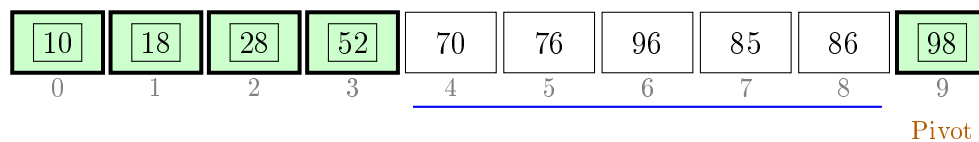
**Schritt 46:**  $i=10$   $j=9 \rightarrow$  Zeiger gekreuzt, Partition beenden



**Schritt 47:** Platziere Pivot: Tausche  $\text{array}[4]=98 \leftrightarrow \text{array}[9]=70$

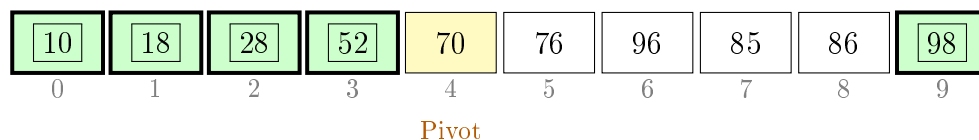


**Schritt 48:** Partition  $[4..9]$  fertig: Pivot 98 an Position 9

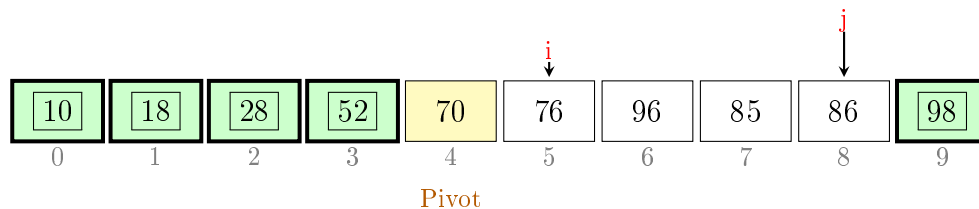


=== Rekursionsebene 3: Partitioniere  $[4..8]$  ===

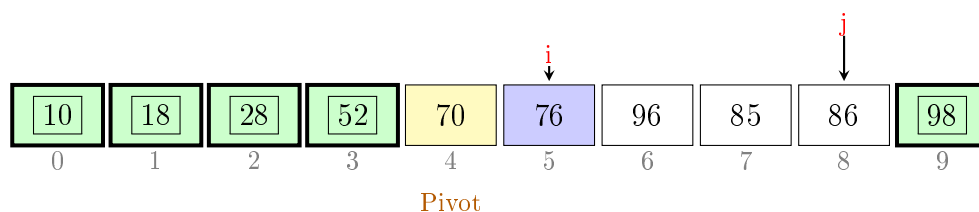
**Schritt 49:** Partition  $[4..8]$ : Pivot = 70 (erstes Element)



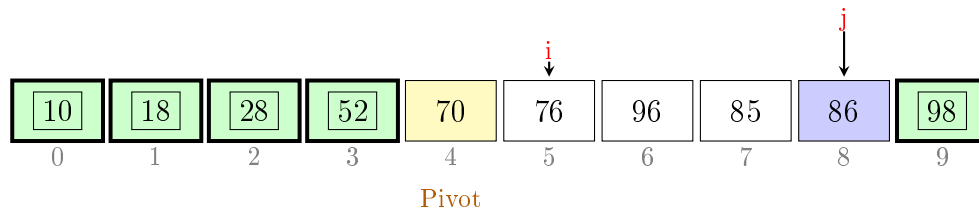
**Schritt 50:** Initialisiere Zeiger:  $i=5$ ,  $j=8$



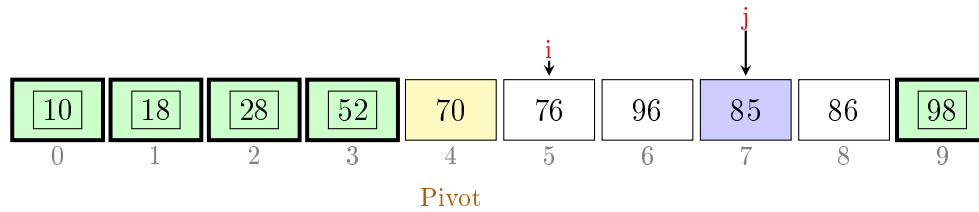
**Schritt 51:**  $\text{array}[5]=76 > \text{Pivot}=70 \rightarrow i$  stoppt



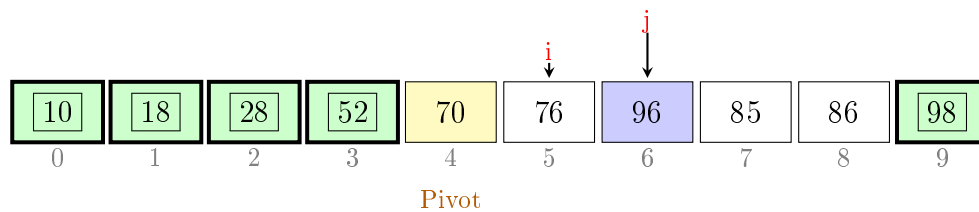
**Schritt 52:**  $\text{array}[8]=86 > \text{Pivot}=70 \rightarrow j$  nach links



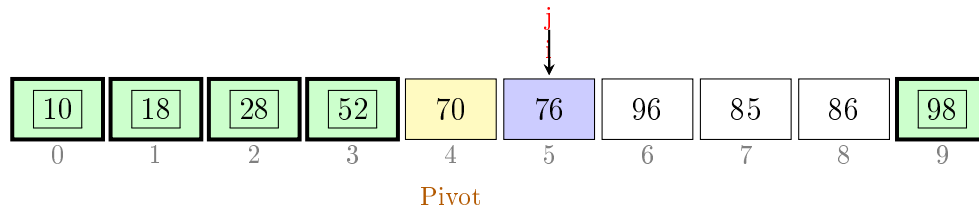
**Schritt 53:**  $\text{array}[7]=85 > \text{Pivot}=70 \rightarrow j$  nach links



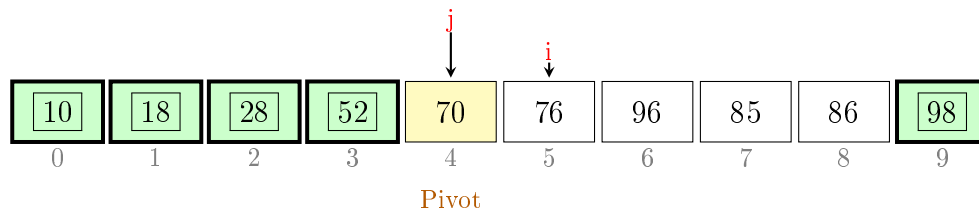
**Schritt 54:**  $\text{array}[6]=96 > \text{Pivot}=70 \rightarrow j$  nach links



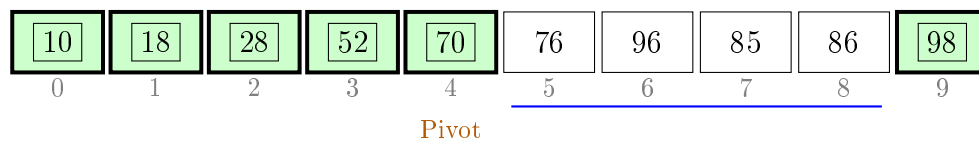
**Schritt 55:**  $\text{array}[5]=76 > \text{Pivot}=70 \rightarrow j$  nach links



**Schritt 56:**  $i=5 \ j=4 \rightarrow$  Zeiger gekreuzt, Partition beenden

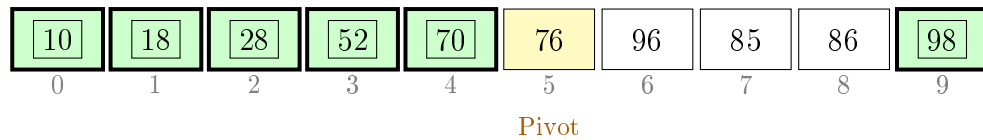


**Schritt 57:** Partition [4..8] fertig: Pivot 70 an Position 4

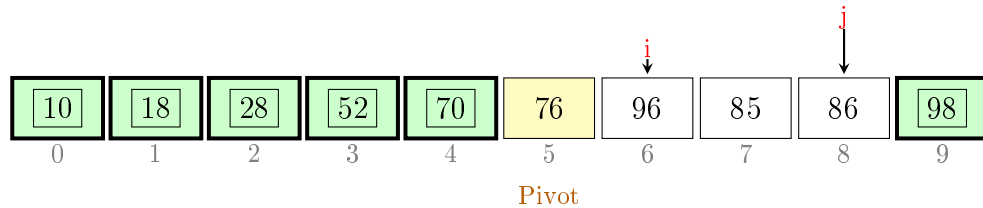


=== Rekursionsebene 4: Partitioniere [5..8] ===

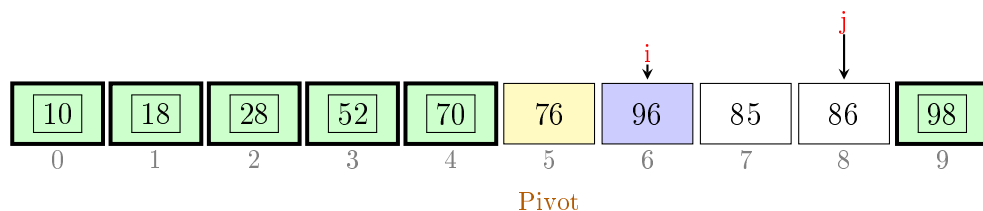
**Schritt 58:** Partition [5..8]: Pivot = 76 (erstes Element)



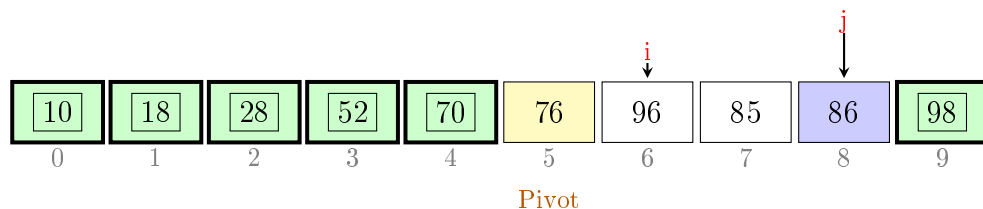
**Schritt 59:** Initialisiere Zeiger: i=6, j=8



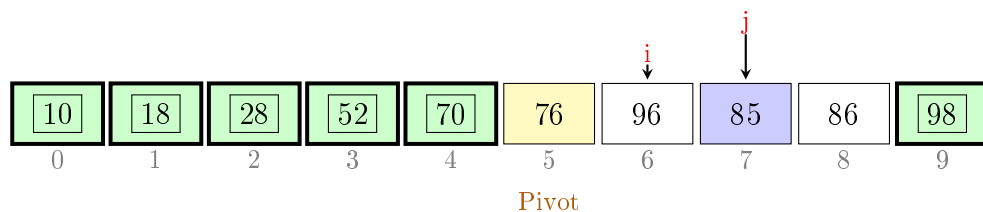
**Schritt 60:** array[6]=96 > Pivot=76 → i stoppt



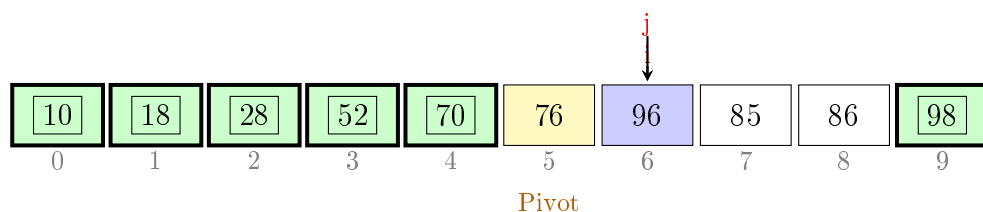
**Schritt 61:** array[8]=86 > Pivot=76 → j nach links



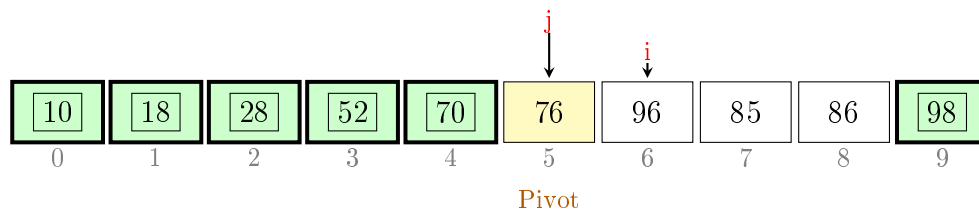
**Schritt 62:** array[7]=85 > Pivot=76 → j nach links



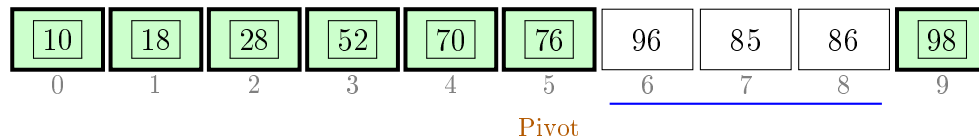
**Schritt 63:** array[6]=96 > Pivot=76 → j nach links



**Schritt 64:**  $i=6$   $j=5 \rightarrow$  Zeiger gekreuzt, Partition beenden

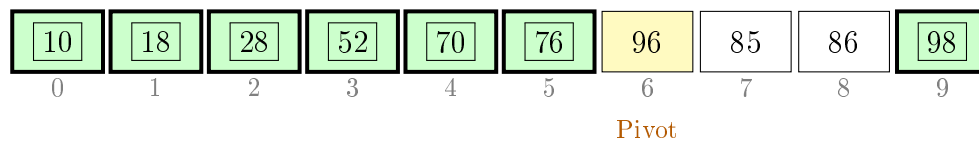


**Schritt 65:** Partition [5..8] fertig: Pivot 76 an Position 5

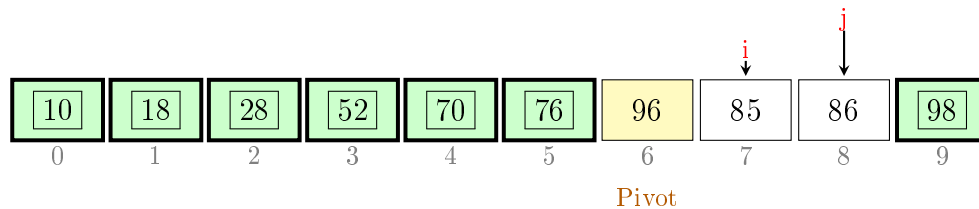


=== Rekursionsebene 5: Partitioniere [6..8] ===

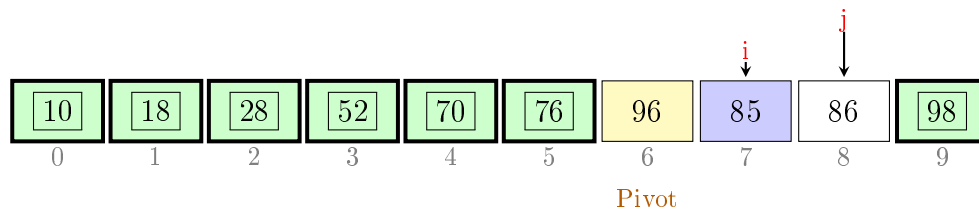
**Schritt 66:** Partition [6..8]: Pivot = 96 (erstes Element)



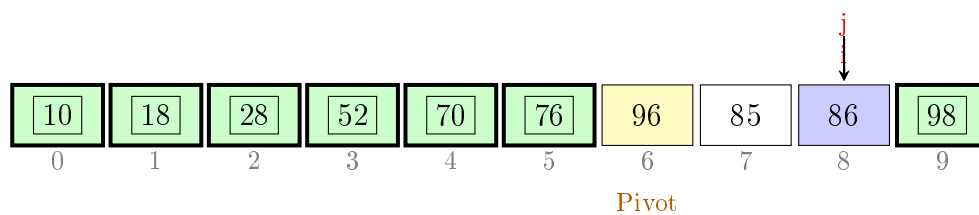
**Schritt 67:** Initialisiere Zeiger:  $i=7$ ,  $j=8$



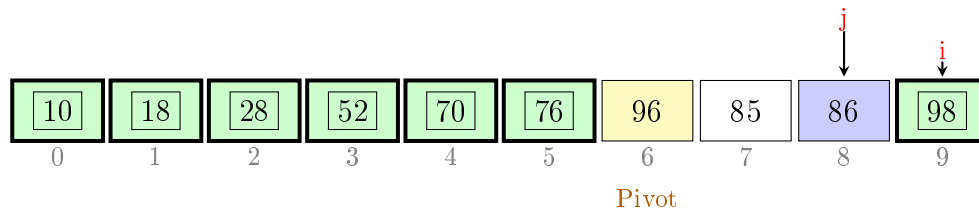
**Schritt 68:**  $\text{array}[7]=85$  Pivot=96  $\rightarrow i$  nach rechts



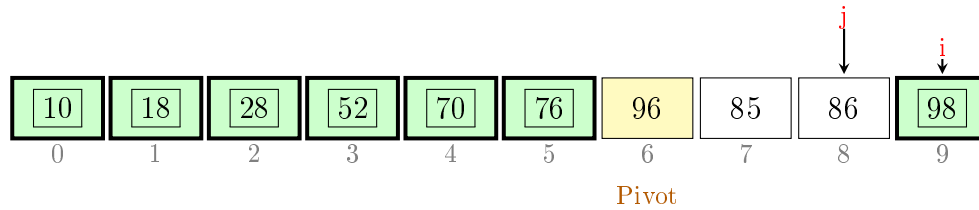
**Schritt 69:**  $\text{array}[8]=86$  Pivot=96  $\rightarrow i$  nach rechts



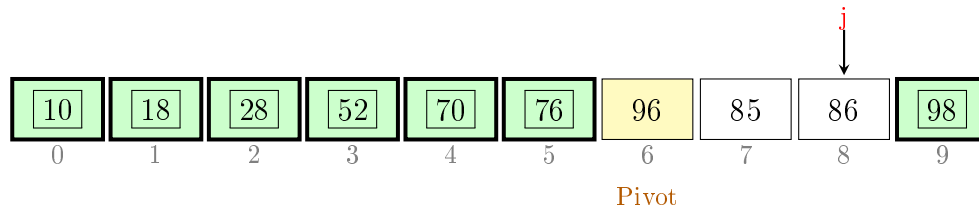
**Schritt 70:** array[8]=86 Pivot=96 → j stoppt



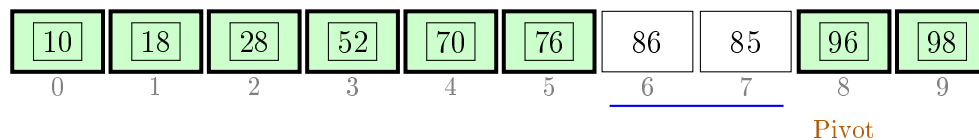
**Schritt 71:** i=9 j=8 → Zeiger gekreuzt, Partition beenden



**Schritt 72:** Platziere Pivot: Tausche array[6]=96 ↔ array[8]=86

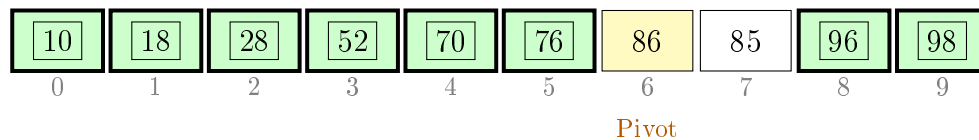


**Schritt 73:** Partition [6..8] fertig: Pivot 96 an Position 8

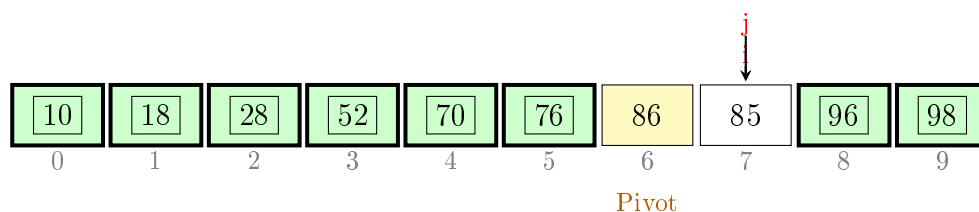


=== Rekursionsebene 6: Partitioniere [6..7] ===

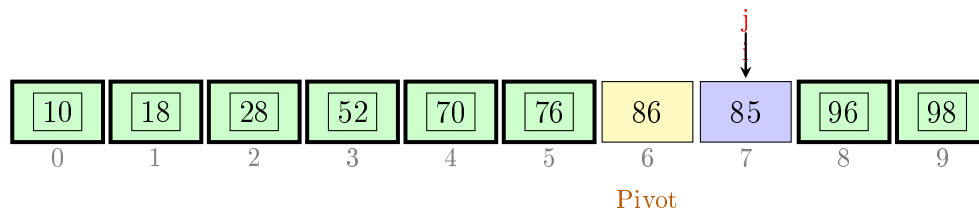
**Schritt 74:** Partition [6..7]: Pivot = 86 (erstes Element)



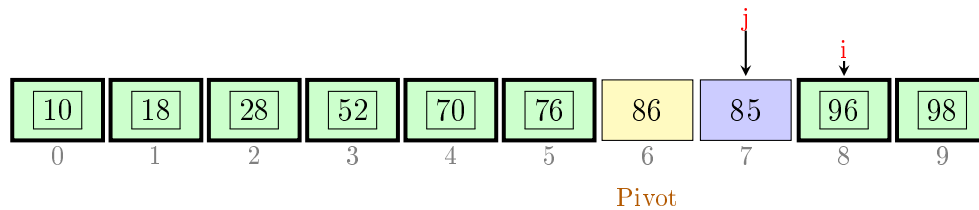
**Schritt 75:** Initialisiere Zeiger: i=7, j=7



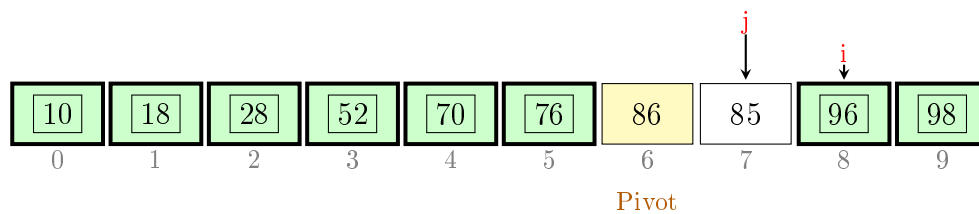
**Schritt 76:** array[7]=85 Pivot=86  $\rightarrow$  i nach rechts



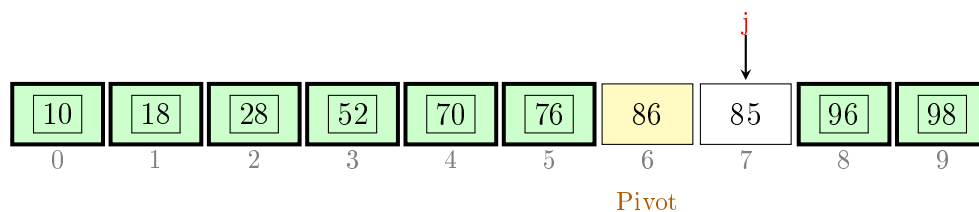
**Schritt 77:** array[7]=85 Pivot=86  $\rightarrow$  j stoppt



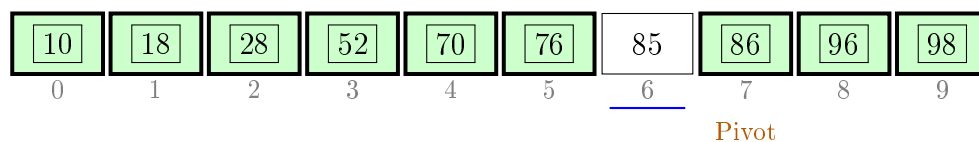
**Schritt 78:** i=8 j=7  $\rightarrow$  Zeiger gekreuzt, Partition beenden



**Schritt 79:** Platziere Pivot: Tausche array[6]=86  $\leftrightarrow$  array[7]=85

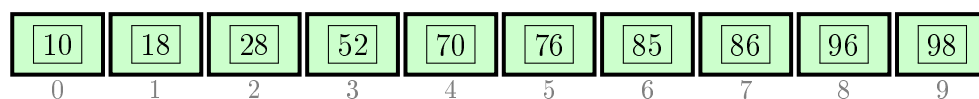


**Schritt 80:** Partition [6..7] fertig: Pivot 86 an Position 7



=== Rekursionsebene 7: Basisfall: [6] (Basisfall) ===

**Schritt 81:** quicksort(6, 6): l == r, einzelnes Element a[6]=85 ist sortiert



**Schritt 82:** Sortierung abgeschlossen

|    |    |    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|----|----|
| 10 | 18 | 28 | 52 | 70 | 76 | 85 | 86 | 96 | 98 |
| 0  | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  |

**Übersicht der Rekursionsschritte:**

Jede Zeile zeigt den Zustand nach einem Rekursionsschritt. Unterstrichene Teillisten werden im nächsten Schritt parallel verarbeitet. Umrahmte Werte haben ihre endgültige Position erreicht.

**Zustand 0:** Ausgangszustand

|    |    |    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|----|----|
| 28 | 10 | 76 | 52 | 98 | 18 | 96 | 85 | 86 | 70 |
| 0  | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  |

**Zustand 1:** Nach Rekursionsschritt 1

|    |    |    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|----|----|
| 18 | 10 | 28 | 52 | 98 | 76 | 96 | 85 | 86 | 70 |
| 0  | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  |

**Zustand 2:** Nach Rekursionsschritt 2

|    |    |    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|----|----|
| 10 | 18 | 28 | 52 | 98 | 76 | 96 | 85 | 86 | 70 |
| 0  | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  |

**Zustand 3:** Nach Rekursionsschritt 3

|    |    |    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|----|----|
| 10 | 18 | 28 | 52 | 70 | 76 | 96 | 85 | 86 | 98 |
| 0  | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  |

**Zustand 4:** Nach Rekursionsschritt 4

|    |    |    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|----|----|
| 10 | 18 | 28 | 52 | 70 | 76 | 96 | 85 | 86 | 98 |
| 0  | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  |

**Zustand 5:** Nach Rekursionsschritt 5

|    |    |    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|----|----|
| 10 | 18 | 28 | 52 | 70 | 76 | 96 | 85 | 86 | 98 |
| 0  | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  |

**Zustand 6:** Nach Rekursionsschritt 6

|    |    |    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|----|----|
| 10 | 18 | 28 | 52 | 70 | 76 | 86 | 85 | 96 | 98 |
| 0  | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  |

**Zustand 7:** Nach Rekursionsschritt 7

|    |    |    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|----|----|
| 10 | 18 | 28 | 52 | 70 | 76 | 85 | 86 | 96 | 98 |
| 0  | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  |

**Antworten:**

1. Nach erster Partition: **[18, 10, 28, 52, 98, 76, 96, 85, 86, 70]**
2. Nach letzter Partition: **[10, 18, 28, 52, 70, 76, 85, 86, 96, 98]**
3. Anzahl Vergleiche: **40**

**Endergebnis:** [10, 18, 28, 52, 70, 76, 85, 86, 96, 98]

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